

Final Year Project Report

Gait Recognition-based Authentication for User-Independent Physical Access Control

Rafferty Isaac Simms

Submitted in partial fulfillment of the requirements for the degree of
BSc Computer Systems (Cyber Security)

2024/25

COMP30151/52 Final Year Project

I am aware of the University's rules on plagiarism and collusion, and I understand that, if I am found to have broken these rules, it will be treated as Academic Misconduct and dealt with accordingly. I understand that if I lend this piece of work to another student and they copy all or part of it, either with or without my knowledge or permission, I shall be guilty of collusion.

In submitting this work I confirm that I am aware of, and am abiding by, the University's expectations for proof-reading.

I understand that I must submit this coursework by the time and date published. I also understand that if this coursework is submitted late, it will, if submitted within 5 working days of the deadline date and time, be given a pass mark as a maximum mark. If received more than 5 working days after the deadline date and time, it will receive a mark of 0%. For referred or repeat coursework, I understand that if the coursework is not submitted by the published date and time, a mark of 0% will be automatically awarded.

I understand that it is entirely my responsibility to ensure that I hand in my full and complete coursework and that any missing pages handed in after the deadline will be disregarded.

I understand that the above rules apply even in the eventuality of computer or other information technology failures.

Abstract

This project explores the development and feasibility of a gait recognition-based authentication system for physical access control, aiming to provide a less obtrusive and more user-independent alternative to traditional biometric and token-based systems. By leveraging Computer Vision (CV) and Machine Learning (ML) techniques, the system identifies individuals based on their lateral walking patterns, enabling passive authentication without the need for direct user interaction. The project proposes and implements a prototype that uses a client-server architecture, where the client captures gait data and the server performs identification and authorisation checks. Through functionality, performance, and user testing, the prototype demonstrates that gait recognition can serve as a viable biometric authentication method, offering advantages in usability and transparency while maintaining an acceptable level of security. The results suggest that with further refinement, gait-based authentication could enhance modern access control systems, particularly in environments where convenience, hygiene, or unobtrusiveness is prioritised.

Acknowledgements

I would like to start by expressing my gratitude to my supervisor, Dr. Nemitari Ajienka for his guidance throughout this project, and for his continued support and encouragement from the idea inception stage through to the final implementation.

I would also like to thank my friends and family for their support. Those that participated in user testing have provided valuable feedback, while general support and encouragement has motivated me to go above and beyond.

Finally, I would like to thank Nottingham Trent University and the School of Science and Technology. I have enjoyed completing my final year project and several other modules during my time here.

Contents

1	Introduction	1
1.1	Introduction	1
1.1.1	Aims and Objectives	2
1.1.2	Report Structure	4
2	Context	6
2.1	Introduction	6
2.2	The State of Physical Access Control Authentication	8
2.2.1	An Overview of Physical Access Control	8
2.2.2	Biometric Authentication Trends in Physical Access Control . . .	10
2.3	Gait Recognition-based Physical Access Control	13
2.3.1	Gait Recognition Overview	13
2.3.2	Computer Vision-based Gait Recognition Techniques	15
2.3.2.1	Model-based Approaches	15
2.3.2.2	Appearance-based Approaches	17
2.3.2.3	Classification Techniques	19
2.3.3	Applications of Gait Recognition in Physical Access Control . . .	21
2.3.4	Perceived Public Opinion (via Survey)	21
2.3.5	Review Summary	23
3	New Ideas	25
3.1	Introduction	25
3.2	Gait Recognition-based Authentication	25
3.2.1	Strength, Weakness, Opportunity, and Threat Analysis	26
3.3	Project Objectives Decomposition	27
3.4	Implementation and Test Planning	30
3.4.1	Gait Recognition Server Requirements	34
3.4.2	Access Control Client Requirements	35

3.4.3	Testing Plan	36
3.4.3.1	Functionality Testing	36
3.4.3.2	Performance and Accuracy Testing	38
3.4.3.3	User Testing	40
4	Implementation	41
4.1	Introduction	41
4.2	Research and Software Development Methodologies	42
4.3	Technologies Used and Techniques Applied	44
4.3.1	Choice of Gait-Recognition Approach	44
4.3.2	Implementation Technology Stack	45
4.4	Design	47
4.4.1	System Architecture Recap	47
4.4.2	Gait Energy Image Creation	47
4.4.3	Access Control Client Design	49
4.4.3.1	Identity Registration Process	49
4.4.3.2	Identity Classification Process	50
4.4.3.3	User Interface Design	52
4.4.4	Gait Recognition Server Design	53
4.4.4.1	Identity Registration Process	54
4.4.4.2	Identity Classification Process	54
4.4.4.3	Access Rule Configuration Process	55
4.4.4.4	Database Design	56
4.4.4.5	User Interface Design	57
4.5	Implementation	59
4.5.1	Image Processing and Gait Energy Image (GEI) Creation	59
4.5.1.1	Capturing Camera Input	60
4.5.1.2	Background Subtraction and Person Detection	60
4.5.1.3	Silhouette Normalisation	62
4.5.1.4	Silhouette Averaging	63
4.5.2	Access Control Client	65
4.5.2.1	Application State Tracking	65

4.5.2.2	Client for Gait Recognition Server API	66
4.5.2.3	Frame Capture, Processing, and Transmission	68
4.5.2.4	Identity Classification Thread	69
4.5.2.5	Identity Registration Process	70
4.5.2.6	Manual Identity Classification	71
4.5.2.7	User-Interface	72
4.5.3	Gait Recognition Server	74
4.5.3.1	Database Modelling	74
4.5.3.2	Feature Extraction, Model Training, and Classification .	75
4.5.3.3	Identity Registration Endpoint	78
4.5.3.4	Identity Classification Endpoint	79
4.5.3.5	Access Control Rule Management	79
4.5.3.6	Identity and Gait Cycle Management	80
4.5.3.7	Statistic and Log Processing	82
4.5.3.8	User-Interface	82
4.6	Implementation Summary	84
5	Results and Discussion	86
5.1	Introduction	86
5.2	Functionality Testing	86
5.3	Performance and Accuracy Evaluation	94
5.4	User Testing Feedback	98
5.5	Evaluation Summary and Feasibility Discussion	101
6	Conclusion and Future Work	103
6.1	Introduction	103
6.2	Conclusion	103
6.2.1	Achievements	103
6.2.2	Criticism	104
6.2.3	Future Work	105
6.2.4	Final Thoughts	106
6.3	Legal, Ethical, Social, and Professional Issues	106
6.3.1	Legal Issues	106

6.3.2	Ethical Issues	107
6.3.3	Social Issues	107
6.3.4	Professional Issues	108
6.4	Personal Reflection and Aspirations	108
Appendices		116
A Forms and Questionnaires		116
A.1	Public Opinion Survey	116
A.2	Participant Information Sheet	117
A.3	Informed Consent Form	120
B Risk Assessment Matrix		121
C Implementation		122
C.1	Code Extracts	122
C.1.1	The 'Camera' Class	122
C.1.2	The 'GaitProcessor' Class	123
C.1.3	Client-Side Identity Classification Thread	126
C.1.4	SQLAlchemy Database Models	126
C.1.5	Activity Log Retrieval and Formatting	127
C.2	Client Code Architecture Diagram	129
C.3	Server Code Architecture Diagram	130
D Training and Testing Automation		131
D.1	Automated Registration Submission Function	131
D.2	Automated Classification Query Function	132
E Project Proposal Document LESPIs		133

List of Figures

1.1	Gait Recognition-based Access Control System Topology	3
2.1	Radio Frequency Identification (RFID) Access Control Example	9
2.2	Access Control Flow Overview	9
2.3	Examples of Biometric Characteristics	11
2.4	2D Pose Estimation of a Still Image using OpenPose	16
2.5	Silhouette and Gait Energy Image (GEI) Image Processing Example . .	18
3.1	Gantt Chart for Implementation of a Gait Recognition System	31
4.1	Agile Software Development Process	42
4.2	System Architecture and Technologies	47
4.3	Steps involved in Gait Energy Image (GEI) Creation	48
4.4	Access Control Client Features Overview	49
4.5	Access Control Client Registration Process Sequence Diagram	50
4.6	Access Control Client Identification Process Sequence Diagram	51
4.7	Access Control Client 'Live Classification' Page	52
4.8	Access Control Client 'Manual Classification' Page	53
4.9	Server Identity Registration Process Sequence Diagram	54
4.10	Server Identity Classification Process Sequence Diagram	55
4.11	Server Access Rule Configuration Process Sequence Diagram	56
4.12	Server Database Entity Relationship Diagram	56
4.13	Gait Recognition Server 'Home' Page	58
4.14	Gait Recognition Server per-'Identity' Page	58
4.15	Gait Recognition Server 'Statistics and Logs' Page	59
4.16	Background Subtraction Process Results	62
4.17	Normalisation Process Results	63
4.18	Example Silhouette Averaging Result	65
4.19	Live Classification Page User-Interface	73

4.20	Manual Classification Page User-Interface	73
4.21	'Server Home' Page User-Interface	83
4.22	'Identity' Page User-Interface	83
4.23	'Statistics & Logs' Page User-Interface	84
5.1	System Testing Hardware Setup	87
5.2	Post-Cycle-Capture Gait Energy Image (GEI) Preview	87
5.3	New Identity Added to Registered Identities Grid	88
5.4	Identity Removal Confirmation Dialog	88
5.5	Volunteer Testing Dataset Preview	89
5.6	Identity Classification of Known Subject Result	89
5.7	Identity Classification of Unknown Subject Result	90
5.8	Access Rule Configuration Toggles	91
5.9	Access Rule Configuration in Database	91
5.10	Identity Classification of Known Subject Result with Access Allowed . .	91
5.11	Errors Displayed by the Client if the Server is Not Running	92
5.12	Live Feed, Silhouette, and Gait Energy Image (GEI) Mid-Cycle Previews	93
5.13	Server Activity Log Database Entries Example	93
5.14	Server UI Activity Log Display	94
5.15	GEI Registration Response Time Distribution	95
5.16	GEI Registration Response Time Trend	96
5.17	GEI Classification Confidence Distribution	97
5.18	GEI Classification Response Time Distribution	97

List of Listings

4.1	OpenPose Configuration for Benchmarking	45
4.2	OpenCV (CV2) 'VideoCapture' Class Usage	60
4.3	Default Access Control Client State Configuration	66
4.4	Server API Client register_gei Method Logic	67
4.5	Server API Client classify_gei Method Logic	67
4.6	Example 'Camera' and 'GaitProcessor' Class usage	68
4.7	'Live', 'Silhouette', and 'GEI' Frame Transmission via WebSocket (WS)	68
4.8	JavaScript WebSocket Listener for Receiving Frames	69
4.9	Front-End JavaScript Listener for Receiving Classification Updates . . .	70
4.10	JavaScript Button-Click Handler for Saving Gait Energy Image (GEI)- based Identity	70
4.11	JavaScript Function for Manual GEI Classification	71
4.12	Server '/api/classify' Endpoint Implementation	72
4.13	Feature Extraction Model Definition and Usage	76
4.14	PCA Model Training Logic	76
4.15	Feature Extraction and PCA Projection Logic	77
4.16	Cosine Similarity Comparison between Input and Samples	77
4.17	Identity Registration Endpoint Logic	78
4.18	Identity Classification Endpoint (Successful) Response	79
4.19	Access Rule Update Logic	80
4.20	Access Rule Toggle Front-End Handler	80
4.21	Identity Deletion Logic	81
4.22	Identity Deletion Front-End Handler	81

List of Tables

1.1	Project Objectives	3
2.1	Key Literature Review Resources	6
3.1	Project Task List	27
3.2	Risk Assessment and Mitigations	32
3.3	Gait Recognition Server Functional Requirements	34
3.4	Gait Recognition Server Non-Functional Requirements	35
3.5	Access Control Client Functional Requirements	35
3.6	Access Control Client Non-Functional Requirements	36
3.7	Gait Recognition Server Core Feature Tests	37
3.8	Performance Testing	39
4.1	Benchmark and Testing System Specifications	44
4.2	Gait Data Extraction Time Benchmark Results	45
4.3	Implementation Technology Stack: Purpose and Justification	46
5.1	Classification Accuracy Metric	96

List of Abbreviations

ACC	Access Control Client
AI	Artificial Intelligence
API	Application Programming Interface
AR	Aspect Ratio
CCTV	Closed-Circuit Television
CNN	Convolutional Neural Network
CV	Computer Vision
CV2	OpenCV
DL	Deep Learning
DRY	Don't Repeat Yourself
EMA	Exponential Moving Average
ERD	Entity-Relationship Diagram
GDPR	General Data Protection Regulation
GEI	Gait Energy Image
GPU	Graphics Processing Unit
GR	Gait-Recognition
GRS	Gait Recognition Server
HTTP	Hypertext Transfer Protocol
JS	JavaScript
JSON	JavaScript Object Notation
LDA	Linear Discriminant Analysis

LESPI Legal, Ethical, Social, and Professional Issues

MFA Multi-Factor Authentication

ML Machine Learning

MVP Minimum Viable Product

NN Nearest Neighbour

OOP Object-Oriented Programming

PACS Physical Access Control System

PCA Principal Component Analysis

PDA Pairwise Distance Analysis

PII Personally Identifiable Information

PIN Personal Identification Number

PoC Proof-of-Concept

PPD Project Proposal Document

REST REpresentational State Transfer

RFID Radio Frequency Identification

RNN Recurrent Neural Networks

SVM Support Vector Machine

UI User-Interface

WS WebSocket

Chapter 1

Introduction

1.1 Introduction

Physical Access Control System (PACS) are used to restrict access to physical areas, with most implementation relying on traditional authentication methods such as Radio Frequency Identification (RFID) key cards or Personal Identification Number (PIN) codes. While effective in enforcing basic access policies, these approaches suffer from several limitations, including their reliance on physical or intangible credentials that can be lost, stolen, or forgotten. The lack of usability and convenience for end users — constantly needing to carry a single-purpose physical token, and interact with an access control panel, for example — leaves much to be desired.

Biometric authentication has grown in popularity in recent years, including within PACS deployments. This type of authentication enables use of inherence-factor-based identification methods like fingerprint, iris, and facial recognition. Although these introduce a significant improvement to security by tying credentials to individual's physiology, most still require active participation and physical interaction with some form of access control interface. Additionally, bias-related challenges, such as unbalanced facial recognition datasets leading to people (especially women) of darker skin tone being misclassified up to 'thirty-five times or more higher' than their white male counterparts, have raised ethical concerns regarding the use of biometric recognition technologies overall.

This project explores the use of Gait-Recognition (GR) as a lesser-used biometric for access control. By assessing the way a person walks — a behavioural biometric — the aim is to enable transparent, contactless authentication that maintains security without disrupting user flow. GR presents an opportunity to reduce friction for PACS users while mitigating the risks associated with token-based systems and more obtrusive or intrusive biometric modalities.

To evaluate this approach, a prototype PACS was developed that uses CV to extract movement information from live video input. This data is abstracted into a GEI — representing statistical and spatio-temporal features of a subject’s gait — which is used to enable classification and identity verification. The system follows a client-server architecture where the client is a lightweight application responsible for capturing video input and abstracting it into a gait cycle representation, while the server is responsible for gait sample storage, classification, and authorisation rule management. The resulting solution aims to demonstrate the feasibility of GR-based authentication and its potential for real-world use to enable secure, unobtrusive physical access control.

The proposed GR-based access control system can be described as enabling ‘pseudo-transparent authentication’, providing an authentication mechanism that operates passively and is not immediately apparent to the user, but continues to maintain security by only permitting access to authorised individuals.

1.1.1 Aims and Objectives

As defined above, this project aims to deliver a prototype access control system that uses GR to passively identify subjects, in an attempt demonstrate the feasibility of assessing gait to perform authentication. A distributed architecture is targeted, with a central application responsible for identification, receiving movement data input from a connected client or clients. The idea is that future development could introduce support for multiple simultaneous clients, enabling gait-based authentication to be implemented across a site.

Figure 1.1 below provides an overview of the desired system architecture, made up of two main components: (a) the Access Control Client (ACC) that captures movement data using CV techniques, sending gait data to (b) the Gait Recognition Server (GRS), which assesses the input and returns the associated identity. After receiving the identity prediction result and associated access control rule for the given input. The ACC will evaluate this information to determine if access should be granted.

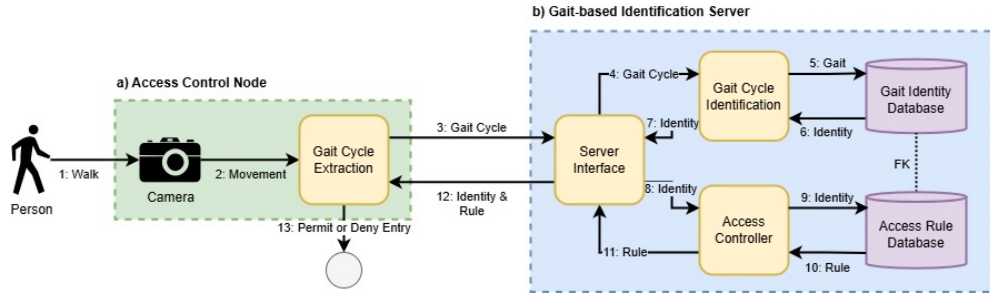


Figure 1.1: Gait Recognition-based Access Control System Topology

GR-based Identification This *could* be achieved using ML to evaluate gait cycle data, or simpler methods of similarity measurement. As discussed in Chapter 2, classification models can be trained on historic data, enabling subsequent identification of individuals from real-time (or historic) footage. Alternatively, Proof-of-Concept (PoC) could be achieved using simpler forms of similarity measurement like Pairwise Distance Analysis (PDA), model-free Nearest Neighbour (NN) classification, or cosine similarity to assess similarities between input and historic data directly. However, this is likely to be less effective than ML model-based approaches, but may still enable demonstration of GR as a PoC.

Access Control System Integration CV-based GR is widely researched but is yet to see widespread adoption in real-world scenarios. The ACC component would leverage the aforementioned GRS to demonstrate authentication, illustrating how it could be used in a PACS context. As shown in Figure 1.1 above, movement data would be captured by the ACC and transmitted to the GRS, access would then be granted or denied in respect to a Boolean rule associated with the identity, mimicking how traditional PACS operate (see Section 2.2.1).

Meeting these aims will require several objectives to be met, some of which are dependent on the prior completion of others. Table 1.1 below outlines five (5) primary objectives of this project and specifies related dependencies.

Table 1.1: Project Objectives

#	Objective	Dependencies
1	Produce a prototype CV-based gait identification system. It should accept movement data as input and return an associated identity if found (e.g., matching with a previously registered identity).	

Project Objectives (Continued)

#	Objective	Dependencies
2	Produce a prototype ACC that leverages the gait-based identification system. The ACC should leverage the result of #1 to enable identity authentication, allow for registration of new identities, support configuration of authorisation rules, and demonstrate provision or denial of access (as a PoC).	1
3	Test and evaluate the performance and accuracy of the gait-based identification. The performance of #1 and #2 should be evaluated in order to determine performance and accuracy.	1
4	Perform user testing (diverse subjects) and demonstrations to receive feedback and opinions. Evaluate the final system to determine how well it meets initial goals and compares to other authentication methods in terms of usability. Address concerns and recommend improvements.	1, 2
5	Evaluate the feasibility of implementing gait-based physical access control in the real-world. Based on the results of #3 and #4, the feasibility of (CV) GR-based PACS being implemented in the real-world should be discussed, alongside recommendations for further developments if necessary.	3, 4

1.1.2 Report Structure

This report is structured into five chapters, each covering a different aspect of research, development, or evaluation:

- **Chapter 1 — Introduction:** Introduces the core problem area, the motivation behind the project, and the concept of GR as a potential solution. The goals of the project are also outlined here.
- **Chapter 2 — Context:** Provides an in-depth review of existing PACS systems, authentication techniques, and the position of GR within the broader scope of biometric authentication. Key issues with current techniques are discussed to justify the project's direction.
- **Chapter 3 — New Ideas:** Outlines the proposed solution in detail, including system aims, objectives, and a breakdown of tasks to be completed. Implementation options are evaluated, a SWOT analysis is conducted, and planning is covered via requirement definitions, a Gantt chart, and test planning.

- **Chapter 4 — Implementation:** Describes how the system was built, covering development methodology, tools, technologies, and the implementation of each component. This includes the gait cycle extraction logic, the ACC, the GRS, and the communication between them.
- **Chapter 5 — Evaluation and Discussion:** Presents the results of functionality, performance, and user testing. Reflections on the effectiveness of the solution are provided, along with identified issues and limitations and considerations for real-world use. The chapter ends with a discussion on future directions.
- **Chapter 6 — Conclusion:** Summarises achievements and criticisms based on the outcome of the project, and provides recommendations for future work if improvements are to be made. Discusses pertinent Legal, Ethical, Social, and Professional issues, and concludes with a personal reflection..

Each chapter builds upon the last, gradually leading from concept to implementation and evaluation, with the aim of clearly demonstrating the feasibility of using GR-based authentication for physical access control.

Chapter 2

Context

2.1 Introduction

This chapter evaluates existing examples of PACS and their associated authentication methods. Security and usability-based issues encountered with traditional PACS are discussed, with comparisons made to systems newer, biometric-based solutions.

Following this, the convenience and obtrusiveness of, as well as ethical concerns relating to common biometric authentication solutions is assessed. A proposition is then made for using a less intrusive biometric factor, GR, to enable less obtrusive, 'transparent' authentication for PACS — aiming to reduce user burden while maintaining security.

Key Resources While many resources contributed to this review, the most relevant articles are listed below in Table 2.1, with details on what they contribute, what may be missing, and how this research could build upon their findings.

Table 2.1: Key Literature Review Resources

Resource(s)	Contributions	Gaps and Issues	New Ideas
<i>The 2022 State of Physical Access Control Report</i> , IFSEC Global, 2022; <i>State of Physical Access Trend Report 2024</i> , IFSEC Global, 2024	Identifies trends in the PACS market and discusses how issues with traditional technologies may have triggered the (slow) migration to biometric solutions by organisations.	Does not acknowledge GR as a viable option for biometric PACS authentication, indicating it is rarely (or never) used in this context.	The gait (walking style) behavioral biometric could serve as an alternative for identity verification compared to facial recognition, fingerprint, and iris recognition.

Key Literature Review Resources (Continued)

Resource(s)	Contributions	Gaps and Issues	New Ideas
<i>Guide to Secure Physical Access Control Systems</i> , Verkada, 2024; <i>What Is Physical Access Control?</i> , Maxsenti, 2023	Covers both traditional and emerging PACS methods such as RFID and smartphone-based authentication, highlighting vulnerabilities and disadvantages like the risk of credential theft and the need to constantly carry physical items.	Recommendations for alternatives remain token-based, albeit more secure; however, biometrics are not discussed in detail.	Inherence factor-based authentication (e.g., GR proposed here) enhances security and usability, reducing the risk of credential theft and eliminating the need for users to carry physical items.
‘Biometric Recognition’, Prabhakar, Pankanti and Jain, 2003; <i>Biometric Data</i> , Parliamentary Office of Science and Technology, Saverat and Brawley, 2024	Covers the types of data that enable different forms of biometric identification, reporting how detail, distinctiveness, and permanence varies between approaches and relates to privacy.	Gait is not covered as a potential biometric identifier by these reports.	Comparisons can be made between methods covered by these reports and GR to highlight the contrast in invasiveness and obtrusiveness, potentially reducing (not eliminating) privacy concerns, and improving usability.
‘NIST Evaluates Face Recognition Software’s Accuracy for Flight Boarding’, National Institute of Standards and Technology, 2021; ‘NIST Study Shows Computerized Fingerprint Matching Is Highly Accurate’, National Institute of Standards and Technology, 2004; ‘High-Accuracy Models for Iris Recognition with Merging Features’, University of Baghdad et al., 2024	Details accuracies of the three (3) most common PACS biometrics: face, fingerprint, and iris. Conveys how accuracy increases with the amount of data available for analysis.	Emphasises how most systems require the active participation of subjects and cannot operate passively from distance — a potential time sink.	Gap in the market for PACS solutions that take enable passive identification. CV based GR could be a solution, authenticating subjects from a distance, without active participation.

Key Literature Review Resources (Continued)

Resource(s)	Contributions	Gaps and Issues	New Ideas
<i>Understanding Bias in Facial Recognition Technologies</i> , Leslie, 2020; ‘How Police Technology Aggravates Racial Inequity’, Moy, 2019; ‘Police Use of Live Facial Recognition Technology’, Uwazuruike et al., 2025	Highlights bias-related issues in biometrics, like facial recognition models trained on unbalanced data, negatively impacting minorities (e.g., darker skin complexions or women). Conveys the importance in collecting diverse datasets to reduce the proliferation of biases by ML.	The most prevalent biometric-related biases involve race and sex, impacting facial and voice recognition systems the most. These reports do not address how or if gait is affected by these biases.	ML for GR still requires datasets to be collected for training. However, there are less opportunities for bias; diversity in race is of less importance, with a greater focus on balancing sex and having a range of ages.
<i>Gait Authentication Security System / Stepscan®</i> , Stepscan, 2020	Introduces a GR based PACS solution that uses under-floor pressure sensors to assess gait.	The cost of hardware increases the greater the size of the area that must be covered, outweighing the benefits of using GR for large-scale applications.	Take advantage of CV to enable GR to be used for identification from greater distances, at a lower cost, providing a Minimum Viable Product (MVP) demonstration using minimal hardware.
<i>PopEye to Strengthen EU Border Biometrics with Gait Recognition Integration / Biometric Update</i> , Chris Burt, 2024	Introduces a European Union consortium proposal for GR-based authentication to be used at border crossings in an attempt to speed up processing times using CV-based solutions.	The experiment is not yet complete with results unpublished. There have been no updates as of October 2024.	This research relates to what the EU consortium is attempting to achieve and could help demonstrate feasibility among other things.

2.2 The State of Physical Access Control Authentication

2.2.1 An Overview of Physical Access Control

The purpose of a Physical Access Control System (PACS) is to restrict access to a location. Individuals that wish to access an area must present credentials (e.g., an employee ID card) which are used to validate their identity. After validation, the individual may be granted access, if authorised (Maxsenti, 2023).

All access control processes consist of two primary stages: verification or authentication,

and authorisation (see Figure 2.2). Identity verification confirms that a 'thing' is valid, like evaluating details on an Radio Frequency Identification (RFID) key card, whereas authentication confirms a *person* is who they say claim. This is done using knowledge, possession, and/or inference factors.

In the subsequent authorisation stage, a set of configured rules are evaluated to determine whether the individual associated with the provided credentials has permission to access the area or resource requested (Sandhu and Samarati, 1994).

IFSEC Global, 2024 found that 63% of enterprise PACS still use key card or token-based controls (see Figure 2.1). RFID-based access control is not new, having seen use since the 1970s (Camilla Ashdown and 2N, 2023). Systems that continue to use traditional controls are more likely to be vulnerable to attack, often through failure to implement adequate encryption. This may allow an attacker to make a copy of a key card, for example — later using it to gain unauthorised access to an area.



Figure 2.1: RFID Access Control Example

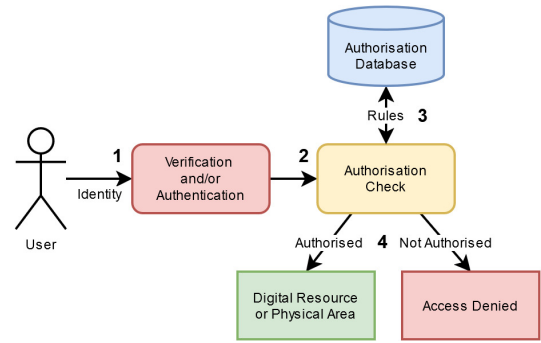


Figure 2.2: Access Control Flow Overview

The physical nature of RFID-based controls means some vulnerabilities remain even after implementation of adequate encryption. For instance, attackers are still able to steal the original key card. Another traditional access control method, Personal Identification Number (PIN) codes, are also vulnerable to theft, despite being intangible. Theft of PIN codes is even less perceptible due to there being no physical evidence (Verkada, 2024). These vulnerabilities remain predominantly due to a lack of authentication, allowing anyone to gain access to an area if in possession of valid credentials.

Trends in similar surveys suggest organisations are slowly moving away from traditional RFID-based access control. An alternative, but still token-based solution is smartphone authentication, which has seen a 7% increase in use between 2022 and 2024 (IFSEC

Global, 2022). Smartphone-based authentication does not suffer from the same vulnerabilities as traditional technologies, with the devices being harder to clone, or gain access to if stolen. The sensors present in many modern phones (e.g., fingerprint scanners) and need for a PIN, enables Multi-Factor Authentication (MFA) — enhancing security.

Usability of PACS could also be improved. As discussed, RFID-based controls require users to carry a physical item holding their unique credentials. In most if not all existing cases, users must direct their attention to or otherwise interact with the respective interface when providing credentials. This could be considered inconvenient for users, especially those who are otherwise authorised and must stop to authenticate multiple times per day (IFSEC Global, 2024)

Although smartphone-based authentication reduces the total number of items a person must carry, time penalties (although small) are still suffered, as users must stop to interact with their smartphone to authenticate. There also remains a risk of theft, or of users misplacing their smartphone, making them unable to authenticate and undermining availability (Maxsenti, 2023).

Many new access control solutions, some discussed in 2.2.2 below, enhance both security and usability by leveraging biometrics for authentication. The nature of these systems and the type of data processed, however, has raised several ethical concerns regarding privacy, bias, and data misuse (Prabhakar, Pankanti and Jain, 2003; Uwazuruike et al., 2025; Parliamentary Office of Science and Technology, Saverat and Brawley, 2024).

2.2.2 Biometric Authentication Trends in Physical Access Control

Biometric authentication methods assess unique biological characteristics — inherence factors — to determine identity. The two main biometric characteristic categories are physiological: the physical parts of a person (e.g., fingerprints, face, or irises) and behavioural: the way a person interacts with an environment or system (e.g., walking style or keystroke pattern).

Figure 2.3 below shows examples of (a) physiological and (b) behavioural characteristics. Most PACS that support biometric authentication leverage physiological factors (IFSEC Global, 2024). Behavioural factors, like gait — explored as a viable alternative by this

project — could improve usability while maintaining the existing level of security we are used to (admin, 2024).

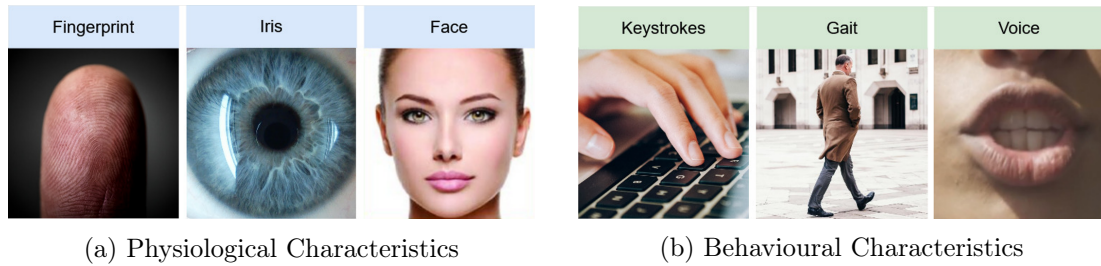


Figure 2.3: Examples of Biometric Characteristics

In the context of PACS, biometrics benefit usability by not requiring users to keep hold of a physical credential token, nor remember a secret value, with identities validated by assessing biological factors. The reduced user burden also benefits security, as it is much harder for a person’s likeness to be imitated convincingly enough to compromise the integrity of a biometric identification system (Terranova Security, 2023).

IFSEC Global, 2024 also reported that 32% of surveyed organisations made use of fingerprint, face, or iris-based biometric authentication. Fingerprint recognition assesses the pattern or ridges on a person’s fingertip(s). A National Institute of Standards and Technology, 2004 study achieved a 98.6% match accuracy using a single finger, rising to 99.9% for two or more fingerprints. Studies evaluating facial and iris recognition reported similar findings: 99.5% and 98.6% accuracy respectively (National Institute of Standards and Technology, 2021; University of Baghdad et al., 2024)

The type of information processed by biometric identification systems — Personally Identifiable Information (PII) — raises concerns regarding privacy. Although required to enable functionality, improper PII handling can have a negative impact. Stolen biometric records could be used for impersonation, or reveal confidential health information, for example (Parliamentary Office of Science and Technology, Saverat and Brawley, 2024). This risk necessitates secure and minimal PII storage, where data handlers must only store what is needed, while it is needed. Governments have published mandates for how PII must be handled (Information Commissioner’s Office, 2024).

Additionally, the use of biometric identification systems by law enforcement, like the facial recognition technology installed as part of security upgrades in preparation for

the London 2012 Olympics (Home Office, 2011), has led to ethical concerns regarding biases that could cause harm through misattribution or impact usability for minorities (Uwazuruike et al., 2025; Leslie, 2020)

Biometric recognition technologies use historic data to classify subsequent input based on similarities in features (e.g., to identify a person using an image of their face). This requires the creation of datasets to train Machine Learning (ML) models (see Section 2.3.2.3) or to be used as a simpler reference database. Failure to collect diverse sets of information — many publically available datasets are unbalanced toward the default white male — can result in embedded biases, impacting a system’s effectiveness and the experience of minority users. Facial recognition is one of the most common PACS biometrics, yet has notoriously suffered.

Leslie, 2020 reported that six out of a set of eight major facial recognition datasets studied contained more male images than female, and that ‘six are composed of over eighty percent light-skinned faces’. Biases like these can lead to misclassification, with darker-skinned women at times misclassified ‘thirty-five times or more higher than for white men’. In the context of access control, misclassification could undermine confidentiality, integrity, and availability by incorrectly identifying an unauthorised user as an authorised user, or vice versa.

Biases in surveillance and law enforcement systems can be more harmful. Misclassification may lead to someone being incorrectly attributed to a crime they did not commit, for example (Moy, 2019). The use of alternative biometric characteristics (i.e., gait) may help reduce these biases. Gait-Recognition (GR) is less affected by varying skin complexion, reducing the number of diversity dimensions to focus on when creating balanced datasets. The most important dimensions for GR appear to be sex and age, influencing body size and shape (Guan, Wei and Li, 2014).

Despite improving security and usability compared to traditional methods, the three most common biometric options do not significantly contribute to making authentication ‘transparent’. Fingerprint, iris, and most facial recognition systems require interaction with a physical interface before entry. Of these, facial recognition is the most transparent, capable of operating from a distance in some implementations (Boon Edam, 2024). Unfortunately, facial recognition often fails to work as expected in dynamic environ-

ments, with varying camera resolution, lighting conditions, and use of face coverings affecting data quality and impacting recognition accuracy (Braje et al., 1998). These issues may be amplified when combined with concerns regarding biases in the datasets used to train the classification models (e.g., identifying darker-skinned individuals in low-light conditions).

Behavioural characteristics like gait (or keystroke analysis in a digital context) may enable less obtrusive authentication, eliminating several drawbacks of existing solutions, while maintaining an expected level of security. These approaches work by assessing activity patterns over space and time to determine identity — typically without requiring user’s cooperation or even knowledge.

2.3 Gait Recognition-based Physical Access Control

2.3.1 Gait Recognition Overview

Gait-Recognition (GR) technology assesses a combination of biometric information to determine identity based on walking patterns. Both physiological (e.g., body shape) and behavioural (e.g., walking speed) attributes can contribute, with data typically collected using Computer Vision (CV) techniques Mallick et al., 2016, or in some cases, under-floor pressure sensors (Stepscan, 2020).

As discussed in Section 2.2.2, biometric identification is enabled by analysing features derived from biological data (e.g., face shape and nose position for face recognition). Gait-related features can be statistical, or spatiotemporal. Statistical features describe attributes over time, like average stride length and speed, and their standard deviation and variability, whereas spatiotemporal features describe a subject’s movement over a defined area and (time) period. Gait analysis often uses a combination of statistical and spatiotemporal information, with assessment of an entire movement cycle typically necessitating both (Güner Şahan, Şahin and Kaya Gülağız, 2024).

The level of security enabled by common biometric authentication techniques (see Section 2.2.2) could be maintained by GR — analysing hard-to-replicate, difficult-to-disguise inference factors (Neves et al., 2017). By assessing body movement to determine identity, GR can work from a distance without user participation, contrib-

uting to what this project aims to achieve. Unlike facial recognition, GR is impacted less by lower sensor resolutions (many appearance-based solutions, see 2.3.2.2, use low-resolution silhouette images) and face coverings — benefiting reliability (Cooper et al., 2022). However, occlusion of other body parts may still have a performance impact (especially for CV-based implementations).

There are not many examples of real-world Physical Access Control System (PACS)s using GR for authentication despite the moderate amount of CV-based GR research that exists. CV-based GR has seen clinical use, however, where gait has been assessed to treat stroke victims and Cerebral Palsy (Davis, 1997). Law enforcement have also demonstrated how GR can be used to asynchronously identify criminals from Closed-Circuit Television (CCTV) footage (Nebula M P, 2024). These examples prove a significant amount of information can be discerned from gait, but do so outside PACS constraints, within which there is a target for how long identification takes to ensure users can be promptly granted access (Sandhu and Samarati, 1994). *This project applies techniques like the above in a real-time context to achieve hands-free user authentication for use in a PACS, performing identification based on movement information captured from a live video feed.*

A rare real-world example of GR being used for authentication is 'Stepscan Secure' — a product described as the 'world's first pressure-based gait-authentication system for access control'. It uses under-floor pressure sensors to assess walking patterns (Stepscan, 2020). While pricing information is not publicly available (and Stepscan did not respond to queries), the assumption can be made that the cost of implementation would be higher than a CV-based solution that uses just a camera. This is because enough pressure sensors must be purchased to span an entrance way, which would also increase costs as the operational distance requirements grow, compared to a CV-based approach which would inherently work at further distances, restricted by only what camera resolution allows.

In reference to the ethical topics discussed in 2.2.2: bias concerns also apply to GR, but unlike facial recognition, relate more to sex and age, calling for on diversity in body shape for datasets (especially in CV-based GR) and not necessarily in skin tone. The CASIA-B dataset (124 subjects captured from 11 angles) is widely used for benchmark-

ing in GR research, but does not specify details about, nor appear to target diversity. However it does visibly include a split between male and female subjects (Fu et al., 2023). Comparatively, the OU-ISIR Large Population Dataset, is advertised to contain subjects aged 'ranging from 1 to 94 years' and a balanced gender distribution (Iwama et al., 2012).

The following sections assess different CV-based GR techniques, each of which can be used to identify and authenticate individuals effectively by PACS.

2.3.2 Computer Vision-based Gait Recognition Techniques

2.3.2.1 Model-based Approaches

Model-based GR systems are a CV-based approach that use attributes extracted from 2 or 3-dimensional 'models' to determine identity. These models, often 'skeleton'-based (collections of joint positions) can be constructed using open-source libraries like 'OpenPose', which analyse footage and 'detect human body, hand, facial, and foot key points'. Figure 2.4 shows (a) an RGB image given as input to OpenPose and (b) the output 2D pose prediction (Cao et al., 2019). Purpose-made hardware like Microsoft's 'Kinect' sensors is also capable of capturing joint information in real-time, this can be used to construct similar models or for analysing the statistical data (e.g., in JavaScript Object Notation (JSON) format). Model-based data enables features like pose, limb lengths, and joint angles to be extracted, contributing to Machine Learning (ML) classification efforts for person identification (Liao et al., 2020).

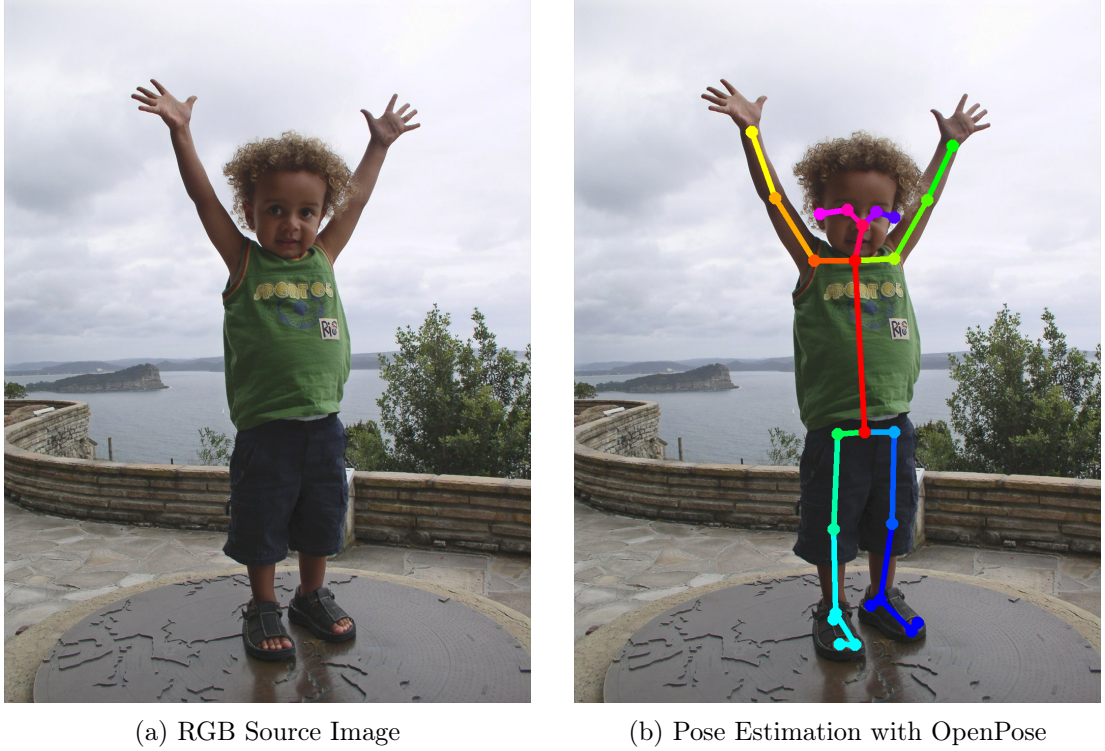


Figure 2.4: 2D Pose Estimation of a Still Image using OpenPose

Some existing examples of model-based GR use extracted features like pose and limb length to create vectors representing individual’s gait cycle — a ‘gait signature’. Liao et al., 2020 proposed ‘PoseGait’: a GR system that constructs vectors of four concatenated features, fed into a Convolutional Neural Network (CNN) for classification. It achieved accuracies around 90% under controlled conditions when tested using the standard CASIA-B dataset. Another proposal — ‘GaitPT’ — achieved similar accuracies using a ‘layered classification’ approach: first (1) assessing spatio-temporal interactions at the joint level, then (2) groups of joints (limbs), groups of limbs, and finally (3) movement of the overall model (Catruna, Cosma and Radoi, 2024). These results prove GR to be a feasible and similarly accurate alternative to facial and fingerprint recognition.

Model-based approaches perform well against varying view-angles. Sensor position relative to the subject has less of an impact on performance. They are also more resilient to occlusion compared to both facial recognition and appearance-based GR, with advancements in ‘occlusion-aware ML models’ enabling high-accuracy joint and pose estimation, even with parts of the body covered (Xu et al., 2024). These advantages can provide value in the context of PACS, where in diverse environments like workplaces, sensors

(for CV-based GR) may be placed in different locations, and parts of subjects may often be obscured (Parashar et al., 2024).

Due to the high reliance on ML and CV to capture footage, construct models, and later perform classification, model-based GR approaches can be highly computationally expensive — requiring powerful, expensive hardware. For instance, in the GaitPT study, an Nvidia A100 GPU (priced at north of US \$10,000) was used during development (Catruna, Cosma and Radoi, 2024). Resource requirements not being met can result in slower model training and classification times, if opting for ML. Considerations must be made as to whether the benefits of model-based GR are worth the increased cost, or if slower identification is acceptable — impacting usability-related aims of this project (Parashar et al., 2024).

2.3.2.2 Appearance-based Approaches

Appearance-based GR systems are another CV approach that assess the movement pattern of a subject using statistical and spatio-temporal feature representations. CV is a core component of appearance-based approaches, usually capturing movement from a lateral view (or multiple angles in advanced implementations). Complexity of techniques used for classification can vary from simpler similarity comparisons to ML/Deep Learning (DL)-based analysis which often enhances accuracy and reliability (Bishop, 2006).

Many appearance-based GR proposals use silhouette and/or Gait Energy Image (GEI)-based feature representations (Güner Şahan, Şahin and Kaya Gülağız, 2024). Figure 2.5 shows (a) an RGB image sequence, (b) a silhouette sequence, and (c) a GEI image (Nunes, Moreira and Tavares, 2019). Silhouette-based GR isolates subjects from video frames, producing a sequence of binary pose images that can be used to create a 'signature' representing an individual's gait (Boyd and Little, 2009). The GEI approach averages the sequence of silhouettes, combining the spatial and temporal information into a single grayscale image describing the complete gait cycle, with pixel brightness correlating with frequency of movement (Han and Bhanu, 2006).

Using GEI has shown to reduce computational cost, as complex algorithms used for feature extraction and historic-input data comparisons must only be performed on the

single image, as suppose to multiple silhouettes. This speeds up processing times and reduces the amount of storage space required per subject (Ling and Jacobs, 2007).

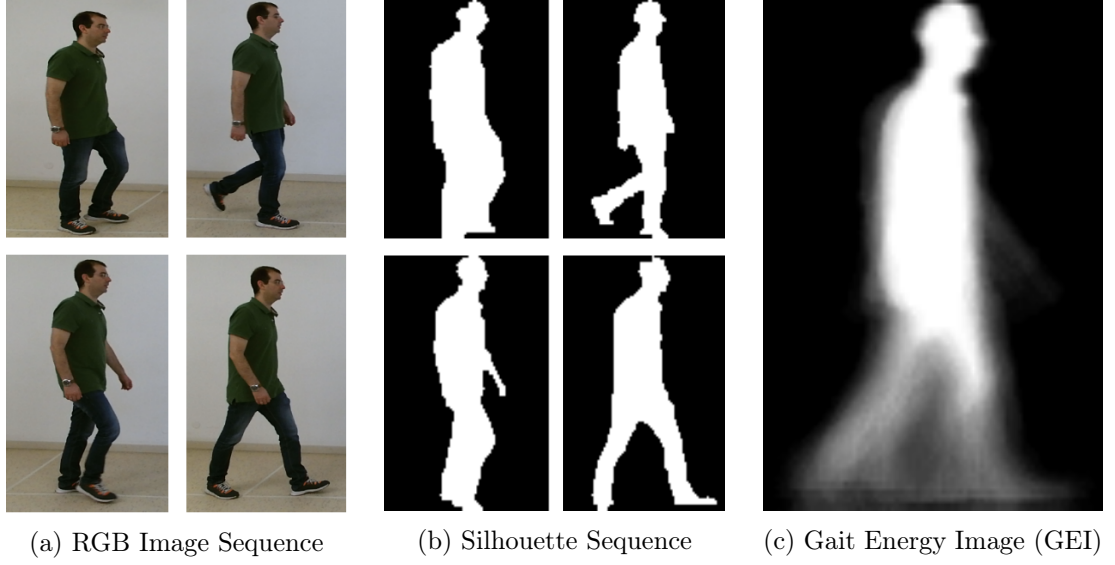


Figure 2.5: Silhouette and GEI Image Processing Example

(Source: Nunes, Moreira and Tavares, 2019)

The current state-of-the-art shows DL-based classification (see Section 2.3.2.3) is optimal for GR tasks, achieving Rank-1 accuracies (the percentage of times the top prediction is correct) north of 90% on the CASIA-B dataset (Güner Şahan, Şahin and Kaya Gülağız, 2024). DL reduces the need for manual feature engineering, enhancing classification efficiency. This benefits accuracy, with traditional ML methods (e.g., Nearest Neighbour (NN) or Support Vector Machine (SVM)) suffering a 10-20% drop on more challenging datasets (Shen et al., 2023). In uncontrolled scenarios, environmental conditions, occlusion, and gait variations will have an impact on accuracy, regardless of classification approach.

Appearance-based GR has in some cases been demonstrated using simpler methods that don't require training of an ML model. In a tutorial created for the 2016 International Conference on Pattern Recognition (ICPR), a GEI-based GR system was demonstrated that calculated differences between pixels in a newly created (from real-time video) GEI image, against a database of historic GEI with assigned identities, finding the closest match based on the resulting statistical similarity scores (Song, 2024). Although suitable for demonstration and Proof-of-Concept (PoC) purposes, simpler solutions forfeit

many advantages achieved by ML-based solutions, therefore their use in production environments may introduce reliability, usability, and security issues, instead of an improvement.

Overall, appearance-based GR systems face some limitations, like reliance on view angle; if a classification model is trained on, or a set of GEI images is created, using lateral movement data, the camera used to capture subsequent gait cycles must be similarly positioned. Anything obstructing a subject’s figure (e.g., bags, baggy clothes, or external objects) will also have a greater impact on appearance-based GR (Han and Bhanu, 2006). However, the resource efficiency compared to model-based solutions is advantageous; speed of identification and lesser resource requirements in a PACS, contributing to a better user experience may influence choices for implementation (Ling and Jacobs, 2007).

2.3.2.3 Classification Techniques

Artificial Intelligence (AI) and ML have revolutionised the feature extraction and classification processes enhancing modern data analysis and biometric identification. They automate traditionally manual tasks of identifying patterns and relationships in raw data to produce structured data, used for efficient classification or predictions based on input (Bishop, 2006). In the context of GR, AI and ML can, and have been employed to extract information about gait patterns from appearance and model-based data, supporting the creation of models and classifiers capable of identifying individuals based on their walking style (Ghilom and Latifi, 2024). Traditional ML or DL approaches can be used to achieve this, with DL alternatively employing more complex algorithms to perform analysis at a more intricate level — resulting in increased accuracy (Shen et al., 2023).

NN classification is a traditional method that categorises input based on the closest matching data point(s) — vectors of features for an object — in a ‘neighbourhood’ (imagine plotting points on a graph). A common method of defining a neighbourhood is ‘ k -NN’, where the closest k number of data points contribute toward classification (Faouzi and Colliot, 2023). NN has proven most suitable for smaller datasets, with performance degrading as the number of data points rises, as this demands computational

effort. For GR, NN could be used to compare static gait features, identifying subjects based on similarity (if any) to labelled historic data (MyEducator, 2024).

Principal Component Analysis (PCA) is a technique used in traditional classification workflows, it optimises, or 'reduces the dimensionality of' complex datasets, producing sets of independent components, while maintaining patterns important for classification. PCA can be used to extract the most pertinent information from images in silhouette and GEI-based GR systems, which is fed into algorithms like Linear Discriminant Analysis (LDA) or SVM, or used by simpler measures like k -NN or cosine similarity. LDA performs further optimisations, enhancing the separability of data points through abstraction, which are then classified by algorithms like k -NN, cosine similarity, or Pairwise Distance Analysis (PDA) (Hongye and Zhuoya, 2015). SVMs perform optimisation by identifying the 'optimal hyperplane' — decision boundary with the greatest margin between linearly separable (achieved by PCA) classes — performing classification itself (Kowalczyk, 2015).

DL has further improved the training and classification process by abstracting manual feature engineering steps (e.g., using a combination of PCA, LDA, and k -NN for dataset optimisation and classification) producing end-to-end learning pipelines capable of identifying patterns in raw, unstructured data. CNN are a type of NN and are the default DL approach to many static image recognition tasks. They automate the feature extraction, dimensionality reduction, and classification of spatial data, enabling unsupervised, iterative learning (Bari and Gavrilova, 2022). Many existing GR systems using CNN are appearance-based, performing classification on static (spatial) features of silhouette images or GEI (Shen et al., 2023).

RNN, another type of NN, support analysis of temporal and sequential data. RNN have been used in addition to CNN for GR purposes, enabling extraction of spatio-temporal information. For example, given a dataset of sequential silhouettes, a CNN would assess characteristics of individual images, whereas a RNN would account for patterns occurring over time or throughout a sequence (Hasan et al., 2022).

The automated framework implemented by NN-based techniques allows for enhanced performance compared to traditional ML techniques; NN learn complex patterns from iterative, in-depth analysis of input (Engati, 2024).

2.3.3 Applications of Gait Recognition in Physical Access Control

Applications of biometric authentication in PACS are not hard to come by, with 32% of organisations surveyed in 2024 reporting their own implementations (see Section 2.2.2). Examples of GR-based biometrics for PACS, however, are not so common. Many real-world examples of CV-based gait analysis technologies are in the clinical or forensic fields (see Section 2.3.1) which although not contributing to access control, demonstrates the feasibility of extracting features from, and performing identification based on gait.

In October 2024, a European Union consortium launched a pilot project — ‘PopEye’ — to study how GR technology can speed up identity checks at border crossings and reduce issues experienced with traditional types of biometrics (e.g., facial recognition, which is widely used at airports across Europe) when environmental conditions and/or appearance vary. While not yet implemented, the project’s success could lead to the adoption of Gait-based authentication technologies across European Union borders and in other environments, like workplaces (Chris Burt, 2024).

One of the rare production examples of a GR-based authentication system for PACS is Stepscan Secure. This system does not use CV, instead assessing walking style using under-floor pressure sensors. The product is advertised as a more ‘secure alternative to traditional PACS Multi-Factor Authentication (MFA)’ methods (Stepscan, 2020). The organisation behind the technology started out producing medical gait assessment systems and appear to have identified the gap in the market for an unobtrusive, transparent authentication solution. It is not clear how varying types of footwear may affect the system’s effectiveness in identifying individuals but may have a similar effect to occlusion in CV-based solutions.

2.3.4 Perceived Public Opinion (via Survey)

To gain insight into the public perception of the concept of GR-based authentication, a questionnaire was distributed to members of the public, requesting they rate their perceived security, comfort, and willingness to adopt GR-based identification technologies. A view of the questionnaire can be seen in Appendix A.1, with twenty-one (21) responses collected from people aged between 18 and 34.

Comfort and Acceptability Most respondents fell in the middle range, specifying they were 'Somewhat Comfortable' or 'Somewhat Uncomfortable' with the concept of GR. Very few respondents expressed strong feelings in either direction, with only one respondent being 'Very Comfortable' and one being 'Very Uncomfortable', stating: 'I'm not comfortable with giving up my personally identifiable data for the simple purpose of access control'. A majority of respondents rated GR's security level as moderate, specifying 'Neutral' to 'Somewhat Secure'.

Perceived Security Respondents generally perceived GR as less secure than fingerprint or facial recognition, with most rating the security level between 2 and 4 on a 5-point scale. However, opinions on traditional methods like Radio Frequency Identification (RFID) cards and Personal Identification Number (PIN) codes were often perceived as less secure than biometrics overall.

Authentication Preferences Only two respondents selected 'passive authentication' as their preferred method of authentication going forward, with the majority preferring traditional methods like RFID cards and PIN codes, or more established biometric methods. The respondents that did select GR as their preferences also cited convenience as a high priority, with one also stating: 'Sometimes my RFID Chip doesn't work on the first try and i have to try again'.

Concerns and Requirements In order to increase comfort with biometric systems including GR, respondents prioritized:

- Clear information about data storage
- Assurance that biometric data remains on-site (not cloud-stored)
- Transparent privacy policies
- Option to delete data when leaving an organization
- Regular system audits
- Alternative authentication methods being available

Demographics The survey primarily captured younger adults, with submission ages ranging from 18 to 34. No clear correlation was observed between demographic factors and attitude toward GR. However, one respondent noted concerns with bias in facial recognition, as they had experienced issues due to a combination of poor lighting and a darker skin complexion in the past.

Interest in Enrollment When asked about willingness to participate in GR enrollment, 4 respondents indicated "Yes", 3 respondents indicated "No", and the remaining respondents were "Unsure". This suggests moderate openness to the technology if properly implemented and proven to be effective.

Overall, these insights show that although there is some interest in GR, there are also significant concerns regarding privacy, data storage, and perceived security levels when compared to alternative biometric methods. If implemented, a GR-based system would need to have proved accuracy and reliability to gain widespread acceptance.

2.3.5 Review Summary

Decades-old technologies remain dominant in the PACS market despite known vulnerabilities and a user experience issues. A level of complacency in physical security is demonstrated by organisations not prioritising moving to more secure solutions. It can be argued that traditional methods are a burden on users, requiring physical tokens and a diversion of attention. This creates opportunity for more convenient, less obtrusive alternatives — *what if authentication for Physical Access Control System (PACS) could be invisible?*

As discussed in Section 2.3, biometric-based authentication solutions have improved both security and usability by using inherent credentials that are more difficult to compromise and reduce user encumbrance. Of the most common solutions, facial recognition stands out by making it possible to determine identity from a distance, reducing the time sink of repeated interaction with access control interfaces. However, its reliability can be influenced by factors like resolution, environment, face coverings, and darker skin tones. Emerging biometrics, like GR, serve as promising solutions to these issues. By assessing walking patterns and body movement to verify identity, GR achieves similar advantages to facial recognition while combatting many weaknesses by

evaluating more than just facial features.

Although thoroughly researched, GR — especially CV-based — has yet to see widespread adoption in PACS. Feedback from the public (see Section 2.3.4) shows that there is a moderate willingness to adopt GR if it can be proven to be secure and reliable, guaranteeing privacy and data security. This provides an opportunity to advance verification and authentication, popularising forms of authentication that eliminate the need for physical tokens and operate 'transparently', using robust and hard-to-compromise credentials, can improve both PACS security and usability.

The following chapter — 'New Ideas' — proposes a GR-based authentication system as an alternative to traditional token-based and biometric authentication methods. It addresses concerns raised in this chapter in relation to security, usability, and ethics, discussing how it may help make improvements to these areas.

Chapter 3

New Ideas

3.1 Introduction

This chapter proposes Gait-Recognition (GR)-based authentication as a solution to security and usability concerns with existing Physical Access Control System (PACS) — outlined in Chapter 2. It aims to demonstrate GR as a less obtrusive authentication option that maintains the level of security provided by traditional authentication methods and the more common biometric authentication options.

3.2 Gait Recognition-based Authentication

Chapter 2 Recap From a security perspective, Gait-Recognition (GR)-based authentication can contribute to eliminating vulnerabilities found in traditional systems by using inherence factor-based (biometric) credentials tied to user physiology — similar to existing technologies like Face and Fingerprint recognition (see Section 2.2.2). Biometrics remove the need for users to carry physical credentials or remember secret values (e.g., Radio Frequency Identification (RFID) Key Cards or Personal Identification Number (PIN) codes in traditional systems). These approaches reduce the risk of the theft of, or users misplacing or forgetting credentials.

While use of biometric authentication for Physical Access Control System (PACS) is rising, with implementations of Face, Fingerprint, and Iris recognition being reported (IFSEC Global, 2024), alternate biometrics like GR may help improve the user experience by eliminating pain points experienced with existing solutions. The most common PACS biometrics are inarguably accurate and reliable (National Institute of Standards and Technology, 2004; University of Baghdad et al., 2024). However, as discussed in Chapter 2, all operate in ways that could be considered obtrusive; users must divert their attention and actively participate (e.g., to scan their fingerprint), or influence accuracy (e.g., face coverings (Cooper et al., 2022) or changes in environment (Braje

et al., 1998) affecting Facial Recognition effectiveness).

This project aims to present Computer Vision (CV)-based GR as a viable alternative for PACS authentication, or authentication in general. It contrasts existing solutions, by primarily being able to operate passively and identify individuals from a distance, while also not suffering from single points of failure, such as face coverings that impact Face Recognition systems.

Although a promising solution to improving the security and usability of PACS, CV-based GR has its own requirements that may sway effectiveness. Physically, there must be enough space to enable a camera to capture subject's walking cycle. It also remains somewhat reliant on environmental conditions, with changes affecting walking style, or camera's ability to capture usable data. Additionally, as mentioned in Section 2.3.2, GR requires hardware powerful enough to process gait cycle data in real-time, which is important when trying to identify subjects unobtrusively.

3.2.1 Strength, Weakness, Opportunity, and Threat Analysis

To evaluate this proposal from a strategic perspective, a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis has been conducted. This provides a view of internal and external factors that could influence the success or openness to adoption of GR-based authentication. The proposal is assessed in respect to performance, security, cost, and ethical concerns discussed in Chapter 2.

- **Strengths:** A CV-based GR authentication system can operate passively, identifying subjects from a distance without their direct participation. It's use of biometric physiological factors (gait), reduces risks relating to lost or stolen credentials. A modular client-server architecture supports future scalability and allows an agile development process.
- **Weaknesses:** A CV-based GR system remains influenced by environmental conditions (e.g., lighting and background contrast) that can affect accuracy, similar to facial recognition. Performance of the system may vary on an individual level due to natural gait variability or temporary changes due to injury, or a change in clothing or footwear. The real-time demands of a PACS may exceed the capabilities of accessible hardware when performing GR processing.

- **Opportunities:** GR-based authentication could be deployed to enable frictionless access control in environments like offices, hospitals, and schools. Interest in contactless authentication relates mostly to convenience, security, and hygiene concerns. Future developments could combine GR with other biometric factors (multi-modal biometrics) for higher accuracy.
- **Threats:** Privacy concerns appear high, with primary research (see Section 2.3.4) highlighting anxiety about the collection and use of their biometric data. The accuracy of the system may prove to be much lower than other biometrics like facial or fingerprint recognition (National Institute of Standards and Technology, 2021; National Institute of Standards and Technology, 2004), impacting feasibility. Spoofing or adversarial attacks against gait recognition may be easier to achieve compared to other biometrics, which could lead to false acceptances or denials, impacting confidentiality, integrity, or availability.

3.3 Project Objectives Decomposition

The project objectives defined in Section 1.1.1 can be decomposed into several smaller tasks which allow for a more detailed plan to be created. These are outlined in Table 3.1 below. This tasks span three main contributory areas: (1) creation of the gait-based identification system (Gait Recognition Server (GRS)), (2) creation of the access control system demonstration (Access Control Client (ACC)), (3) testing and evaluation of the final product.

Table 3.1: Project Task List

#	Task	Dependencies
1	Identify a suitable gait-data extraction method and Gait-Recognition (GR) approach to be implemented. From the approaches discussed in Section 2.3, a suitable option should be chosen that will allow for completion of a Proof-of-Concept (PoC) within the timeframe of this project, with hardware that is accessible.	
2	Acquire the required hardware for the chosen GR approach. A Computer Vision (CV)-based approach will be used, but the chosen method may influence the type of camera required and the computing power of the device running the application(s).	1

Project Task List (Continued)

#	Task	Dependencies
3	Identify the required software resources to implement the chosen GR approach. A programming language and required libraries should be chosen that will allow for implementation of the this systems functionality, this will likely be Python.	1
4	Produce code to collect movement data using CV and abstract it into a suitable representation. Utilising the acquired hardware and chosen programming language, functionality to capture movement data from a camera feed and abstract it into a suitable gait cycle representation should be implemented.	1, 2, 3
5	Produce GRS code to compare input gait cycle data (Task #4) to historic data and predict identity. Code should be created that compares fresh input data to historic data in order to classify it and attempt to identify the associated individual.	4
6	Migrate gait cycle collection code (Task #4) to the ACC codebase. The ACC will ultimately be responsible for collecting gait cycle data in real-time, therefore this functionality should be moved here.	4
7	Produce code to allow the ACC and GRS to communicate. To allow the ACC to send gait data to and receive responses from the GRS, network communication functionality must be implemented.	
8	Create structure of the GRS' access control rule database. As shown in Figure 1.1, the GRS will use a database that stores access control rules to determine which identities have access.	
9	Produce code to check access control rules and permit or deny entry based on the identity returned by the GRS. The GRS must be able to query the database created in Task #8 and retrieve the rule associated with a given identity. This will be returned to the ACC with identificaiton results.	8
10	Create a ACC User-Interface (UI) to allow the registration of gait-based identities. To boost usability enable gait-based identity registration a suitable user interface (e.g., graphical) should be created for the ACC.	
11	Add a form of binary output to the ACC interface that demonstrates access being permitted or denied based on identification. Either physical or digital output should be implemented to demonstrate the provision or denial of access based on identification results returned by the GRS.	10
12	Create a GRS UI for managing registered identities and associated access control rules. The UI should allow administrators to view and delete registered identities, as well as manage their associated access control rules. This will facilitate easier management of the system and ensure that access can be dynamically adjusted as needed.	

Project Task List (Continued)

#	Task	Dependencies
13	Create or acquire a historic dataset relevant to the chosen GR approach. Depending on the chosen approach (e.g., silhouette or model-based, and Machine Learning (ML) or model-free), a dataset must be acquired or created to enable reliable testing.	1
14	Create historic gait data store (and/or train a predictive ML model) using Task #13 data. For a model-free approach, the acquired dataset should be formatted (e.g., in a folder) suitably, or if ML is being used, a model must be trained using the data that will allow for subsequent classification.	13
15	Complete testing of the product of Tasks #1-13 using Task#14 data in respect to Objective #3. The final product should be adequately tested to ensure it is capable of identifying individuals based on gait cycle both efficiently and accurately.	1-14
16	Perform demonstrations and collect feedback and opinions on the resulting GR-based access control system. Receiving feedback where possible is important, as one of the goals is to improve user experience with Physical Access Control System (PACS) authentication.	15
17	Evaluate the final product, address problems, and identify areas for improvement. Based on self-assessment and feedback received from others, assess how the project has met the initially defined goals, and where improvements could be made if developed further.	16
18	Discuss the feasibility of implementation in a real-world setting and what must change to make that happen. After producing a Proof-of-Concept Gait-based authentication system, it should be evaluated to determine how it could be implemented in a real-world setting, associated risks, and changes that may need to be made to improve robustness or change functionality.	17

An agile development approach will be taken during implementation of this PoC. This will enable the separation of concerns, and for tasks to be prioritised depending on what is needed to progress. It will also allow for adjustments to be made to the project approach if needed, and for tasks to be worked on in parallel, such as elements of the ACC and GRS that are dependent on one another.

3.4 Implementation and Test Planning

As discussed in 3.3 the Proof-of-Concept (PoC) will be implemented following an agile development strategy. Work has been separated into tasks, which will be completed in order of priority, but can be worked on in parallel or revisited if needed.

Although contributing to the overall project deliverable, most tasks relates to one of three primary component of the PoC. Time has been allocated to tasks in day-sized chunks, with some set to be completed in parallel where dependencies allow. For example, the access control rules database could be modelled immediately, while other components of the Access Control Client (ACC) and Gait Recognition Server (GRS) are being developed. Tasks may also be completed in parallel or consecutively where they contribute to the other's completion, like communication functionality between the ACC and GRS.

The Gantt chart displayed in Figure 3.1 on the next page illustrates milestones and the time allocated for each task's completion — referencing the eighteen (18) tasks listed in Table 3.1. Time for development has been set between the dates of March 1st and April 18th 2025, with the final product due for submission by April 25th at the latest, allowing a week of contingency.

Most information regarding planning or design tasks, or work unrelated to this project (e.g., other university modules) has been abstracted, but can be seen in the original Gantt chart included in the Project Proposal Document (PPD).

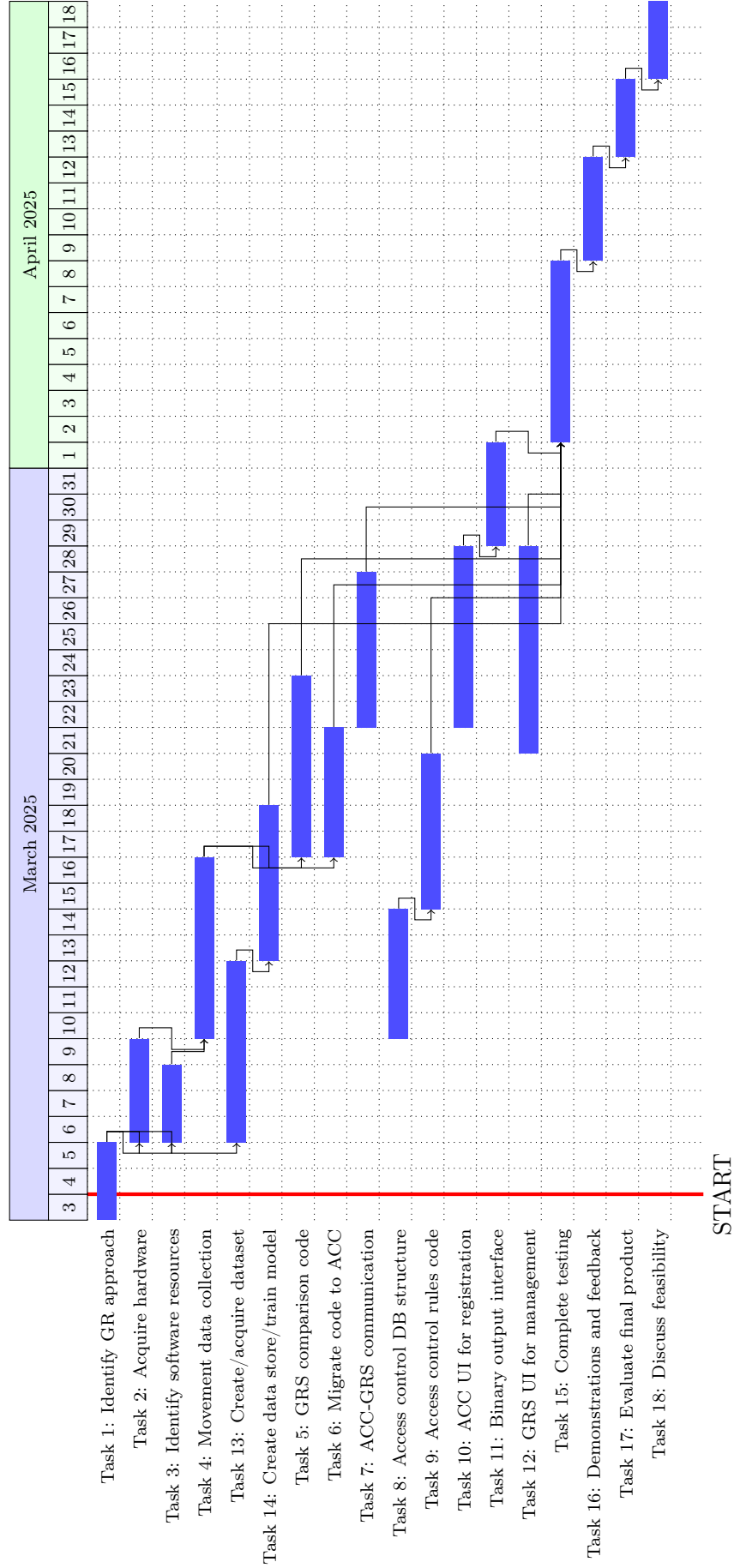


Figure 3.1: Gantt Chart for Implementation of a Gait Recognition System

Development was initially proposed to start in November 2024 as defined in the PPD, but other, non-project-related tasks (e.g., other modules) have been prioritised. This has resulted in a restructure of the Gantt chart, with milestones pushed back. Findings during design and implementation may also lead to a Gait-Recognition (GR) method being chosen that does not directly align with the original PPD proposal, which specified the use of both Machine Learning (ML), and a (gait) model-based GR approach. However, if changes are made, the project deadline and goal of producing a gait-based identification for use in a Physical Access Control System (PACS) context should not be impacted.

Using the risk assessment matrix displayed in Appendix B, a list of the most significant risks to the project were classified based on their probability and impact.

Table 3.2: Risk Assessment and Mitigations

#	Risk	Probability	Impact	Rating	Mitigation
1	Machine Learning Knowledge and Familiarity	3	5	15	Consult with academic staff experienced in ML
2	Distributed Computing and Networking Familiarity	2	3	6	Research networking principles and utilise available documentation for support
3	Project Size and Scope being too large/wide	2	3	6	Define clear project boundaries and implement phased approach with regular scope reviews
4	Hardware Functionality and Compatibility	3	5	15	Test hardware compatibility early, maintain backup hardware options
5	Computer Processing Power Issues	1	4	4	Optimise algorithms for efficiency and secure access to additional computing resources if needed
6	Integration Challenges between Components	2	4	8	Document Application Programming Interface (API) specifications early and create detailed integration test plans
7	Insufficient data capture or dataset resolution	2	4	8	Implement data quality checks and plan for supplementary data collection if needed
8	Limited historic research or dataset availability	2	3	6	Identify alternative data sources and consider secondary datasets or synthetic data generation
9	Unsuitable machine learning models for this use case	2	5	10	Research multiple model types and perform A/B testing

Risk Assessment and Mitigations (Continued)

#	Risk	Probability	Impact	Rating	Mitigation
10	Unsuitable speed of algorithms for smooth operation	2	2	4	Benchmark gait data extraction algorithms early and optimise critical paths
11	Data Processing Liability (Legality Related)	1	5	5	Consult legal expertise or documentation to ensure compliance with relevant regulations
12	Issues with subject availability for data collection	2	4	8	Create flexible scheduling (agile development) and have backup subjects identified
13	Issues with performing adequate tests to assess functionality	3	3	9	Develop comprehensive test plan with a measure of effectiveness if functionality is not achievable
14	Unforeseen delays to project for personal reasons	2	5	10	Build buffer time into schedule and have contingency plans for critical tasks (don't plan tasks until right before due date)
15	Unforeseen availability issues with project tutor	2	2	4	Establish multiple communication channels and schedule regular check-ins, produce documentation and plans to follow unsupervised
16	Unforeseen data loss of documentation, report or code	1	5	5	Implement regular backups and version control

Based on the tasks defined in Table 3.1 and laid out in the Gantt chart illustrated in Figure 3.1, a set of requirements have been drafted in an effort to explicitly define what the deliverable of this project should provide in terms of functionality and operability. These are outlined in Sections 3.4.1 and 3.4.2 and have been defined using the MoSCoW prioritisation framework (Brennan, 2009) categorising requirements as follows:

1. **(M)ust Have** Requirements that are critical to the deliverable meeting the project's aim(s).
2. **(S)hould Have** Requirements that are important and add value but can be delayed if needed.
3. **(C)ould Have** Requirements that would improve the project quality but are not necessary.

4. **(W)on't Haves** Things that will not be added in this round of development but may in future.

Additionally, the type of each requirement; Functional or Non-Functional, has been specified. Functional requirements define what the deliverable(s) should be capable of, whereas non-functional requirements define expected performance quality and usability (Brennan, 2009). Each requirement has been assigned an ID for reference in Chapters 4 and 5. IDs are suffixed with 'GR' if primarily relating to the GRS, or 'AC' for the ACC. An 'F' is added to the suffix if the requirement is functional, or 'NF' if non-functional.

3.4.1 Gait Recognition Server Requirements

Functional Requirements Table 3.3 below displays functional requirements relating primarily to the GRS component of this project, listed in priority order.

Table 3.3: Gait Recognition Server Functional Requirements

ID	Description	Priority
GRF1	MUST support the addition (registration) of new identities and gait cycles.	M
GRF2	MUST be able to predict identity using movement data captured in real-time.	M
GRF3	MUST produce suitable log entries dictating events (e.g., registration, deletion, identification).	M
GRF4	MUST store a dataset of historic movement data to enable comparisons and classification.	M
GRF5	MUST include a database holding access control rules that can be associated with an identity.	M
GRF6	MUST contain a User-Interface (UI) allowing configuration of access rules and management of identities.	M
GRF7	configuration functionality SHOULD operate independently of the ACC.	S
GRF8	SHOULD receive movement data data from the ACC over a network connection.	S
GRF9	SHOULD display the identity prediction or specify 'unknown' after an identification query is made.	S
GRF10	COULD use a trained ML model to classify gait data.	C
GRF11	COULD support input from multiple ACC 'nodes' in parallel.	C
GRF12	WON'T use a multi-camera-angle GR architecture.	W

Non-Functional Requirements Table 3.4 below displays non-functional requirements relating to the GRS component of this project, listed in priority order.

Table 3.4: Gait Recognition Server Non-Functional Requirements

ID	Description	Priority
GRNF1	MUST NOT retain movement data received during non-registration operations (i.e., passive identification).	M
GRNF2	SHOULD be able to demonstrate functionality irrespective of the gender or age of a subject.	S
GRNF3	SHOULD be able to discern identity with minimal delay (e.g., max 15 seconds) after receiving movement data.	S
GRNF4	SHOULD be able to operate effectively using consumer (low-medium power) processing and vision (camera) hardware.	S
GRNF5	SHOULD be able to complete data processing for identity registration with minimal delay (e.g., max 5 seconds) after receiving movement data.	S
GRNF6	WON'T be able to identify the same individual in vastly different contexts compared to those in which registration was performed (e.g., walking gradient, camera angle).	W

3.4.2 Access Control Client Requirements

Functional Requirements Table 3.5 below displays functional requirements relating to the ACC component of this project, listed in priority order.

Table 3.5: Access Control Client Functional Requirements

ID	Description	Priority
ACF1	MUST use Computer Vision (CV) to capture movement (gait) data.	M
ACF2	MUST abstract captured data into a suitable format before sending to the GRS.	M
ACF3	MUST wait for an identity to be returned by the GRS before permitting/denying access.	M
ACF4	WILL depend on the GRS and send data over a network.	M
ACF5	SHOULD function effectively using consumer hardware (processing, camera).	S
ACF6	SHOULD contain a UI displaying the movement data capture and processing feeds.	S
ACF7	SHOULD contain a UI allowing registration of new gait-based identities.	S
ACF8	SHOULD contain a form of output to demonstrate access being permitted or denied as a PoC.	S
ACF9	COULD support asynchronous submission of historic gait data for identification.	C
ACF10	WON'T have a backup form of authentication as a fail-safe if GRS is not possible.	W

Non-Functional Requirements Table 3.6 below displays non-functional requirements relating to the ACC component of this project, listed in priority order.

Table 3.6: Access Control Client Non-Functional Requirements

ID	Description	Priority
ACNF1	MUST NOT retain movement data captured during non-registration operations (i.e., passive identification).	M
ACNF2	WILL NOT be capable of handling the identification of multiple individuals simultaneously.	M
ACNF3	WILL NOT be attached to, nor control a physical entry way or door.	M

3.4.3 Testing Plan

After completion of the implementation stage (see Chapter 4), the solution will be tested in multiple ways to ensure Functional and non-functional requirements are met, issues (if any) are identified, and user feedback can be received. Testing will complete objectives #2 and #4, and tasks #15, #16, and #17 listed in Tables 1.1 and 3.1 respectively.

Testing actions will fall into one of the three (3) categories (specified below), and will occur after implementation has concluded, with the results discussed in Chapter 5.

1. **Functionality Testing** will evaluate functional and non-functional requirements of the GRS and ACC as outlined in sections 3.4.1 and 3.4.2.
2. **Performance Testing** will evaluate the performance and accuracy of the final solution (relating to some non-functional requirements of the GRS), in relation to the project aim of demonstrating GR-based identification for real-time PACS access control.
3. **User Testing** will involve performing live testing and demonstrations with ACC and GRS to receive feedback and insights which will be compared to project aims and objectives, and concerns raised in the public opinion survey 2.3.4.

3.4.3.1 Functionality Testing

Table 3.7 below lists ten (10) tests addressing requirements of the GRS and ACC pertaining to expectations under the **MUST** and **SHOULD** have categories of the MoSCoW prioritisation framework in Tables 3.3, 3.4, 3.5, and 3.6. It covers requirements GRF1-GRF9, ACF1-ACF4, ACF6-ACF8, GRNF1, and ACNF1.

Table 3.7: Gait Recognition Server Core Feature Tests

ID	Req	Description and Methodology	Expected Result
FT1	GRF1, GRF8, ACF1, ACF2, ACF7	Test the ability to register a new gait-based identity from the ACC UI. From the ACC, capture and send movement data to the GRS with a specified label.	The GRS should store the gait cycle data under the given label.
FT2	GRF5, GRF6	Test the ability to remove a previously registered identity from the GRS UI. From the GRS UI, delete the identity registered in T1.	All data associated with the identity should be removed from persistent storage.
FT3	GRF2, GRF4, GRF8, GRF9, ACF1, ACF2, ACF8	Test the response of the GRS to an identification query for valid gait data. From the ACC, input movement data of a previously registered identity to be sent to the GRS for classification.	The GRS should return the correct identity information and access rule.
FT4	GRF2, GRF4, GRF7, GRF8, GRF9, ACF1, ACF2, ACF8	Test the response of the GRS to an identification query for invalid gait data. From the ACC, input movement data of a non-registered identity to be sent to the GRS for classification.	The GRS should return unknown or a low confidence score, indicating that the identity is not recognized.
FT5	GRF5, GRF6	Test the ability to configure Boolean access rules per identity from the GRS UI. From the GRS UI, configure an access rule for an individual identity from disallow to allow.	The GRS should update the access rule for the identity to match the new configuration.
FT6	GRNF1, ACNF1	Confirm that movement data sent during identification is not retained by the GRS. From the ACC, input movement data of a registered identity to be sent to the GRS for classification, then verify the data is not stored post-classification.	The GRS should not retain the movement data received for classification after the process is complete.
FT7	ACF3	Confirm that Boolean access rule output is not updated on the ACC UI until the classification result is returned. From the ACC, input movement data of a registered identity to be sent to the GRS for classification, then verify that the access status remains unchanged until the result is returned.	The ACC UI should not update the access output until the classification process is complete.

Continued from previous page

ID	Req	Description and Methodology	Expected Result
FT8	ACF4	Confirm that the ACC does not function when the GRS is not running. Attempt to send movement data from the ACC to the GRS while it is not operational.	The ACC should indicate an error or failure to connect, confirming that it cannot process identification requests without the GRS.
FT9	ACF6	Confirm that the live camera feed and gait cycle extraction is displayed in the ACC UI. Verify that the ACC UI shows a real-time video feed from the camera and extracted gait cycle data.	The ACC UI should display the real-time camera feed and extracted gait cycle data without significant delay.
FT10	GRF3	Confirm that a log of identification requests is kept and displayed on the GRS UI. From the GRS UI, send multiple identification requests and verify that each request is logged and displayed in the user interface.	The GRS UI should show a complete log of all identification requests made, including timestamps and results.

Tests evaluating non-functional requirements relating to performance, accuracy, and usability (e.g., identification speed or of difference sexes, or the speed of gait cycle extraction for registration), are not included in this table and will be assessed in performance and accuracy testing (Section 3.4.3.2) or user testing (Section 3.4.3.3) where appropriate.

3.4.3.2 Performance and Accuracy Testing

To evaluate the system's ability to accurately and efficiently identify individuals based on their gait cycle, tests will be conducted to assess performance and accuracy in relation to some of the expectations set by non-functional GRS requirements defined in Table 3.4.

Performance tests will evaluate the speed at which the system can identify individuals and handle registration of new gait-based identities. Table 3.8 below lists tests designed to evaluate the performance of the final GR system, referencing relevant requirements GRF1, GRF2, GRNF2-GRNF4, and ACNF1.

Table 3.8: Performance Testing

ID	Req	Description
PT1	GRF2, GRNF2	The accuracy of the system should be tested against a range of subjects (varying sex, age, and body type) to determine if it can correctly discern identity regardless of these factors.
PT2	GRNF3, GRNF4	The system should be tested on consumer hardware (e.g., laptop and webcam) to determine if it operates efficiently at a reasonable level on low-power, budget-friendly hardware.
PT3	GRNF3	The raw speed of identification should be assessed to determine if it is capable of identifying individuals in real-time (within 10 seconds).
PT4	GRF1, ACNF1	The speed of registering a new identity should be assessed to ensure that the system can process new registrations within an acceptable time frame (under 5 seconds).

In addition to the performance tests defined above, accuracy tests will be conducted to evaluate how well the system can correctly identify individuals, conducted based on the following methodology:

"The final system will be populated with a number of identities, each with associated gait cycle data. Fresh gait cycle data will be captured and input, being submitted to the GRS for classification. The results will be compared to the manually verified identity of the subject, allowing a measure of accuracy to be determined."

Since the system will focus on recognising known individuals, accuracy evaluations will assess correct and incorrect *positive* identifications only, with no consideration given to *false negatives* at this stage. Results of this analysis will be communicated using the accuracy: *The percentage of identification attempts that resulted in the correct identity being returned.*

$$\text{Accuracy} = \frac{TP}{TP + FP}$$

Where:

- TP = True Positives (correctly identified individuals)
- FP = False Positives (incorrectly identified individuals)

Note: As true and false negatives are not calculable in this (prediction-based) context,

recall, precision, and F-measure are not applicable.

3.4.3.3 User Testing

User testing will be conducted to evaluate the usability of the system, conducted based on the following methodology:

"The final system will be tested several subjects, with varying levels of experience with access control systems. Subjects will be taken through the identity registration (on-boarding) process, before the system's ability to identify them without manual intervention is demonstrated.

Following on-boarding and demonstration, subjects will be asked to fill out a questionnaire (see questions below), providing feedback on the ease of registration, perceived speed of identification, and interest of using a system of this nature in a real-world context. Subjects will also be given the opportunity to raise any concerns or issues they may have with the system or concept of passive identification."

This methodology aims to provide insights into the following areas:

- **Usability and onboarding experience:** How intuitive and straightforward users find the registration process, including the clarity of instructions and the responsiveness of the system during interaction.
- **Perceived performance and convenience:** Users' impressions of the system's identification speed and their perception of the convenience afforded by passive, contactless authentication.
- **User preference and acceptance:** Whether users would prefer gait-based, passive authentication over traditional methods such as key cards, PIN codes, or fingerprint scanners, and their willingness to adopt such a system in real-world contexts.
- **Concerns and perceived risks:** Any opinions users have regarding data privacy, potential for misuse, or ethical considerations surrounding behavioural biometrics, particularly in environments where such technologies might be implemented.

Chapter 4

Implementation

4.1 Introduction

Based on the aims and objectives of this project outlined in Chapter 3, options were evaluated to settle on the best course of action for implementing a solution that: met requirements, and could be completed within the allocated time frame, using available resources.

The first half of this chapter: outlines the software development methodology followed, lists and justifies technologies and techniques used, and provides an overview of the intended system architecture using UML, Sequence, and Wireframe diagrams; conveying the structure of code, flow of data, and User-Interface (UI) layout.

Following this, the second half of this chapter covers the development process in relation to the expected tasks and expected functionality defined in Chapter 3. Implementation sections are not be ordered in exact accordance to the tasks defined in Table 3.1 or allocated time in Figure 3.1, but are instead ordered in a way that conveys the system's overall functionality in the best way possible.

The list below provides a simplified overview of the contents and order of the development documentation in this chapter:

1. **Gait Recognition Approach Benchmarking and Selection:** Model and appearance-based gait data extraction methods are benchmarked and a selection is made for this Proof-of-Concept (PoC) system.
2. **Camera Input Processing and Gait Energy Image (GEI) Creation (spoiler!):**
The functionality of processing camera input and extracting and abstracting gait features is developed.
3. **Access Control Client Implementation:** The Access Control Client (ACC) is developed, leveraging the functionality implemented in #2 above to capture,

process, and transmit gait features for registration and classification.

4. **Gait Recognition Server Implementation:** The Gait Recognition Server (GRS) is developed, implementing Gait-Recognition (GR), allowing registration of identities, and subsequent classification using Convolutional Neural Network (CNN), Principal Component Analysis (PCA), and cosine similarity.
5. **Implementation Technology Stack:** A list of technologies and techniques used to implement the system is provided, along with justifications for their use.
6. **System Architecture:** An overview of the system architecture is provided using UML, Sequence, and Wireframe diagrams.
7. **Gait Energy Image Creation:** The process of creating gait energy images (GEIs) is detailed, including the steps taken and the technologies used.

Note: The final testing and evaluation tasks detailed in Chapter 3.4 are not covered here, and are instead discussed in Chapter 5.

4.2 Research and Software Development Methodologies

As discussed in Chapter 3, the software development methodology followed during implementation was agile, chosen due to its support for rapid iteration and flexibility, allowing previously completed areas to be revisited if issues are encountered, for example. Figure 4.1 below provides an overview of the agile development process:

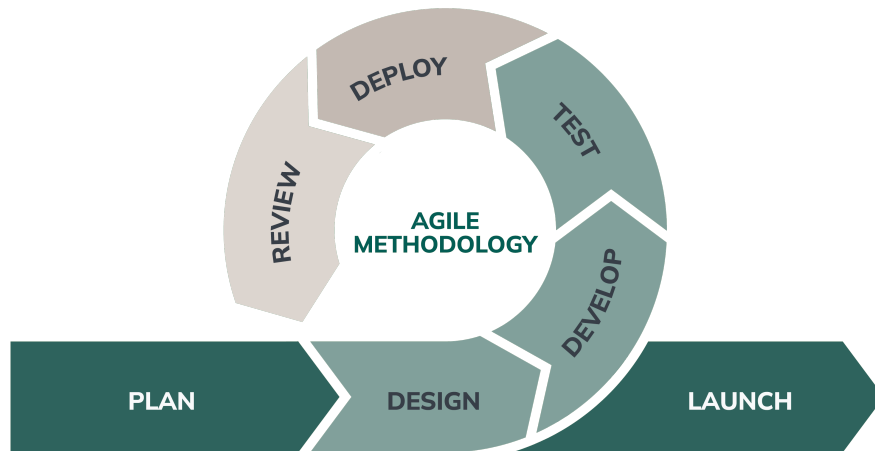


Figure 4.1: Agile Software Development Process

While software development 4.5 was completed following the agile methodology, the

overall research project is structured as follows:

1. **Research:** The current state-of-the-art in Physical Access Control System (PACS), biometrics, and Gait-Recognition (GR) is researched, with comparisons made between technologies to identify security and usability-related trade-offs.
2. **Survey:** A survey was conducted to gather feedback from the public (potential users) on their perceptions and openness to GR technology in the context of access control.
3. **Proposal:** A 'new idea' — the software deliverable of this project — is proposed, with a high-level overview of what the system will do, with aims, objectives, and requirements defined.
4. **Design:** The system architecture is designed based on the proposal, with a general understanding gathered of how the problem will be tackled, and how components will operate and interact with each other and external entities.
5. **Implementation:** The system is implemented, with decisions made on what technologies to use, and the exact way in which the system will be constructed. Following the agile methodology, this happens incrementally.
6. **Testing:** The system is tested based on functionality and performance requirements defined in the proposal, relating to project objectives.
7. **Evaluation:** The final solution is evaluated based on the aims and objectives defined in the proposal, with a focus on security and usability. A feasibility study is also conducted to determine if the system is ready for deployment in a real-world scenario.

The following sections outline the software and hardware technologies used when implementing the solution, followed by the design, and implementation documentation.

4.3 Technologies Used and Techniques Applied

4.3.1 Choice of Gait-Recognition Approach

Two main forms of Gait-Recognition (GR) are discussed in Chapter 2 and 3: Appearance (e.g., silhouette, GEI), and Model-based (e.g., 2 or 3-dimensional skeletons). To settle on the most suitable approach to use for this project (not just for efficacy; the project needed to be completed within the given timeframe and able to operate on available hardware) benchmarks were performed to evaluate the technologies required to implement either approach.

The same system was used to perform all benchmarking, ensuring results could be compared directly and would represent system performance. As a target of this project is for the solution to be efficient enough to run on 'consumer'-grade (low-medium cost and power) hardware, the system used for these tests — an LG 16Z90P laptop — was of no significant cost, and was only equipped with an integrated Graphics Processing Unit (GPU).

Specifications of this system relevant to performance are detailed in Table 4.1 below.

Table 4.1: Benchmark and Testing System Specifications

Component	Specification
Processor	11th Gen Intel Core i7-1165G7 @ 2.80GHz, 2803 Mhz, 4 Cores, 8 Logical Processors
Physical Memory	16.0 GB
Graphics	Intel Iris Xe Graphics (CPU-integrated)
Operating System	Microsoft Windows 11 Home 64-bit

A single data sample was used for benchmarking — a set of 106 lateral movement frames at 1080p resolution taken from the publically accessible GRIDDS sample dataset (Nunes, Moreira and Tavares, 2019) — with both approaches applied to the same sample. The aim here was to measure the time taken by each approach to perform feature extraction for an entire movement cycle (i.e., a person walking through the camera's field of view laterally).

Model-based GR The OpenPose framework was used here. It enables extraction of 2-dimensional (or 3-dimensional with multiple cameras) joint information from images

of people, with resolution options of 15, 18, or 25 joints (for full-body pose estimation). The framework’s `OpenPoseDemo.exe` application was invoked to perform feature extraction, using the `pose_iter_584000` model. The configuration used when invoking the application via `CMD` is shown in Listing 4.1 below.

```
1 OpenPoseDemo.exe ^
2   --image_dir input_images ^
3   --write_json output ^
4   --display 0 ^
5   --render_pose 0 ^
6   --scale_number 1
```

Listing 4.1: OpenPose Configuration for Benchmarking

Appearance-based GR A self-authored Python script, based on an example from Song, 2024, was used to perform (1) Silhouette extraction for each frame and (2) gray-scale GEI image creation and output. The temporal results of these benchmarks are shown in seconds in Table 4.2 below.

Table 4.2: Gait Data Extraction Time Benchmark Results

Approach	Time (s)
Model-Based	2986.8 (49.7 minutes)
Appearance-Based	3.1

At face-value, these tests show Gait Energy Image (GEI)-based feature extraction to be far more efficient than the model-based alternative. However, research explored in Chapter 2 suggests model-based approaches are more accurate and reliable if given enough resources (e.g., a more powerful GPU).

Because of the desire for this system to run on low-medium powered hardware, the decision was made to proceed with GEI-based GR. This strays from what was defined in the Project Proposal Document (PPD), which proposed a model-based (i.e., skeleton) GR approach. The intended functionality of the overall system however, remains the same.

4.3.2 Implementation Technology Stack

The Proof-of-Concept (PoC) solution was primarily built using the Python programming language HTML, and JavaScript, with a number of libraries and frameworks

leveraged throughout, enabling: Computer Vision (CV) and image processing, math and matrix transformations for statistical analysis, Machine Learning (ML)-based feature extraction and classification, and the creation of User-Interface (UI)s based on designs illustrated in Section 4.4.

A detailed list of the technologies used, their purpose, and justifications for their use are shown in Table 4.3 below.

Table 4.3: Implementation Technology Stack: Purpose and Justification

Technology	Purpose and Justification
Python	The primary programming language for backend development. Ideal for rapid development and prototyping-, as well as math and ML
Flask	Web framework for building the backend Application Programming Interface (API) and routing front-end. Flask is lightweight and flexible, allowing for easy routing and handling of requests with minimal overhead.
OpenCV	Library for CV tasks, including image processing. OpenCV (CV2) provides robust tools for image manipulation, making it suitable for tasks like silhouette extraction and background subtraction.
TensorFlow	Framework for ML tasks. TensorFlow supports efficient model training and deployment, particularly for tasks involving neural networks like Convolutional Neural Networks (CNNs).
EfficientNetB0	Pre-trained CNN model for feature extraction from images. EfficientNetB0 offers a good balance between performance and computational efficiency, suitable for real-time applications.
SQLAlchemy	ORM for database interactions. SQLAlchemy simplifies database operations and provides a high-level abstraction for managing database records.
SQLite	Lightweight database for storing identities and gait samples. SQLite is easy to set up and requires no additional server, making it ideal for development and small-scale applications.
HTML/CSS/JavaScript	Technologies for building the frontend UI. These are standard web technologies that allow for responsive and interactive web applications.
Socket.IO	Library for real-time, low-overhead communication between back and front ends of both components. Socket.IO enables efficient data transmission, allowing for real-time updates without the overhead of traditional Hypertext Transfer Protocol (HTTP) requests.
Jinja2	Templating engine for rendering HTML pages. Jinja2 allows for dynamic content generation in HTML, with variables able to be passed from the back-end to the front-end, making it easier to create interactive web pages.

The list of hardware resources used throughout this project was much shorter and doesn't warrant a table, only requiring a laptop computer (details specified in Table 4.1) for development and testing, an external webcam (Logitech C920) to capture video input, and a tripod (Manfrotto Compact Light) for mounting the webcam at an accept-

able height and angle.

4.4 Design

4.4.1 System Architecture Recap

As discussed in Chapter 3, the system produced in this project should follow a client-server architecture, where the client is responsible for collecting Gait Energy Image (GEI) information (for registration or recognition) and sending it to the server for processing.

Figure 4.2 below provides a high-level overview of the system architecture, detailing how the client, server, and users are connected, and which technologies have been deployed and where.

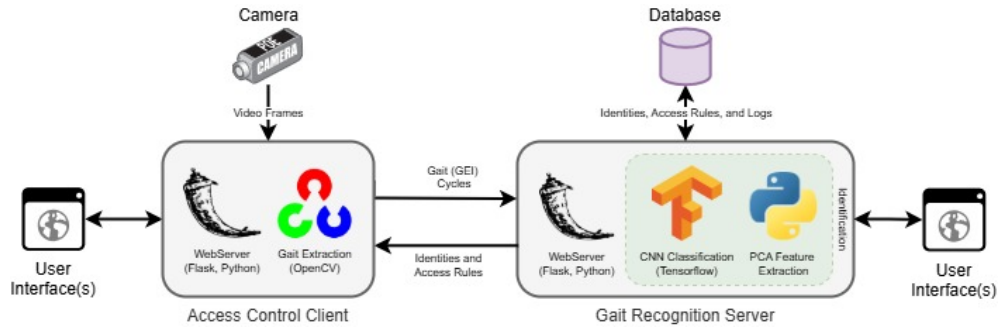


Figure 4.2: System Architecture and Technologies

The Flask web framework is used on both the Access Control Client (ACC) and Gait Recognition Server (GRS) to render the User-Interface (UI) and serve Application Programming Interface (API) endpoints. The ACC receives camera input and employs Computer Vision (CV) and image processing techniques using the OpenCV (CV2) Python library to produce GEIs, these are then transmitted to the GRS which uses Convolutional Neural Network (CNN) and Principal Component Analysis (PCA) for feature extraction and identity classification. The following sections detail the design of each component.

4.4.2 Gait Energy Image Creation

Real-time, appearance-based Gait-Recognition (GR) requires pre-processing of camera input to create GEIs which are used for both identity registration and subsequent

classification.

Existing research indicates GEIs creation consists of four main steps, performing subject detection, background subtraction, silhouette normalisation, and silhouette averaging. The GEI creation steps are illustrated below in Figure 4.3, followed by a description of each stage.

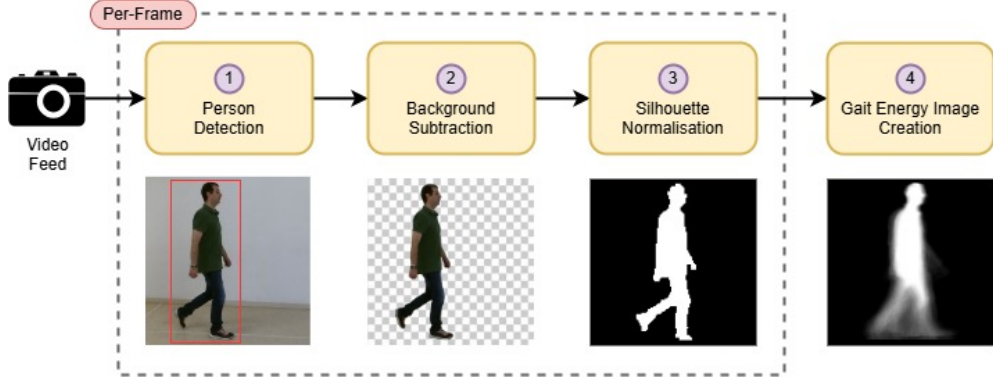


Figure 4.3: Steps involved in Gait Energy Image (GEI) Creation

1. **Person Detection:** A person detection algorithm will be used to localise the person present in each given frame. The goal here is to remove any unnecessary parts of the image which may add noise, isolating the subject.
2. **Background Subtraction:** The person will be isolated from the remaining background in the frame using statistical differencing (e.g., comparing the frame to a static background), leaving just the subject's silhouette describing their pose at frame k .
3. **Silhouette Normalisation:** The silhouette image will be normalised to a standard resolution (e.g., 128x88), aligned to the centre of the frame, and given a solid colour to ensure uniformity required for creation of a GEI with pixel averaging.
4. **Silhouette Averaging:** All normalised silhouette images will be combined and averaged using an implementation of $GEI = \frac{1}{N} \sum_{k=1}^N I_k$ where I_k is the k -th image in the sequence. This results in a grayscale image where pixel intensity (brightness) denotes how often a region was occupied by the subject (a concept outlined in Chapter 2).

In the final solution, this process will be implemented within the ACC, where GEIs will be created and transmitted to the GRS, which will use the GEI to either register a new identity, or to perform comparisons to what is stored in the database for identity

matching. However, because of how critical this process is, it will be worked on in isolation before being implemented in a wider context within the ACC.

4.4.3 Access Control Client Design

The ACC will be responsible for processing camera input to create GEIs that are sent to the GRS for identity registration or matching. It will have two main features, listed below, while also allowing asynchronous manual GEI classification.

1. **Identity Registration:** A 'Save Identity' button can be clicked at any time which will allow the user to assign a label to the current GEI and send it to the GRS to be added to the identification model.
2. **Person Identification:** Continuously polls the GRS with updated GEIs, awaiting a response of the predicted person, confidence score, and access rule.

The ACC will be controlled using a web-application-based UI, allowing the user to view the (1) Live Video Feed, (2) Isolated Silhouette, and (3) Current GEI, in addition to controlling new identity registration.

Fundamentally, the ACC uses the GEI creation process defined in Section 4.4.2 to perform all image processing and averaging actions.

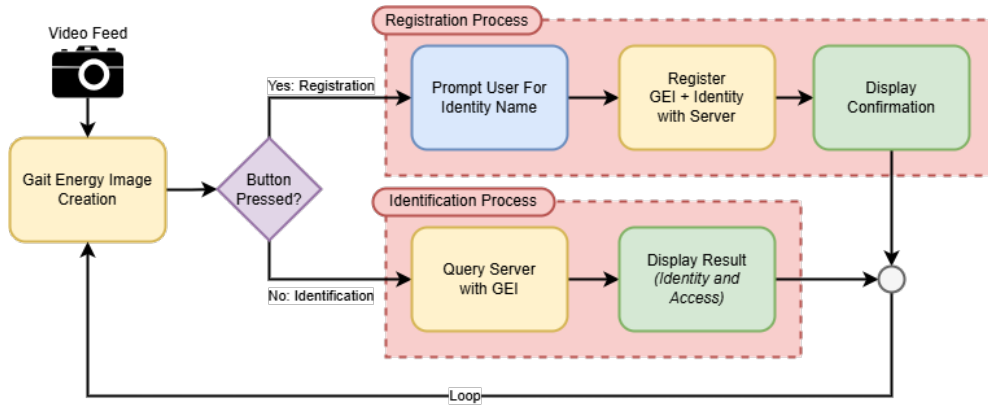


Figure 4.4: Access Control Client Features Overview

Figure 4.4 above provides an overview of the ACC and its two main features, discussed further in the following sections.

4.4.3.1 Identity Registration Process

The ACC registration process will allow new GEIs to be stored, enabling subsequent identification. It will use the same frame buffering and GEI creation process as Section

4.4.2, but will allow the user to assign a label (e.g., name) to the GEI through the UI before it is sent to the GRS where feature extraction and classification model training will take place.

A sequence diagram, Figure 4.5 below, illustrates the flow of data when a user is registering a new, or updating an existing identity with a new GEI.

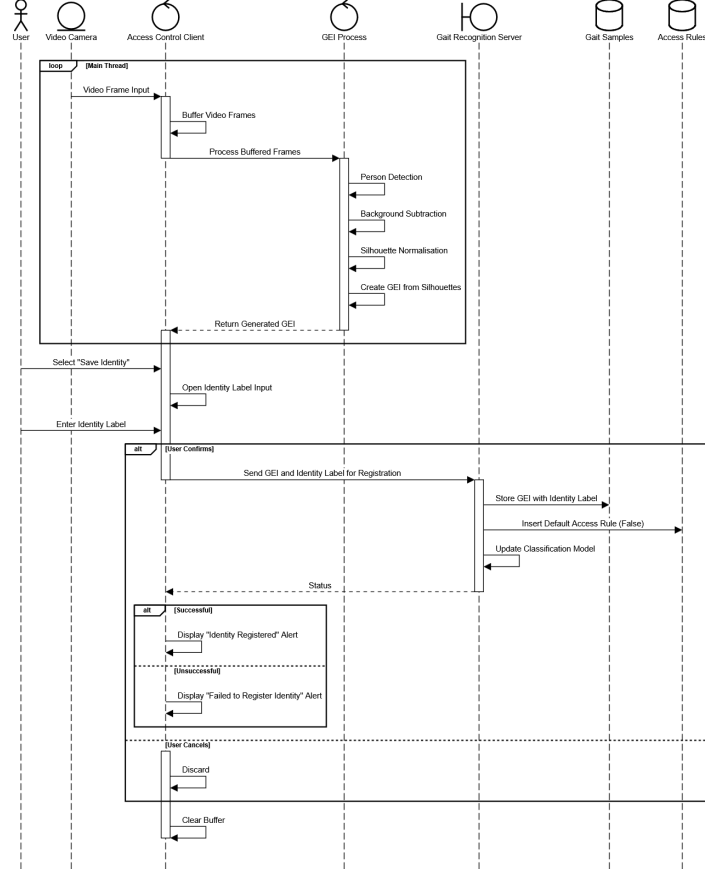


Figure 4.5: Access Control Client Registration Process Sequence Diagram

4.4.3.2 Identity Classification Process

The ACC classification will involve cyclically buffering an amount of camera frames before using them to create a GEI which is sent to the GRS for classification. The ACC will await a response containing a prediction of the identity, confidence score, and access rule. It will display a permit or deny message based on the access rule, *given the confidence score is also above a set threshold*.

A sequence diagram, Figure 4.6 below, illustrates the flow of data from the perspective of the ACC's classification thread; showing the actions performed to transform camera input into GEIs, and queries made to the GRS in an attempt to discern identity, as well

as the GRS's interactions with relevant tables in the database.

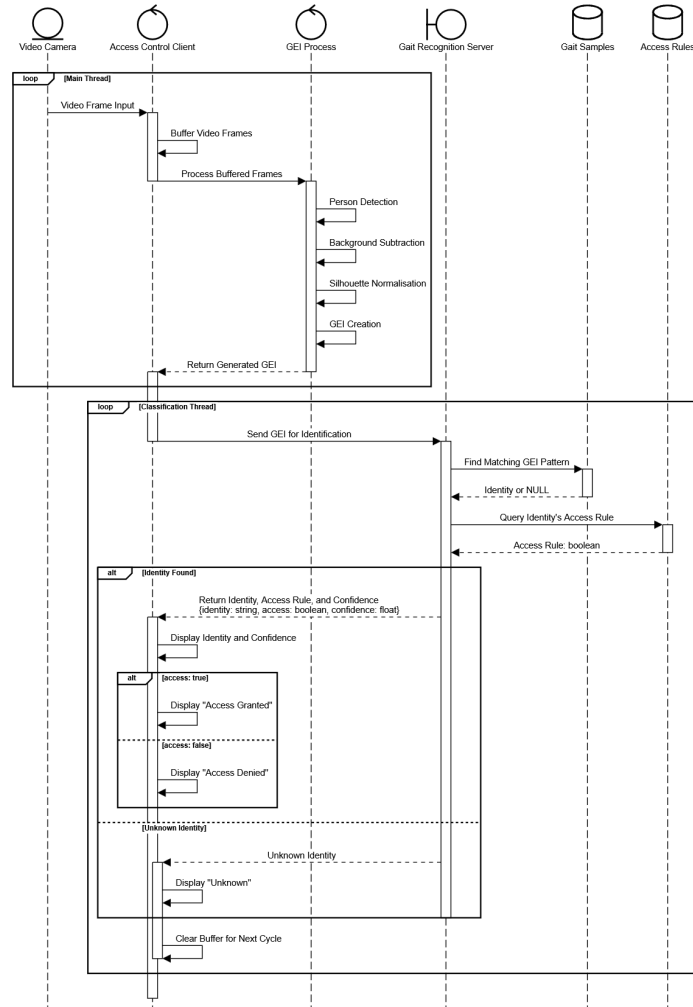


Figure 4.6: Access Control Client Identification Process Sequence Diagram

This diagram shows: (1) The ACC receiving the camera feed input, proceeding to (2) buffer a number of frames before using them to generate a GEI with the process defined in Section 4.4.2. After the GEI is created, (3) the GRS is queried which (4) performs classification of the given GEI and returns the resulting identity or a message stating an identity could not be determined. If an identity is received, (5) the ACC displays the identity label (i.e., the person's name) and uses the Boolean 'access' parameter to display either access granted or denied, if the prediction confidence is above a set threshold. Finally, (6) the buffer is cleared and the process repeats, occurring indefinitely until the application is stopped.

Inter-process messaging is to be performed following REpresentational State Transfer (REST) principles using Hypertext Transfer Protocol (HTTP) (predominantly POST) requests. Image data is to be encoded prior to transmission in a common format

(Base64) allowing serialisation for inclusion in JavaScript Object Notation (JSON) objects. All messages are to be sent in the JSON format.

4.4.3.3 User Interface Design

The ACC's UI will be a simple, two-page web-application. The default view, shown in Figure 4.7 below, will display the live camera feed, alongside the isolated silhouette, and the most recently created GEI. Underneath these views will be text displaying the label of the most recently identified person, the identification confidence percentage, and a Boolean access rule (set False if confidence is below a set threshold, otherwise respective of the configured access rule). For refined control, buttons will exist to reset the background image used for background subtraction, and to register the current GEI to an identity (discussed in Section 4.4.3.1).

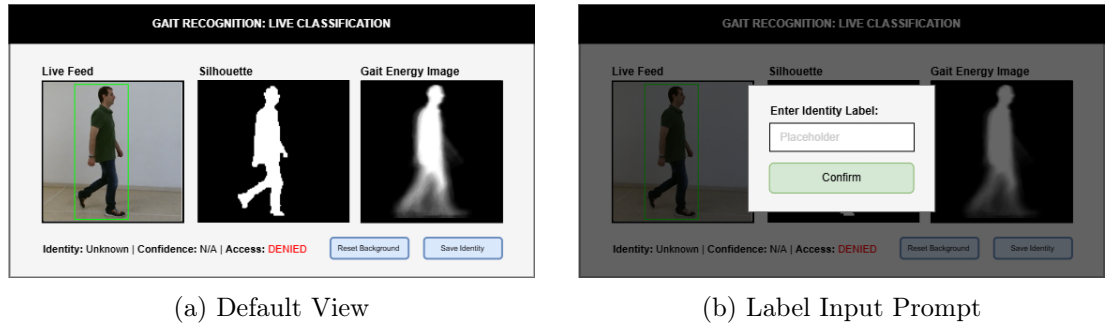


Figure 4.7: Access Control Client 'Live Classification' Page

The second ACC page will allow for manual image uploads which will be sent to the server for classification, using the same process defined in Section 4.4.3.2. This is predominantly for demonstration and testing purposes but enables asynchronous GEI classification. The concept layout of this page can be seen in Figure 4.8 below, consisting of a drag-and-drop upload box, a preview of the GEI being classified, and the same text details as Figure 4.7 above.

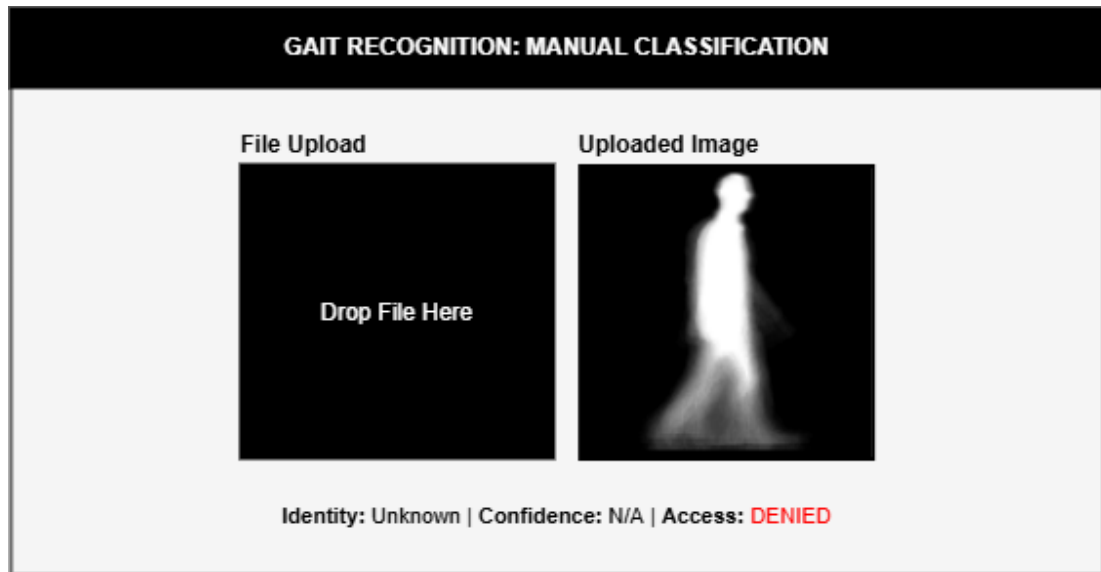


Figure 4.8: Access Control Client 'Manual Classification' Page

4.4.4 Gait Recognition Server Design

The second component: the Gait Recognition Server (GRS), will be responsible for GEI-based identity registration and recognition based on input received from the ACC. It will also support configuration of Boolean (grant or deny) access control rules tied to each identity, communicated to the ACC alongside classification results. The three primary functions of the GRS are defined below:

1. **Identity Registration:** A new identity is registered by sending a GEI to the GRS which will store it in the database.
2. **Identity Matching:** A GEI is sent to the GRS which will compare it to what is stored in the database, returning a prediction of the identity found, confidence score, and access rule.
3. **Access Rule Configuration:** An access rule is configured by sending a request to the GRS which will update the rule (linked to an identity) in the database.

This component will be controlled using a web-application-based UI, enabling configuration of access control rules and a visualisation of registered identities and their associated GEIs (see Section 4.4.4.5). A UI is not required for the GEI-based registration and matching features as their purpose is to operate continuously in the background and will be accessible to the client via API endpoints.

4.4.4.1 Identity Registration Process

The GRS' identity registration process will support the ACC's process defined in Section 4.4.3.1; awaiting registration messages containing a GEI and an identity label specified by the user. No additional image processing will be required here, and the provided GEI will be stored in the GRS' database alongside the identity label (e.g., person's name). A sequence diagram has been used to communicate the flow of data between the ACC and GRS when an identity is being registered, shown in Figure 4.9 below.

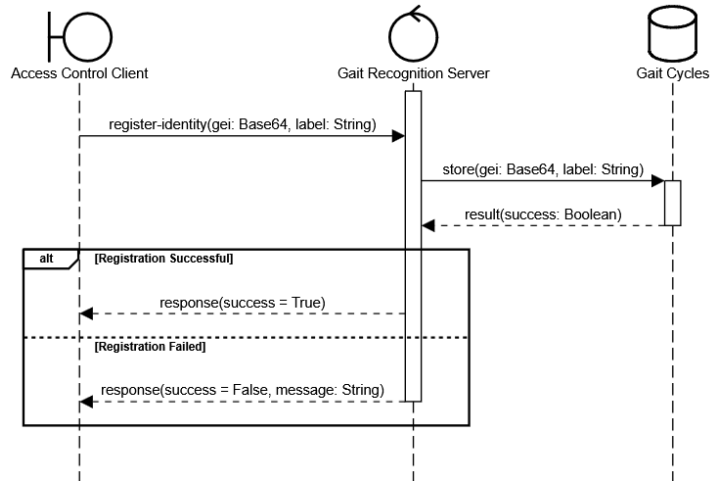


Figure 4.9: Server Identity Registration Process Sequence Diagram

This sequence diagram shows: (1) a message being received from the ACC on the 'register-identity' endpoint containing a GEI and identity label, followed by (2) the GRS storing the GEI and identity label in the gait database, and (3) the GRS giving a dynamic response to the ACC based on whether the identity was successfully registered.

4.4.4.2 Identity Classification Process

The GRS' classification process will require no direct interaction from the user. It will be accessible to the ACC on an API endpoint, allowing the ACC to send classify messages at set intervals (e.g., every k number of frames) containing the GEI created with the most recent k frames captured. The GRS will then use its classification algorithm to compare the provided GEI with what is stored in the database, returning the predicted identity, confidence score, and associated access rule.

A sequence diagram has been used to communicate the flow of data between the ACC and GRS when gait cycle information is being submitted for identification, shown in

Figure 4.10 below.

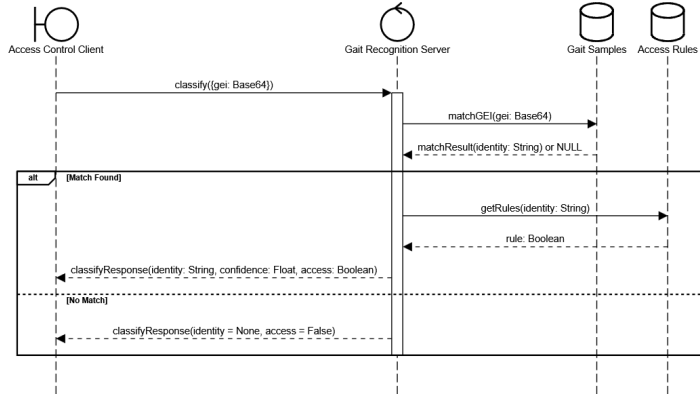


Figure 4.10: Server Identity Classification Process Sequence Diagram

This sequence diagram shows: (1) a message being received from the ACC on the 'classify' endpoint containing a GEI, followed by (2) the GRS calling its classification algorithm to find a matching identity, and (3) the GRS responding with the matching identity, confidence score, and associated access rules.

4.4.4.3 Access Rule Configuration Process

Most user interaction on the server side will occur when viewing existing identities and configuring their associated access control rules (see Section 4.4.4.5). Users will be able to view the list of registered identities and select any from the list to view the associated GEIs and access rule configuration. From there, they will be able to modify the identity's access, with the configuration stored in the access rule table within the database, or delete the identity or individual GEIs.

The sequence diagram shown in Figure 4.11 below illustrates the procedures that will be executed when a user selects an identity from the list, and when they submit a request to update the identity's associated access rule.

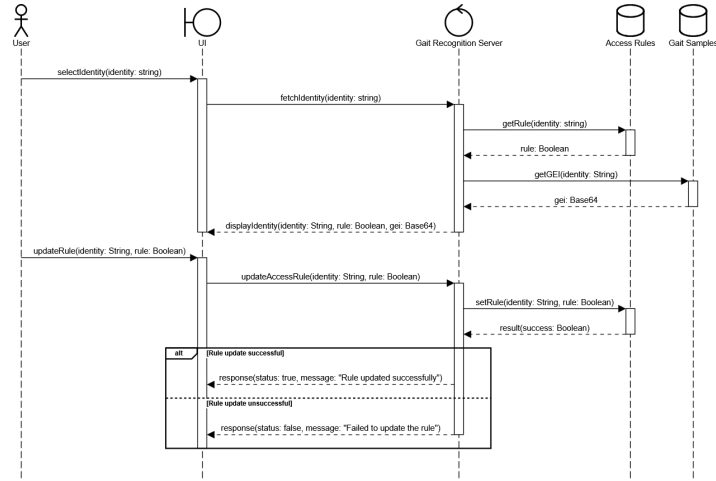


Figure 4.11: Server Access Rule Configuration Process Sequence Diagram

4.4.4.4 Database Design

The database for the GRS will enable storage of identities, GEI samples, access control rules, activity logs, and a trained classification model — all stored in their own respective tables and linked through relationships where necessary. The following Entity-Relationship Diagram (ERD) in Figure 4.12 illustrates the relationships between these tables:

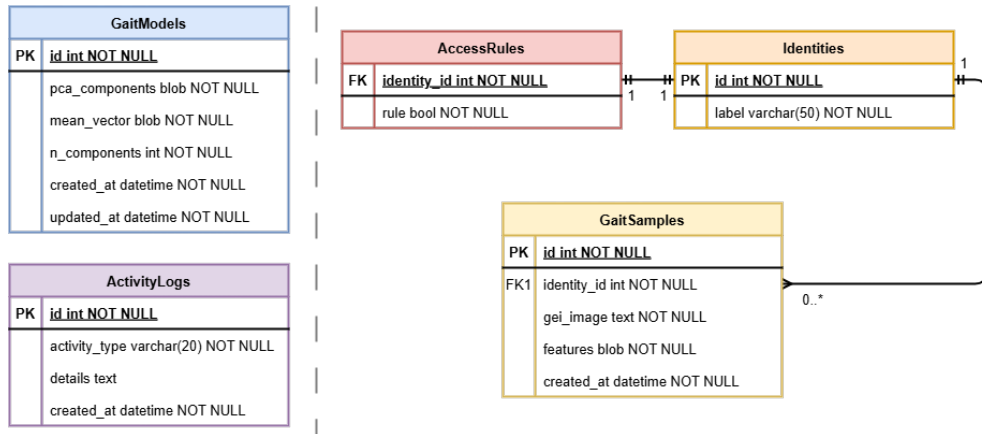


Figure 4.12: Server Database Entity Relationship Diagram

Identity Registered identities will be stored in this table, each with a unique identifier and a label (e.g., name). Each identity can have multiple GEI samples associated with it, with each sample linked to a specific identity through the `identity_id` field. A one-to-one relationship will also exist between identities and access rules, with each identity having a single Boolean access rule.

GaitSample GEI samples will be stored in this table, each associated with an identity through the `identity_id` field. GaitSample table entries will also store the GEI image as Base64 encoded data, and features extracted from the GEI as binary data, while also storing the timestamp of when the sample was created.

AccessRule Boolean access control rules will be stored in this table, each record linked to an identity through the `identity_id` field. The `rule` field will store the Boolean value indicating whether access is granted (`True`) or denied (`False`).

GaitModel A trained classification model will be stored in this table, containing the `pca_components` and `mean_vector` which will be regularly updated as new GEI samples are added to the database. The model will be stored as binary data, with the timestamp of when the model was created or last updated also being stored. The model will be used to perform classification on submitted GEIs as defined in Section 4.4.3.2.

ActivityLog System activity history will be stored in this table, including the type of activity, timestamp of when the activity occurred, and dynamic details about the activity (e.g., identity label, access rule, confidence score).

This database design ensures the GRS can efficiently manage identities, GEI samples, access control rules, logging, and a classification model in the same place. The relationships defined between tables will also ensure referential integrity is maintained.

4.4.4.5 User Interface Design

The GRS' UI will consist of three pages: a home page, an individual per-identity page, and a statistics and logs page. The home page, shown in Figure 4.13 below, will display the list of registered identities in a grid, with each identity's name, the most recent GEI sample. Clicking on an identity card will open the individual identity page for the respective identity.

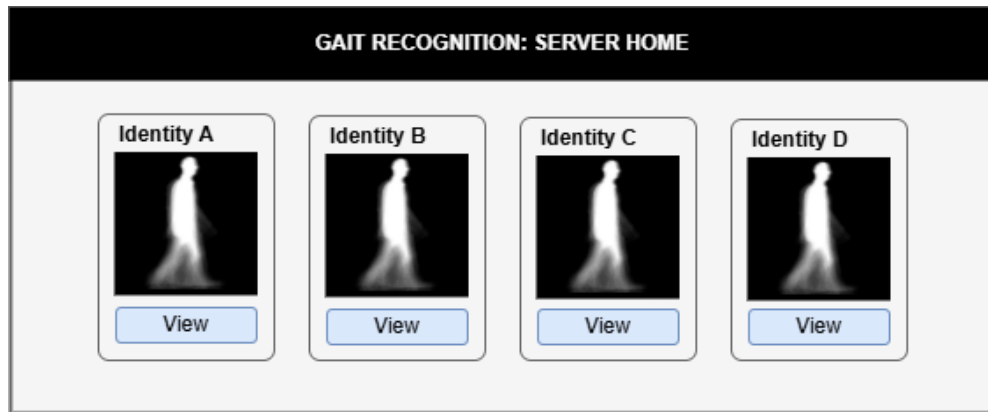


Figure 4.13: Gait Recognition Server 'Home' Page

The identity page(s), shown in Figure 4.14 below, will display a selected identity's details, including their label, number of associated GEI samples, creation date, and a toggleable access control rule. Below this information, a list of the identity's associated GEI samples will be displayed, each with an global ID number and a button to delete the individual sample.

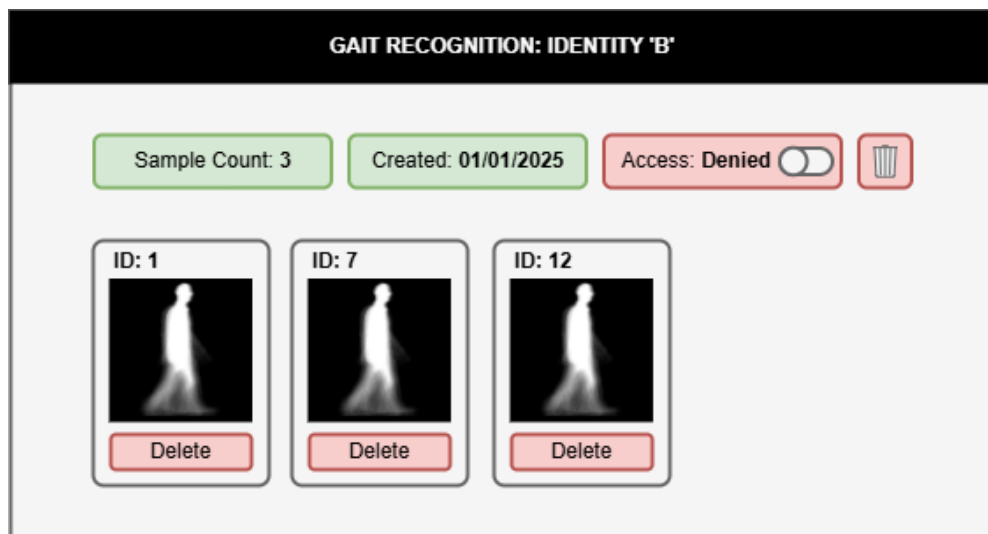


Figure 4.14: Gait Recognition Server per-'Identity' Page

Finally, the statistics and logs page, shown in Figure 4.15 below, will display the total number of identities registered, the total number of GEI samples stored, and the average number of GEI samples per identity. Underneath this information, a chronological list of system activities will be displayed in descending creation order.

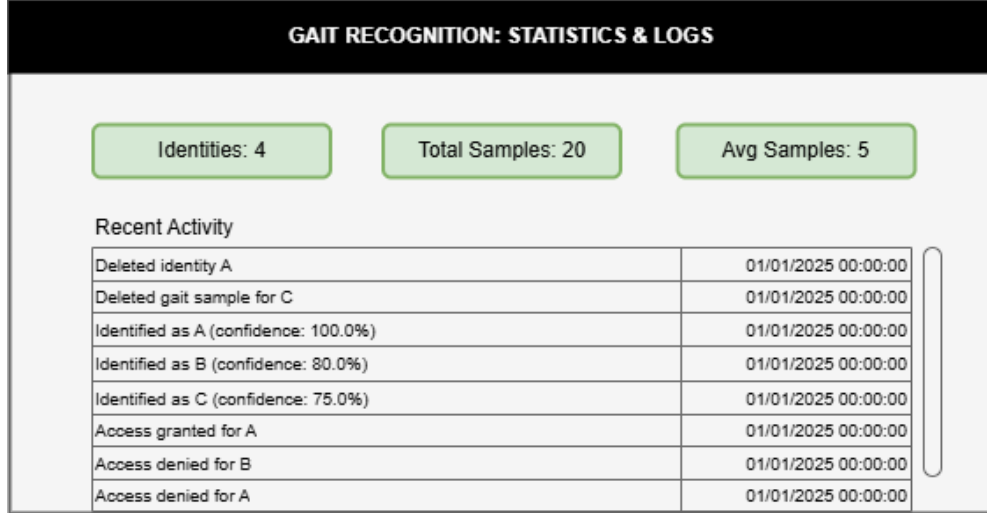


Figure 4.15: Gait Recognition Server 'Statistics and Logs' Page

4.5 Implementation

The following sections detail the implementation of the Gait Recognition Server (GRS) and Access Control Client (ACC) components, resulting in a complete Proof-of-Concept (PoC) of the Gait-Recognition (GR)-based access control system.

Several reference are made to things including buffer sizes, counters, and threshold values throughout. The following list provides a summary of what these variables were set to in this implementation, for reference.

- References to 'Buffer Size', 'Max Frame Count' or similar indicate a value of 30
- References to 'Minimum Contour Area' indicate a value of 500
- References to 'Confidence Threshold' or similar indicate a value of 75%
- References to 'Normalised Width' and 'Normalised Height' indicate a value of 88 and 128 respectively

4.5.1 Image Processing and Gait Energy Image (GEI) Creation

The Gait Energy Image (GEI) creation process, designed in Section 4.4.2, was the first to be implemented. This is a component of the ACC (designed in Section 4.4.3 and implemented in Section 4.5.2) but was developed first due to being the core component of this project; producing the abstracted gait cycle representations that enable person identification.

Following the design steps outlined in Section 4.4.2, this process was implemented to

enable GEI generation from video footage. The following subsections describe how each stage of the process was implemented.

4.5.1.1 Capturing Camera Input

To capture input from the camera using Python, the OpenCV (CV2) library is utilized. It provides functionality allowing frames to be read using the `VideoCapture` class and its `read` method. A basic example of this functionality is shown in Listing 4.2 below:

```
1 capture = cv2.VideoCapture(camera_id=0)
2
3 def get_frame():
4     ret, frame = capture.read()
5     if ret:
6         return frame
7     return None
8
9 while True:
10     frame = get_frame()
```

Listing 4.2: OpenCV (CV2) 'VideoCapture' Class Usage

To improve robustness, adhere to Object-Oriented Programming (OOP) principles, and maintain separation of concerns, this functionality is encapsulated in a `Camera` class. This allows frame capture to run in a separate thread, initiated using the class' `start` method, with the most recent frame retrieved using its `get_frame` method. *The final implementation of the `Camera` class can be seen in Appendix C.1.1.*

4.5.1.2 Background Subtraction and Person Detection

The person detection and background subtraction stages defined in Section 4.4.2 have been combined, with efficient person detection actually becoming easier after irrelevant information (the background) is removed.

Before background subtraction is performed, a simple 'background model' is created describing what the average background looks like. This is done by buffering a number N frames (collected using the `Camera` class' `get_frame` method), converting them to grayscale, and using the 'NumPy' library's `median` function to compute the background model by averaging the pixel intensity of buffered frames. For example, let $I_k(x, y)$ be the intensity of the k^{th} frame at pixel (x, y) for $k = 1, 2, \dots, N$. The background model

$B(x, y)$ is computed as:

$$B(x, y) = \text{median}(I_1(x, y), I_2(x, y), \dots, I_N(x, y))$$

Where:

- N is the background frame buffer size
- $B(x, y)$ is the average intensity (0-255) at pixel (x, y)

After computing the background model, the frame buffer is cleared, and the model is stored as a variable `background_model`. Background subtraction is then performed on subsequent frames, with the first steps involving converting the input frame to grayscale and applying a light Gaussian blur. This simplifies processing and reduces computational load by reducing noise. Following this, more CV2 functions are used to:

1. Calculate the 'absolute difference' between the background model and input frame:

```
diff = cv2.absdiff(background_model, gray)
```

2. Apply a 'binary threshold' of 25 to the difference to isolate the likely foreground:

```
_, binary = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)
```

3. Perform 'morphological closing' to fill small gaps in the foreground (noise reduction):

```
silhouette = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, np.ones((5, 5), np.uint8))
```

4. Identify all of the contours (shapes) present in the foreground:

```
contours, _ = cv2.findContours(silhouette, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
```

After separating foreground from background and identifying contours, any contours below a specified size `MIN_CONTOUR_AREA` are removed from the list to further reduce noise. A new frame is then created containing the remaining contours.

Finally, the largest contour in the list (most likely to be a person) is identified, and its dimensions are evaluated to determine whether it has an acceptable Aspect Ratio (AR), where $AR = \frac{width}{height}$. In this implementation, a valid AR is defined as being between 1.5 and 3.5, with a bounding box defined if so, indicating the region of interest that the person is assumed to be in. If the largest contour does not have an acceptable AR, the process does not return a bounding box, indicating that no person is detected in the frame.

This process allows for simple and efficient background subtraction and person detection *per frame*. While it is not guaranteed that the largest contour with an acceptable AR

is a person, the assumption can and has been made in this controlled environment. Figure 4.16 below illustrates the output achieved using the information provided by this process.



Figure 4.16: Background Subtraction Process Results

In the final ACC implementation, this process is encapsulated within the private `_extract_silhouette` method of the `GaitProcessor` class — see Appendix C.1.2.

4.5.1.3 Silhouette Normalisation

Following silhouette extraction, normalisation is applied to crop and resize silhouettes to a standard resolution and horizontally align them based on the subject's centre of mass (i.e., the head and torso, which are most consistently positioned). *Note:* this and subsequent stages are only executed if a region of interest is defined by the silhouette extraction stage.

The first step in normalisation is to crop the silhouette. This is done using the bounding box values calculated previously to 'slice' the NumPy array representation of the silhouette based on the bounding box's `width`, `height`, `y`, and `x` position (top left corner):

```
person_region = silhouette[y:y+h,x:x+w] .
```

Once the figure has been isolated from the greater silhouette frame, the bounding box is tightened by identifying non-zero pixels and redefining the boundary to the smallest possible size that still encloses the silhouette, removing unnecessary space.

The silhouette's centre of mass `cx` is then calculated by averaging the x-coordinates of

all non-zero pixels, providing an estimate of the subject's geometric centre along the x-axis:

$$cx = \left\lfloor \frac{1}{N} \sum_{i=1}^N x_i \right\rfloor$$

Where:

- N is the total number of non-zero pixels in the region
- x_i is the x-coordinate of the i^{th} non-zero pixel

Based on the estimated centre of mass, the silhouette is inserted into a new frame (of equal height) and horizontally aligned. This ensures consistent alignment across all silhouettes before they are used to produce the GEI, allowing for more consistent feature extraction and identity classification.

Finally, the NumPy array representing the silhouette is sliced to resize and crop it to a standard resolution — defined by the `NORMALISED_HEIGHT` and `NORMALISED_WIDTH` constants. Figure 4.17 below shows (a) the initial silhouette, and (b) the normalised output. This step is repeated for every frame captured, with each normalised silhouette appended to a buffer maintained using a fixed size list `aligned_buffer`, and subsequently used to create the GEI.



Figure 4.17: Normalisation Process Results

In the final ACC implementation, this process is encapsulated within the private `_align_silhouette` method of the `GaitProcessor` class — see Appendix C.1.2.

4.5.1.4 Silhouette Averaging

The final step in GEI creation is to average the set of normalised silhouettes created by the steps defined above. Every time a normalised silhouette is produced, it is appended

to the sliding `aligned_buffer`. Once at the required size, the silhouettes are stacked and averaged to form the GEI, representing spatial and temporal information about the subject's gait pattern.

If no prior GEI exists in memory (i.e., on the first run), the GEI is computed as the mean of all silhouettes in the buffer:

$$GEI = \frac{1}{N} \sum_{k=1}^N I_k$$

Where:

- GEI is the stack of averaged silhouettes - the Gait Energy Image (GEI)
- I_k is the aligned silhouette at frame k
- N is the number of silhouettes in the buffer

However, if there is an existing GEI, a 'weighted update' is performed using a configurable Exponential Moving Average (EMA). This helps the GEI adapt at a defined rate over time as new silhouettes are captured and the buffer is updated:

$$GEI_{new} = (1 - \alpha) \cdot GEI_{prev} + \alpha \cdot \bar{S}$$

Where:

- $\bar{S}$ is the mean silhouette from the buffer
- $\alpha \in [0, 1]$ is the weighted update rate (lower = slower)
- GEI_{prev} is the previous GEI
- GEI_{new} is the new GEI

In this implementation, a low (0.2) α value is used which prioritises older silhouettes over new ones, minimising the potential impact of intermittent noisy or inconsistent frames. The closer the α value is to 1.0, the faster the GEI is adapted to new silhouette input.

The output of this step is a GEI image representing the subject's gait cycle over k frames. These are used by the GRS to perform feature extraction and identity classification. Figure 4.18 below shows an example of a GEI created by applying this process to a sequence of normalised silhouettes from the GRIDDS dataset (Nunes, Moreira and Tavares, 2019).

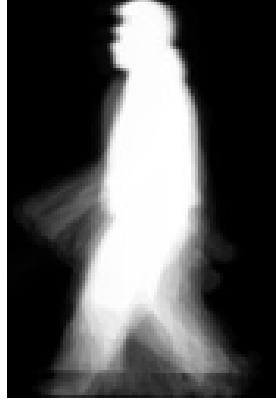


Figure 4.18: Example Silhouette Averaging Result

In the final ACC implementation, this process is encapsulated within the private `_update_gei` method of the `GaitProcessor` class — see Appendix C.1.2. All methods used in the GEI creation process are called from within the `GaitProcessor`'s public `process_frame` method. This method is invoked by the ACC's main thread (see Section 4.5.2) when a new camera frame is retrieved, returning both the non-normalised silhouette frame and the current GEI.

4.5.2 Access Control Client

The ACC leverages the camera frame capture and GEI creation process to provide the user with a view of what is occurring, following the designs outlined in Section 4.4.3.3, where a preview of the live feed, isolated silhouette, and GEI are displayed. It also communicates with the GRS to support registration of GEI-based identities and receive classification results of created or input GEIs, using the results and associated access rule for access provision (rule = True AND confidence > 75%).

The (Python) Flask web framework is used to serve the front-end User-Interface (UI), and handle routing and user requests on the back-end. WebSockets (WSs) (Socket.IO) are used to enable efficient data transmission between the front and back-end, continuously updating the three video views and classification results with less network overhead than traditional HTTP requests.

4.5.2.1 Application State Tracking

To streamline updates across the client, such as resetting the average background model for background subtraction from the front-end, tracking the number of frames processed,

or the current GEI, a `state` module has been implemented which simply enables shared use of values across back-end functions.

```
1 state = {
2     "background_init": False, # True when background model has been
    ↪ created
3     "reset_requested": False, # True when reset button is pressed
4     "frame_counter": 0, # max 30 in this implementation
5     "gei_buffer": None, # numpy array representing the current GEI
6     "last_saved_gei": None, # last GEI sent for classification
7 }
```

Listing 4.3: Default Access Control Client State Configuration

The values in the state tracking dictionary shown in Listing 4.3 above are utilized throughout the client as follows:

- `background_init`: Set to `True` by the main thread when the `GaitProcessor` class has finished creating the averaged background model (see)
- `reset_requested`: Set to `True` by the web server thread when a request is received from the front-end due to a user pressing the 'Reset Background' button
- `frame_counter`: Incremented by the main thread every time a new frame is processed, reset to zero by the identification thread (see Section 4.5.2.4) every time a GEI is sent for classification.
- `gei_buffer`: Set by the main thread every time a frame is processed and GEI returned by the `GaitProcessor`, accessed by the identification thread when retrieving the current GEI for classification
- `last_classified_gei`: Stores the last GEI sent for classification, used by the identification thread to determine whether the current GEI has already been classified

4.5.2.2 Client for Gait Recognition Server API

The GRS designed in Section 4.4.4 and implemented in Section 4.5.3 provides two Application Programming Interface (API) endpoints allowing the client to either: (1) Register GEI-based identities or (2) Submit a GEI for classification. To abstract the use of this in the ACC, a class `ServerAPIClient` was created, providing two methods: `classify_gei` and `register_gei` which are used to submit requests to the respective endpoints.

The `register_gei` method, shown in Listing 4.4, accepts a GEI and identity label string before encoding the GEI as Base64 and `POST`ing a JavaScript Object Notation (JSON) payload containing the GEI and label: `gei: ..., label: ...` to the GRS' `/api/register` endpoint.

```
1 def register_gei(self, gei_image: bytes, label: str):
2     try:
3         gei_base64 = base64.b64encode(gei_image).decode("utf-8")
4         payload = {"label": label, "gei": gei_base64}
5         response = requests.post(self.register_endpoint, json=payload)
6
7         if response.status_code == 200:
8             return True, None
9         else:
10             ...
11     except:
12         ...
```

Listing 4.4: Server API Client `register_gei` Method Logic

The `classify_gei` method, shown in Listing 4.5, accepts a Base64-encoded GEI as input, before `POST`ing a JSON payload containing the GEI: `{gei: ...}` to the GRS' `/api/classify` endpoint. The GRS responds with a JSON object containing the predicted label (person), confidence (percentage), and access rule, which the method returns.

```
1 def classify_gei(self, gei_base64: str):
2     try:
3         response = requests.post(self.classify_endpoint, json={"gei":
4             ↪ gei_base64})
5
6         if response.status_code == 200:
7             return response.json(), None
8         else:
9             ...
10     except:
11         ...
```

Listing 4.5: Server API Client `classify_gei` Method Logic

Implementation of this class enabled the separation of concerns and ensured adherence to the Don't Repeat Yourself (DRY) coding principles, as these methods are used in

several places throughout the client codebase.

4.5.2.3 Frame Capture, Processing, and Transmission

The *main* thread of the ACC loops continuously and leverages the `Camera` and `GaitProcessor` classes defined in Section 4.4.2 to capture frames, isolate silhouettes, and create GEIs, updating the front-end view with every new frame.

When the ACC's `main` function is called, a new instance of the `Camera` class is created, and video capture is started by calling the class' `start` method. Following this, a new instance of the `GaitProcessor` class is created with the `buffer_size` set to thirty (30) frames, as illustrated in Listing 4.6 below:

```
1 cam = Camera(camera_id=0)
2 cam.start()
3 processor = GaitProcessor(buffer_size=30)
```

Listing 4.6: Example 'Camera' and 'GaitProcessor' Class usage

The main loop begins by retrieving the most recent video frame (`frame0`) and calling the `GaitProcessor`'s `process_frame` method which returns the non-normalised silhouette (`frame1`), and the current GEI (`frame2`). All three frames are then encoded as JPEG `cv2.imencode(".jpg", current_gei)`, then as Base64 `base64.b64encode(buffer).decode("utf-8")` before being transmitted to the front-end over the WS using the endpoint defined by their frame name: `frame[0,1,2]` — see Listing 4.7 below:

```
1 for i, img in enumerate([frame, silhouette, gei]): # frame0, frame1,
   ↪ frame2
2     _, buffer = cv2.imencode(".jpg", img) # encode as jpeg
3     img_encoded = base64.b64encode(buffer).decode("utf-8") # encode as
   ↪ base64
4     socketio.emit(f"frame{i}", img_encoded) # transmit to front-end
```

Listing 4.7: 'Live', 'Silhouette', and 'GEI' Frame Transmission via WS

On the front-end, JavaScript (JS) is used to listen for data received on either of the three frame endpoints (see Listing 4.8), updating the associated front-end HTML `img` tag's `src` attribute with the Base64 encoded frame in the format

```
data:image/jpeg;base64,<base64_frame> .
```

```
1 socket.on('frame0', data => {  
2   frame0.src = 'data:image/jpeg;base64,' + data;  
3 });  
4 ...+2 for frame1 and frame2
```

Listing 4.8: JavaScript WebSocket Listener for Receiving Frames

The final action in the main loop is to sleep for 0.03 seconds. This targets a frame-rate of around 30 frames per second; $0.03 \cdot 30 \approx 1$. A check also exists to ensure a user is present on the main page before camera frames are processed, saving unnecessary use of resources.

This fulfills the ACC ACF1 and ACF2 requirements, abstracting movement data, captured using Computer Vision (CV) techniques into GEIs before they are transmitted.

4.5.2.4 Identity Classification Thread

The identity classification process defined in Section 4.4.3.2 has been implemented as a separate thread within the ACC. This is because it requires no user interaction and is intended to operate passively.

The classification loop starts by checking the 'state' of the ACC, evaluating the state's `frame_counter` value to determine whether it is greater than or equal to the defined interval of thirty (30) frames. If that is the case, the current GEI and last checked GEI are retrieved from state storage, and a check is performed to determine whether or not they differ using the `is_different` function which performs a simple not-equal-to comparison.

If the GEI has been updated since classification was last performed, the state is updated with the new GEI set to the last checked GEI, the frame is encoded as a JPEG image and then into Base64, before a `classify_gei` request: `response, message = api_client.classify_gei(gei_base64)` is made to the GRS using the `ServerAPIClient` class (defined in Section 4.5.2.2). If a valid response is received, the details are transmitted to the front-end over the `status` WS endpoint: `socketio.emit("status", response)`. The front-end evaluates the response, and if the prediction confidence is at or above the threshold of 75%, the classification result is

displayed to the user (see Section 4.5.2.7) using the JS shown below in Listing 4.9.

```
1 socket.on('status', (data) => {
2   if (data.confidence < 75) {
3     confidence.textContent = 'N/A';
4     access.textContent = 'Denied';
5     access.classList.toggle('access-denied', true);
6     person.textContent = 'Unknown';
7     return;
8   }
9
10  person.textContent = data.person;
11  access.textContent = data.access ? 'Granted' : 'Denied';
12  access.classList.toggle('access-denied', !data.access);
13  confidence.textContent = `${data.confidence}%`;
14 });
```

Listing 4.9: Front-End JavaScript Listener for Receiving Classification Updates

Finally, the `frame_counter` state value is reset to zero (0) before the loop restarts. The identity classification thread in its entirety can be seen in Appendix C.1.3. This fulfills the ACC ACF3 requirement, providing passive and continuous identity classification.

4.5.2.5 Identity Registration Process

The identity registration process is triggered from the front-end but is handled by a WS endpoint on the back-end. When the user presses the 'Save Identity' button, a JS function is triggered which prompts the user to enter a label for the identity, before transmitting the label and GEI over the WS to the back-end, shown in Listing 4.10 below.

```
1 saveBtn.addEventListener('click', () => {
2   const label = prompt('Enter label for this GEI:');
3   if (label) {
4     socket.emit('save', { label: label, gei: window.latestGEI });
5   }
6 });
```

Listing 4.10: JavaScript Button-Click Handler for Saving GEI-based Identity

The request to the back-end is handled by the client's `handle_save_gei` Socket.IO WS endpoint, which sends a `register_gei` request `api_client.register_gei(gei_bytes, label)` to the GRS using the `ServerAPIClient` class. The GRS-side logic for this process is

discussed in Section 4.5.3.

This fulfills the ACC ACF7 requirement, allowing the user to submit a captured GEI for registration with.

To support control of identity registration, a 'Reset Background' button also exists on the front-end. When pressed, it triggers a `reset` request to the back-end via the `handle_reset_background` Socket.IO WS endpoint, which modifies the state's `reset_requested` value to `True`. A check is performed once per loop in the main thread which invokes the `GaitProcessor`'s `reset_background` method (see Appendix C.1.2) if `reset_requested` is `True`. This clears and reinitialised the background model based on the camera's current view.

4.5.2.6 Manual Identity Classification

The manual identity classification feature enables isolated demonstration of this system's GEI matching functionality. The user is able to upload an image on the front-end's 'Manual Classification' page which is sent to the back-end before the `ServerAPIClient`'s `classify_gei` method is used to request classification results from the GRS in the same way as live classification.

On the front-end, image submission is handled by a JS function, `handleImageInput`. This function reads the uploaded image file, converts it to Base64, and sends it to the back-end's `/api/classify` endpoint via a `POST` request. Upon receiving the response, it updates the classification result fields and the displayed image preview. The key functionality for this is shown in Listing 4.11 below:

```
1 const response = await fetch('/api/classify', {
2   method: 'POST',
3   body: JSON.stringify({
4     gei: base64Image
5   })
6 });
7 const result = await response.json();
8
9 person.textContent = result.person;
10 confidence.textContent = `${result.confidence}%`;
11 access.textContent = result.access ? 'Granted' : 'Denied';
12 access.classList.toggle('access-denied', !result.access);
13
14 previewImage.src = reader.result;
```

Listing 4.11: JavaScript Function for Manual GEI Classification

Note: This purposefully does not implement the prediction confidence threshold check as it is intended to be used for demonstration purposes in isolation, where it may be more valuable to display the result regardless.

The ACC's back-end implementation of the `/api/classify` endpoint shown in Listing 4.12 below, accepts the Base64-encoded image, passes it to the server-side `/api/classify` endpoint, and returns the results as JSON.

```
1 @routes.route("/api/classify", methods=["POST"])
2 def classify_gei():
3     try:
4         data = request.get_json()
5         response, _ = api_client.classify_gei(data["gei"])
6         return jsonify(response), 200
7     except Exception as e:
8         ...error handling
```

Listing 4.12: Server `/api/classify` Endpoint Implementation

This feature supports testing and demonstration of the system's classification pipeline without relying on live camera input, meeting the ACC ACF9 requirement. It also aids in debugging and validation by allowing the user to test individual GEI samples manually and enables the efficacy evaluation detailed in Chapter 5.

4.5.2.7 User-Interface

The ACC's front-end — the two pages designed in Section 4.4.3.3 — is built using HTML (with the Jinja2 templating engine) and CSS, with JS used to modify page content and interface with the back-end where necessary; resetting the `GaitProcessor` background model, saving a GEI-based identity, or manually submitting a GEI for classification.

The 'Live Classification' (index) page, shown in the screenshot in Figure 4.19, features three views displaying the stages of GEI creation, which are updated using JS. It also features text areas that display classification results when received from the GRS, also updated using JS. Additionally, the page includes buttons for resetting the `GaitProcessor` background model on the back-end and saving a GEI sample, both of which are handled by JS event listeners, detailed in Section 4.5.2.

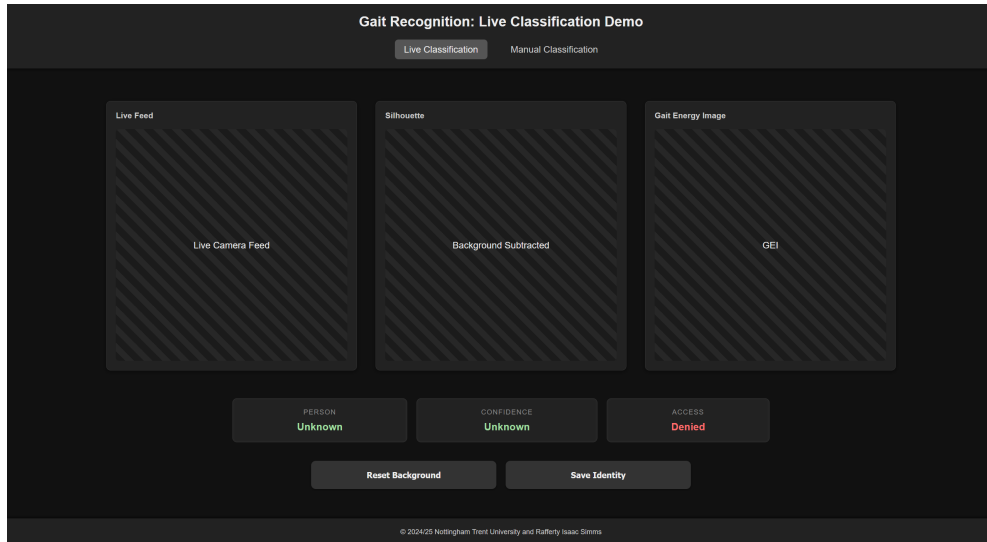


Figure 4.19: Live Classification Page User-Interface

The use of WSs on this page allows for efficient data streaming from the back-end to update frame data or classification results without the overhead of additional HTTP headers.

The 'Manual Classification' page shown in the screenshot in Figure 4.20, enables manual GEI-based identity classification, support for which was implemented in Section 4.5.2.6. It features a 'drag-and-drop' area for image file submission, alongside a display of the uploaded image. Below this, the same text views from the main page are used to present classification results returned by the GRS.

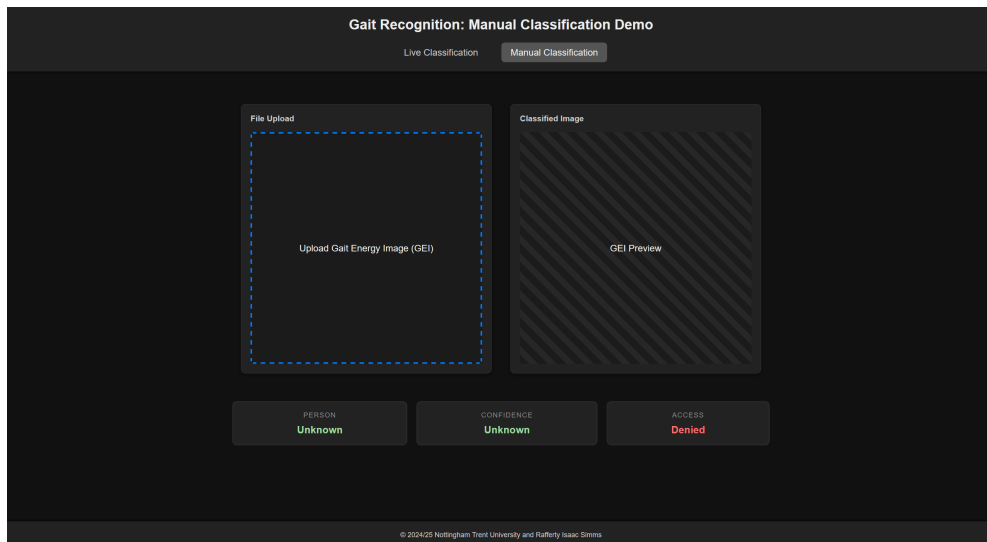


Figure 4.20: Manual Classification Page User-Interface

Implementation of the front-end UI concluded development of the ACC. It also fulfills

the ACC ACF6 and ACF8 requirements, providing a UI displaying the camera and GEI processing feeds, and displaying access decisions. An evaluation of this component’s adherence to the design and requirements is detailed in Section 5.

4.5.3 Gait Recognition Server

4.5.3.1 Database Modelling

Following the database structure defined in Section 4.4.4.4, an SQLite database model has been implemented using the SQL Alchemy Python package. A custom Python module named ‘models’ has been created, within which the below five classes are defined, inheriting from the SQL Alchemy `Model` class. The database models module can be seen in full in Appendix C.1.4.

The `Identity` class models individual identities registered in the system. Each identity entry contains a unique `id`, a label, and establishes relationships with both the gait sample and access control rule tables. Referential deletes have been defined with these relationships, maintaining integrity when records are removed.

The `GaitSample` model stores individual GEI samples, each linked to an identity. Each sample includes a GEI in Base64-encoded text format, a binary representation of its extracted features (see Section 2.3.2.2), and a timestamp marking its creation. Implementation and this and the `Identity` table meets the GRS GRF4 requirement, maintaining historic movement data for each registered identity.

The `AccessRule` model associates Boolean access permissions (`True` for allow, `False` for deny) with a given identity. It enforces a one-to-one relationship with the `Identity` model by setting `identity_id` as its primary key. This meets the GRS GRF5 requirement, allowing the system to manage access control rules for each registered identity.

The `GaitModel` class stores serialised data representing the trained Principal Component Analysis (PCA) model used for GEI feature dimensionality reduction. It includes the binary-encoded PCA components, the mean vector, the current number of components, and the timestamps denoting model creation time and last update. In this implementation, only one model is stored in this table, with the values contributing to GEI classification.

The `ActivityLog` class is used for recording system activity, when identification occurs, or identities or gait samples are created or deleted, records are added to this table. A

page on the GRS' front-end then displays the logs for the user. Each log entry denotes the type of activity (e.g., 'identify'), optional descriptive details, and a timestamp.

Implementing the database model in a module allows it to be accessed and used in many areas throughout the GRS application by importing the database instance:

```
from models import db
```

The following sections will refer to the use of this model when inserting, updating, or deleting database records.

4.5.3.2 Feature Extraction, Model Training, and Classification

Like the ACC's gait processing features, gait-related functions in the GRS have been encapsulated into a single 'gait' module. This module enables GEI feature extraction using a Convolutional Neural Network (CNN), PCA dimensionality reduction and model training, and GEI-based identity classification; processes discussed in more detail below.

Issue Encountered! A model-free k -Nearest Neighbour (NN) approach was originally implemented at this stage. The idea being that for a PoC, the classification did not have to be perfect, and a model-free matching algorithm could be used for demonstration purposes. Unfortunately, testing showed this was highly susceptible to overfitting, meaning the chance of misclassification, especially with smaller datasets was higher than acceptable. Because of this, the decision was made to commit to implementing an Machine Learning (ML) model-based classification scheme, with a CNN, PCA, and cosine similarity pipeline implemented in the final solution.

Feature extraction is achieved using a CNN with the pretrained `EfficientNetB0` model through the 'TensorFlow' Python package (see Listing 4.13) after being transformed to expected resolution and colour format (224x224 and RGB) using CV2. This model provides a good trade-off between performance and efficiency compared to more demanding options like `VGG16` — ideal for this demonstrative scenario.

```
1 base_model = EfficientNetB0(weights="imagenet", include_top=False)
2 feature_extractor = Model(inputs=base_model.input,
   ↪ outputs=base_model.get_layer("top_conv").output)
3 ...
4 batch = np.expand_dims(gei, axis=0)
5 features = feature_extractor.predict(batch, verbose=0)
```

Listing 4.13: Feature Extraction Model Definition and Usage

After processing, the CNN output is flattened forming a 'feature vector' representing the input GEI: `features.flatten()`. These features are stored in the database alongside other sample information in the `GaitSample` table when a new GEI is registered (see Section 4.4.4.4).

To improve classification speed and accuracy, PCA is applied to reduce the dimensionality of CNN-extracted features (see Section 2.3.2.3). A 'trained' PCA model is stored in the database and updated every time a new GEI sample is added using the `update_model` function. This retrieves all samples from the database before standardising features using the 'scikit-learn' Python module's `StandardScaler` — ensuring equal structure and fair PCA evaluation. The number of PCA components is chosen dynamically based on the size of the dataset: $\min(128, \text{dataset_size} - 1)$. Listing 4.14 below demonstrates the creation of a scaled list of all features, and training of a PCA model based on the standardised features.

```
1 features = np.array([pickle.loads(sample.features) for sample in
  ↪ gait_samples])
2
3 scaler = StandardScaler()
4 features_scaled = scaler.fit_transform(features)
5
6 n_components = min(128, len(features) - 1)
7 pca = PCA(n_components=n_components)
8 pca.fit(features_scaled)
```

Listing 4.14: PCA Model Training Logic

Dimension-reduced PCA components and its the 'mean vector' are then serialised using the Python 'pickle' module and stored in `pca_components` and `mean_vector` fields of the database's `GaitModel` table, improving future classification performance by reducing redundant computation of each sample when performing comparisons.

The third function provided by this module — `classify` — enables classification of GEIs. It accepts a single GEI as input before calling the `extract_features` function to retrieve its CNN features. Following this, the PCA model is retrieved from the `GaitModel` table and decoded from binary. The features of the input GEI are then scaled and projected

according to the PCA model's components and mean vector, as shown in Listing 4.15.

```
1 features = extract_features(gei)
2 gait_model = GaitModel.query.first()
3
4 pca_components = pickle.loads(gait_model.pca_components)
5 mean_vector = pickle.loads(gait_model.mean_vector)
6
7 features_scaled = (features - mean_vector) / np.sqrt(np.var(features))
8 projected_features = np.dot(features_scaled, pca_components.T)
```

Listing 4.15: Feature Extraction and PCA Projection Logic

After the input GEIs features have been extracted and the PCA model loaded, all existing samples are loaded and iterated through with the 'cosine similarity' (calculated by calculating the cosine of the angle between the feature vector of the input and each sample) to the input calculated for each, using the scikit-learn module's `cosine_similarity` function once they have been scaled and projected to fit the PCA model structure. This process is illustrated below in Listing 4.16.

```
1 for sample in gait_samples:
2     sample_features = pickle.loads(sample.features)
3     sample_features_scaled = (sample_features - mean_vector) / np.sqrt(
4         np.var(sample_features)
5     )
6     sample_projected = np.dot(sample_features_scaled, pca_components.T)
7
8     similarities.append(
9         (
10            cosine_similarity(
11                projected_features.reshape(1, -1),
12                sample_projected.reshape(1, -1),
13            )[0][0],
14            sample.identity.label if sample.identity else None,
15        )
16    )
```

Listing 4.16: Cosine Similarity Comparison between Input and Samples

Finally, the best match is selected from the list of similarities, and the confidence is calculated as a percentage of the best match's similarity. The function then returns the predicted label and confidence. This meets the GRS GRF2 and GRF10 requirements, allowing the system to classify (predict the identity of) GEI samples using a trained

ML model.

4.5.3.3 Identity Registration Endpoint

GEI-based identity registration functionality is provided to the ACC via the GRS' API endpoint at `/api/register`, implemented using the Flask web framework. This endpoint accepts `POST` requests containing a JSON object with two fields: a Base-64-encoded GEI and the identity label entered by the user on the client side.

Upon receiving a request, the handler function first checks whether an identity with the specified label already exists in the database. If not, a new `Identity` entry is created and saved, with a default `AccessRule` entry also initialised for the new identity, set to `False` (i.e., access denied by default).

The GEI is then decoded from its Base64 representation and passed to the `extract_features` function (see Section 4.5.3.2), which extracts a feature vector using the EfficientNetB0 CNN. The extracted features are serialised using the 'pickle' module, and a new `GaitSample` record is created and populated accordingly. At the same time, a new `ActivityLog` entry is created of type 'register' to log the registration event. This is shown in Listing 4.17 below.

```
1 gei = base64.b64decode(base64_gei)
2 nparr = np.frombuffer(gei, np.uint8)
3 gei = cv2.imdecode(nparr, cv2.IMREAD_GRAYSCALE)
4
5 features = extract_features(gei)
6 serialized_features = pickle.dumps(features)
7
8 new_sample = GaitSample(identity_id=identity.id, gei_image=base64_gei,
   ↪ features=serialized_features, created_at=datetime.now())
9 db.session.add(new_sample)
10
11 activity_log = ActivityLog(activity_type="register",
   ↪ details=str({"label": label}), created_at=datetime.now())
12 db.session.add(activity_log)
```

Listing 4.17: Identity Registration Endpoint Logic

Finally, the `update_model` function is called to retrain the PCA model using the updated dataset (see Section 4.5.3.2), before a JSON response is returned to the client containing the ID of the affected identity. This meets the GRS GRF1 requirement, supporting the

registration of new identities and GEI samples.

4.5.3.4 Identity Classification Endpoint

The GRS handles GEI classification requests from the ACC via the `/api/classify` API endpoint. The endpoint accepts `POST` requests containing a JSON object with a single Base64-encoded GEI.

Upon receiving a request, the handler function decodes the GEI from Base64 and passes it to the `classify` function (see Section 4.5.3.2) to retrieve the predicted identity label and confidence score reflecting prediction certainty:

```
prediction, confidence = classify(gei).
```

Following identity prediction, the system retrieves the corresponding access rule for the predicted identity from the AccessRule table. An `ActivityLog` entry of type 'identify' is created to record the classification event, including both the predicted identity and the associated confidence score.

Finally, the GRS responds to the client with a JSON object containing three fields: the predicted identity, the confidence score, and the access rule. A successful response is shown in Listing 4.18 below. This meets the GRS GRF3 and GRF9 requirements, responding to ACC queries with the predicted identity, and logging the event.

```
1 "success": True,  
2 "person": prediction,  
3 "confidence": float(confidence),  
4 "access": access_granted,
```

Listing 4.18: Identity Classification Endpoint (Successful) Response

4.5.3.5 Access Control Rule Management

Access rule configuration is enabled via toggle switches in the front-end UI (see Section 4.4.3), and implemented using Socket.IO-based WebSocket (WS).

On the back-end, access rule updates are handled by the `handle_access_rule_update` Socket.IO listener. This expects transmissions to contain two fields: an `identity_id` integer and an `allow_access` Boolean. The GRS uses the provided `identity_id` to query the corresponding identity from the database and updates the associated `AccessRule`

entry's `rule` field to reflect the new status. This is shown in Listing 4.19 below.

```
1 identity = Identity.query.get_or_404(identity_id)
2 identity.access_rule.rule = allow_access
```

Listing 4.19: Access Rule Update Logic

On the front-end, each identity's page includes a toggle switch linked to a JS function (shown in Listing 4.20) that is triggered whenever the toggle is clicked. This function emits a WS message containing the toggle's Boolean state and the relevant identity's ID to the back-end.

```
1 const accessToggle = document.getElementById('accessToggle');
2
3 accessToggle.addEventListener('change', function (e) {
4   const identityId = this.dataset.identityId;
5   const allowAccess = this.checked;
6
7   socket.emit('update-access-rule', {
8     identity_id: identityId,
9     allow_access: allowAccess
10  });
11 });
```

Listing 4.20: Access Rule Toggle Front-End Handler

The GRS front-end is designed to function independently of the ACC, allowing configuration changes to be made regardless of whether the ACC is running, meeting the GRS GRF7 requirement.

4.5.3.6 Identity and Gait Cycle Management

The GRS' UI provides functionality for deleting either entire identities or individual gait cycles associated with an identity. This functionality is also enabled through WS-based communication and supported on the back-end via two Socket.IO event handlers: `handle_delete_identity` and `handle_delete_gait_sample`.

The `handle_delete_identity` function expects a transmission containing an `identity_id` integer. This is used to retrieve the corresponding `Identity` entry from the database. An `ActivityLog` entry of type 'delete' is then created to record the deletion event, including

the label of the identity. The identity is then removed from the database, and due to the configured database model relationships (see Section 4.4.4.4), the associated `AccessRule` and `GaitSample` records are also removed. This is shown in Listing 4.21 below.

```
1 identity = Identity.query.get_or_404(identity_id)
2
3 activity_log = ActivityLog(
4     activity_type="delete",
5     details=str({"label": identity.label}),
6     created_at=datetime.now(),
7 )
8 db.session.add(activity_log)
9
10 db.session.delete(identity)
```

Listing 4.21: Identity Deletion Logic

On the front-end, a delete button is placed in the top right corner of each identity's page. When clicked, a JS function (shown in Listing 4.22) is triggered, prompting the user to confirm the deletion. If confirmed, the `identity_id` is emitted to the back-end over the WS connection.

```
1 const deleteIdentityBtn = document.getElementById('deleteIdentity');
2
3 deleteIdentityBtn.addEventListener('click', function () {
4     if (confirm('Are you sure you want to delete this identity?')) {
5         const identityId = this.dataset.identityId;
6         socket.emit('delete-identity', { identity_id: identityId });
7     }
8 });
```

Listing 4.22: Identity Deletion Front-End Handler

For individual GEI deletion, the `handle_delete_gait_sample` function is used. It accepts a `sample_id` integer, retrieves the relevant `GaitSample` record from the database, and performs similar actions as with identity deletion: logging the deletion event via a new `ActivityLog` entry and removing the sample record from the `GaitSample` table: `db.session.delete(sample)`.

On the front-end, each GEI preview on the identity page includes a delete button. Clicking this triggers a JS function similar to Listing 4.22 that prompts the user for

confirmation, then emits the corresponding `sample_id` to the back-end via the WS connection.

The final step for both handlers is to call the `update_model` function (see Section 4.5.3.2) to retrain the PCA model on the updated dataset, ensuring no redundant data is included that may affect accuracy.

4.5.3.7 Statistic and Log Processing

The GRS includes an independent 'Statistics & Logs' page, providing an overview of system activity. When the page is accessed, the back-end queries the total number of entries in the `Identity` and `GaitSample` tables and retrieves entries from the `ActivityLog` table in descending creation order.

Once retrieved, the logs are processed into a `formatted_logs` list, where each entry is transformed into a human-readable message summarising the action performed (e.g., identity registration, classification, or deletion).

Finally, the stats page (see Section 4.5.3.8) is rendered with the identity count, GEI sample count, and the list of formatted log messages passed as parameters to the front-end, where they are displayed in an accessible format.

The logic for retrieving and formatting the statistic and log data can be seen in Appendix C.1.5.

4.5.3.8 User-Interface

The GRS's front-end — comprising the three pages designed in Section 4.4.4.5 — is built using the same technologies as the ACC's front-end. It enables users to manage registered identities and their associated GEI samples, configure access control rules, and view system statistics and logs.

The 'Server Home' (index) page, shown in the screenshot in Figure 4.21, presents a grid view of all registered identities. Selecting an identity navigates the user to the 'Identity' page outlined below. Each identity card includes the identity's label, the most recent GEI sample added, the creation date, and a Boolean flag indicating access status.

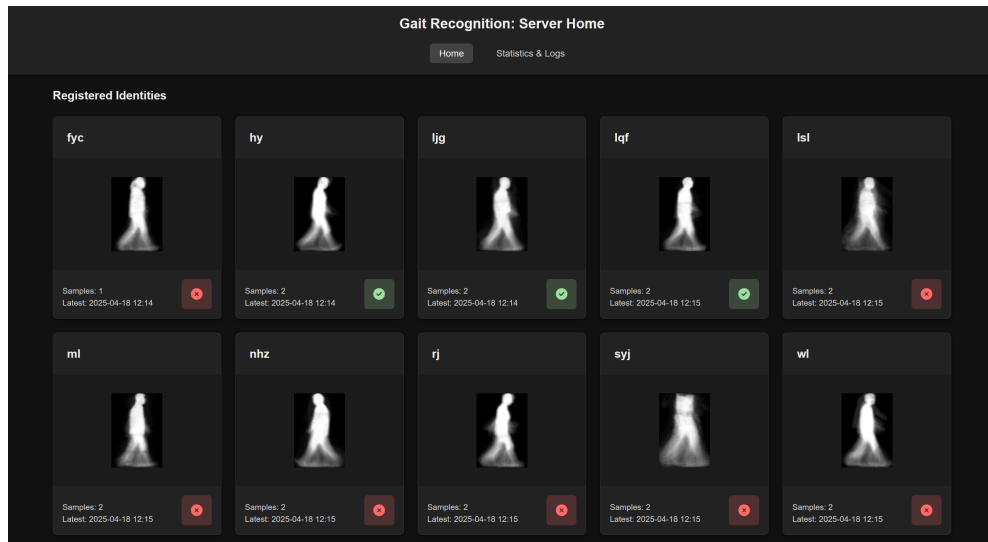


Figure 4.21: 'Server Home' Page User-Interface

The 'Identity' page, shown in Figure 4.22, displays detailed information about the selected identity, including the identity ID, number of GEI samples, creation date, and last update timestamp. A toggle switch allows configuration of the access rule for the identity, and a delete button enables removal of the identity (see Section 4.5.3.6 for deletion logic). Below this, a grid displays all GEI samples associated with the identity, with individual delete buttons allowing selective removal of specific samples.

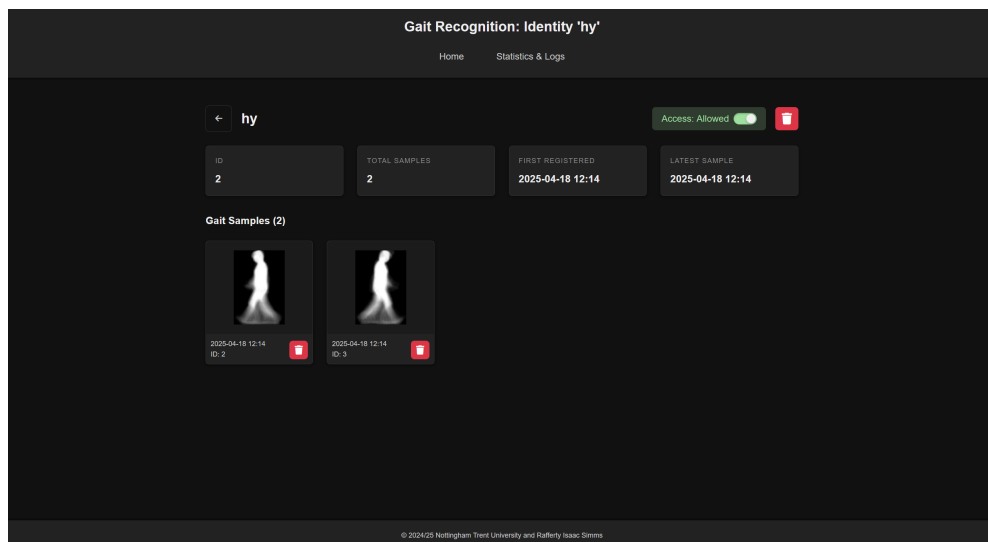


Figure 4.22: 'Identity' Page User-Interface

The 'Statistics & Logs' (stats) page, shown in Figure 4.23, presents a summarised view of the system's state. A row of three metrics — the total number of registered identities, total number of GEI samples, and average number of GEI samples per identity — is

displayed at the top. Below that, a chronological list of activity logs is provided, in descending creation order.

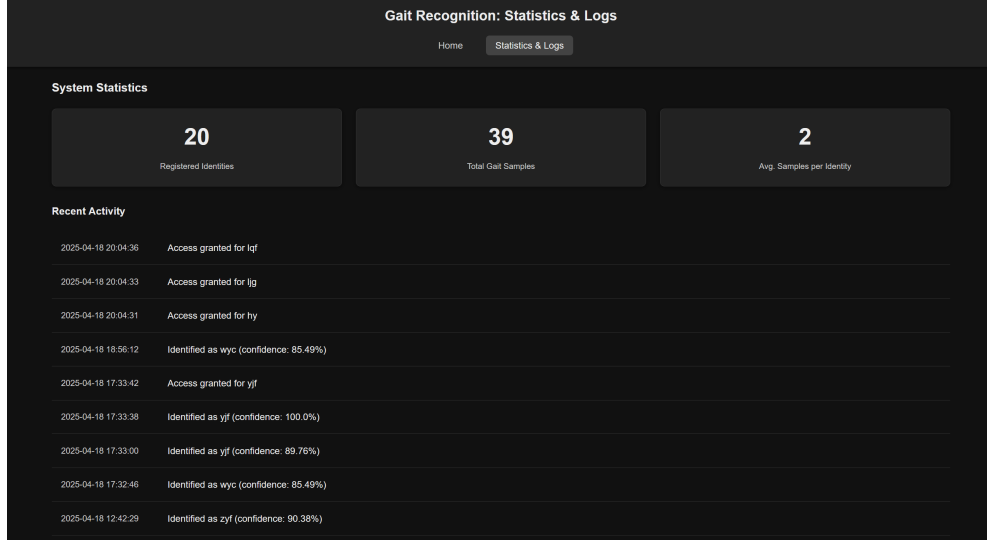


Figure 4.23: 'Statistics & Logs' Page User-Interface

Implementation of the server-side UI concluded development of the gait recognition server and overall project. Adherence of this component to the functional requirements defined in Chapter 3, along with evaluation results regarding classification performance and accuracy, are discussed in Section 5.

Overall, only one major issue was encountered during implementation: the original choice of a model-free k -NN approach for GEI classification was highly susceptible to overfitting (discussed in Section 4.5.3.2), meaning the chance of misclassification, especially with smaller datasets was higher than acceptable. However, this issue was quickly resolved thanks to the software development methodology being used, allowing revisions to be made quickly.

4.6 Implementation Summary

This chapter has presented the design and implementation of a Proof-of-Concept (PoC) Gait-Recognition (GR) authentication based access control system. The system was designed to be a modular, scalable, and secure solution that could be easily deployed as a Physical Access Control System (PACS).

The solution consists of two components: the Access Control Client (ACC) and the Gait Recognition Server (GRS), with the ACC responsible for movement data collection and

abstraction into Gait Energy Images (GEIs), and the GRS responsible for performing classification, and managing identities and their associated access configuration, with their final structure from a code perspective presented in Appendix C.2 and Appendix C.3 respectively.

The following section will cover functional, performance, and user testing of the system, based on the plans defined in Section 3.4. This is followed by an assessment of how objectives were met, and an evaluation of the feasibility of implementation in a real-world setting.

Chapter 5

Results and Discussion

5.1 Introduction

This chapter presents the results and evaluation of the overall gait-based access control system Proof-of-Concept (PoC) developed as a result of this project.

The following sections use the test plans defined in Chapter 3 to evaluate both the Access Control Client (ACC) and Gait Recognition Server (GRS) components of the system against functional and non-functional requirements. Performance and accuracy testing is also performed to assess the system’s capabilities, with results compared to the accuracies of existing biometric identification technology.

User feedback, collected from six (6) volunteers during testing and demonstrations of the PoC is also presented, with comparisons made to the public opinion survey conducted in Chapter 2.

Finally, the objectives of the project are revisited to determine if they have been met, concluding with an assessment of the feasibility of implementing a similar system in a real-world setting.

5.2 Functionality Testing

Following the functionality testing methodology defined in Section 3.4.3.1, that assesses core Functional and Non-Functional requirements of the Gait Recognition Server (GRS) and Access Control Client (ACC), the following was conducted:

FT1 — Identity Registration The GRS and ACC applications were started, with the connected camera — a Logitech C920 HD Pro Webcam — set up on a tripod facing a blank wall (for contrast). Figure 5.1 below shows an example setup of the camera and tripod facing a blank wall.



Figure 5.1: System Testing Hardware Setup

Following software initialisation and hardware setup, a subject walked in a straight line through the camera's field of view, from three meters away (ensuring the entire subject was within frame). The subject was instructed to walk at a normal pace, and was asked to stop after three seconds. Figure 5.2 below shows the resulting Gait Energy Image (GEI) preview displayed in the ACC User-Interface (UI) after the subject (this has been substituted for the report author to protect confidentiality of volunteer biometrics) had exited the frame.

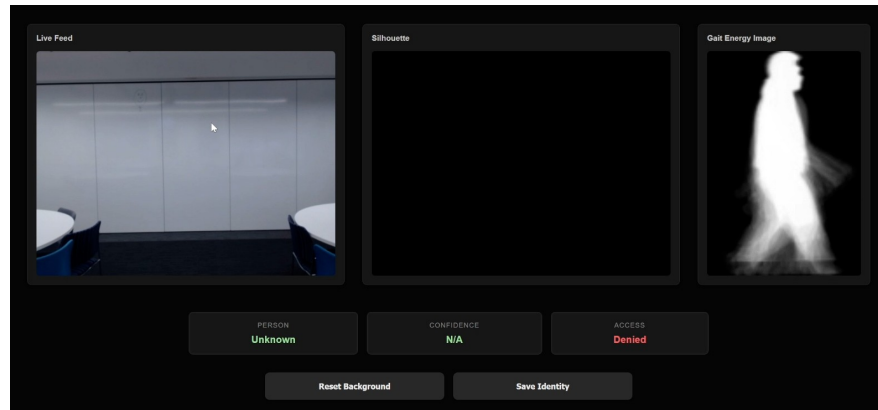


Figure 5.2: Post-Cycle-Capture GEI Preview

After the subject had exited the frame, the 'Save Identity' button was clicked, a label was entered, and the ACC application submitted the combination of the GEI and label to the GRS' `/api/register` endpoint.

The GRS UI was then refreshed, with the new registered identity displayed in the 'Registered Identities' grid, with the subject's label, ID number, and GEI preview image

— shown in Figure 5.3 below.

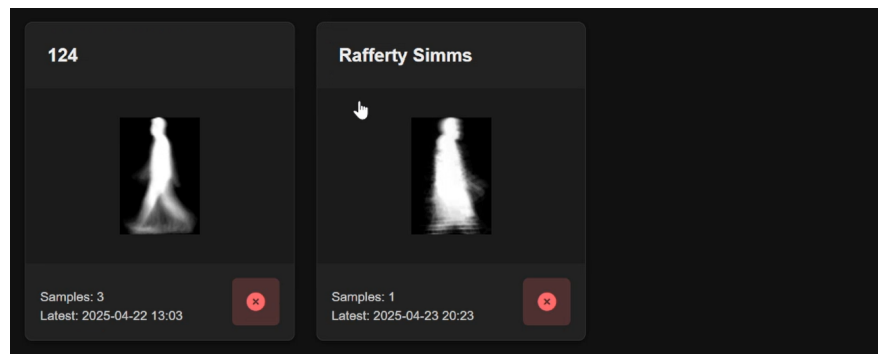


Figure 5.3: New Identity Added to Registered Identities Grid

This result aligns with what was expected for #FT1, with users able to capture gait cycles, producing a GEI, which can then be saved to the GRS database. This is a core requirement for this project as a whole.

Issues Testing of gait cycle capture and GEI creation was not smooth sailing. Although generally successful, results suffered when the background was low contrast to the subject (e.g., a dark background and dark clothing). This would result in areas of the silhouette being misidentified as background, and therefore being excluded from the silhouettes and resulting GEIs.

FT2 — Identity Removal The identity registered in #FT1 and displayed in the 'Registered Identities' grid was selected, loading its 'Identity' page. The delete button in the top right corner of the page was then clicked, displaying a confirmation dialog shown in Figure 5.4 below.

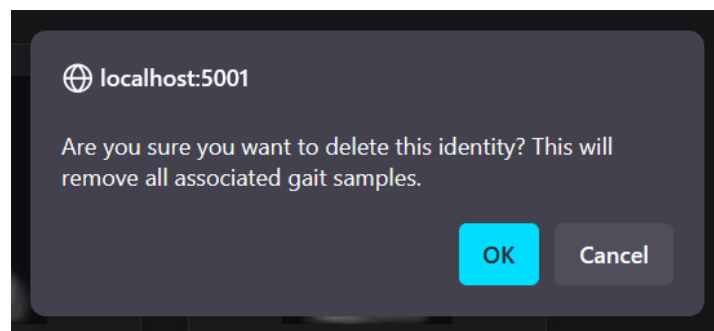


Figure 5.4: Identity Removal Confirmation Dialog

The 'OK' button was then clicked, removing the identity from the database along with

any associated GEI samples and access control rules, with the GRS UI redirecting to the home page where the grid of registered identities was empty — as expected.

This result aligns with what was expected for #FT2, with the identity and all associated data removed from the GRS database. This is a critical requirement for systems handling sensitive (biometric) data, ensuring legislations such as the General Data Protection Regulation (GDPR) are adhered to.

FT3 — Identity Classification of Known Subject Due to the prediction-based nature of the GRS' classification model, testing the efficacy of the system required a number of samples to be captured and registered before this test could be conducted. Five (5) identities were registered (using volunteers from the University's student population), with one (1) sample captured for each subject. *Note: Data was stored under anonymised labels to ensure confidentiality and was deleted after testing concluded.* Figure 5.5 below shows a redacted preview of the GRS 'Registered Identities' grid.

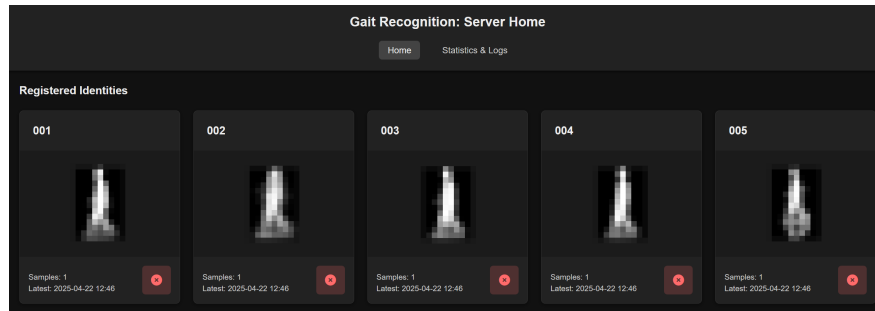


Figure 5.5: Volunteer Testing Dataset Preview

Following this, one of the subjects was instructed to walk through the camera's field of view, with the ACC application capturing another GEI sample of the subject. A few seconds later, the ACC's UI was updated with the classification result; displaying the subject's anonymised label, the prediction confidence score, and their associated (currently False) access rule — shown in Figure 5.6 below.

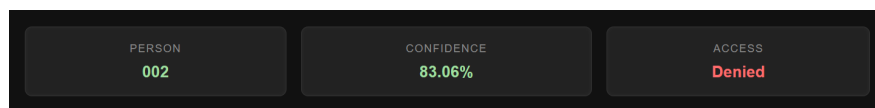


Figure 5.6: Identity Classification of Known Subject Result

This result aligns with what was expected for #FT3, proving that the system is able to correctly classify a known subject. The nature of true-positive predictions implies increased accuracy with a larger number of registered samples. The accuracy of the system's classification model is evaluated in greater depth in Section 5.3.

FT4 — Identity Classification of Unknown Subject Following population of the GRS with the dataset defined in #FT3, a sixth subject was instructed to walk through the camera's field of view, with the ACC application producing a GEI sample of the subject. The output displayed in the ACC's UI indicated the predicted identity of this subject was 'Unknown', as expected — shown in Figure 5.7 below.

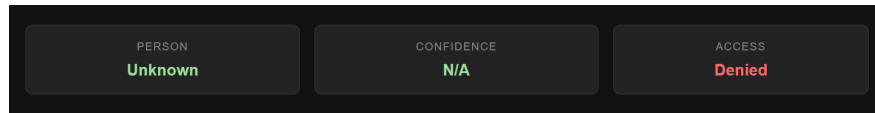


Figure 5.7: Identity Classification of Unknown Subject Result

Checking the back-end console output of the ACC actually indicated that the subject was misclassified as subject #003, but with a confidence score of 62.03% ("access":false,"confidence":62.03,"person":"003"). Because the prediction confidence was below the 75% threshold defined when implementing the ACC's access provision conditions, the UI was not updated to display the (incorrect) identity.

This result aligns with what was expected for #FT4, with the system correctly dealing with a classification prediction below the set threshold and specifying the subject's identity was unknown. This is a critical requirement for security-sensitive applications like Physical Access Control System (PACS), as misattribution of identity could result in a security breach.

FT5 — Access Rule Configuration From the GRS' UI, a registered identity was selected, loading its 'Identity' page. The toggle displayed in the top right of the page was then clicked, changing the access rule from 'Disallow' to 'Allow', providing feedback by changing the colour of the toggle from red to green — shown in Figure 5.8 below.



Figure 5.8: Access Rule Configuration Toggles

To confirm the effect of this action, the GRS database was manually checked, with the 'rule' field in the AccessRule table associated with the identity having been updated to 1, representing a True value — shown in Figure 5.9 below.

	identity_id	integer		rule	boolean
1	1			0	
2	2			0	
3	3			1	

Figure 5.9: Access Rule Configuration in Database

The steps defined in #FT3 were then repeated, with the ACC application now displaying that the (correctly) classified subject was 'Allowed' access, shown in Figure 5.10 below.

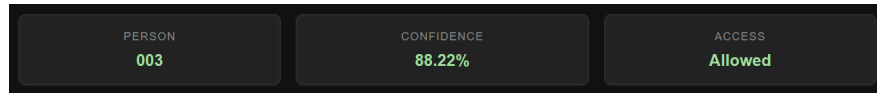


Figure 5.10: Identity Classification of Known Subject Result with Access Allowed

This result aligns with what was expected for #FT5, with the system correctly updating the access rule for the identity. This demonstrates the end-to-end functionality of Gait-Recognition (GR) being used as an authentication mechanism for access control.

FT6 — (No) Data Retention In line with legislation such as the GDPR, it is important that data is not retained for longer than necessary, meaning data used classification and not registration must not be retained by the ACC or GRS.

From observing system behaviour, the ACC' file directory, and GRS file directory and database, it was found that nothing captured, transmitted, received, or processed during classification operations was ever written to persistent storage.

This result aligns with what was expected for #FT6, with the system correctly not retaining data used for classification. There is no visual evidence to display for this test, as the requirement is that something not happen.

FT7 — Access Control Atomicity Observing the GRS database, it was found that access rule element in the ACC UI was updated after (and only after) a response to a classification request was received by the ACC from the GRS.

This result aligns with what was expected for #FT7, with the system only proceeding to access rule provision after a classification result was received.

FT8 — Client’s Dependency on the Server In a similar fashion to tests #FT1 and #FT3, the ACC was used to submit registration and classification requests while the GRS was not running. This resulted in the ACC displaying an error message stating that an identity could not be registered, or could not be classified respectively — shown in Figure 5.11 below.

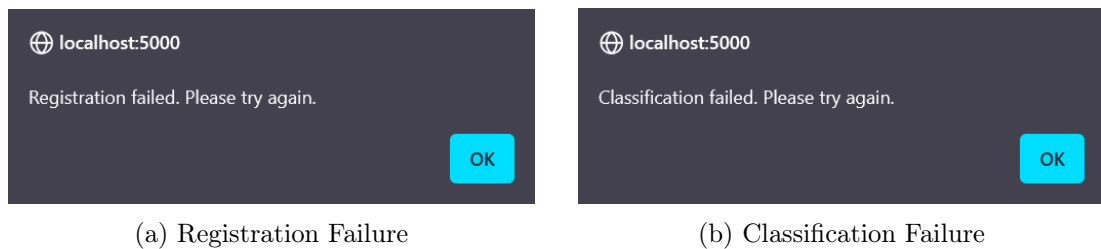


Figure 5.11: Errors Displayed by the Client if the Server is Not Running

This result aligns with what was expected for #FT8, with the system correctly not proceeding to access rule provision until a classification result was received.

FT9 — Client Camera Feed and Gait Cycle Previews The ACC was used in tests #FT1, #FT3, #FT6, #FT7, and #FT8 to capture and submit GEI samples to the GRS. Throughout these tests, the camera feed and gait cycle previews on the ACC’s UI were displayed and continuously updated as expected. Figure 5.12 below shows a screenshot of the ACC’s UI displaying the camera feed and gait cycle previews during sample capture.

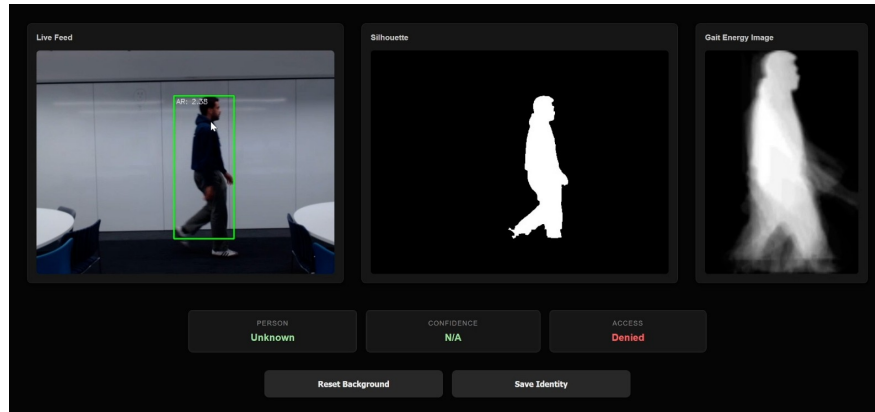


Figure 5.12: Live Feed, Silhouette, and GEI Mid-Cycle Previews

This result aligns with what was expected for #FT9, with the system correctly displaying the camera feed and gait cycle previews.

FT10 — Server Activity Logging and Display Throughout these tests, when identities were registered, removed, and classified, or when access rules were updated, the GRS' 'ActivityLog' database tables was updated with log entries as expected. Figure 5.13 below shows a screenshot of the GRS' 'ActivityLog' database table, listing several log entries relating to tests conducted in this section.

	id	activity_type	details	created_at
1	386	identify	{'confidence': np.float64(53.93), 'la...	2025-04-22 18:02:44.044876
2	393	identify	{'confidence': np.float64(58.5), 'lab...	2025-04-22 18:28:00.513817
3	387	identify	{'confidence': np.float64(62.03), 'la...	2025-04-22 18:06:36.078948
4	389	identify	{'confidence': np.float64(62.03), 'la...	2025-04-22 18:16:06.110846
5	385	identify	{'confidence': np.float64(68.53), 'la...	2025-04-22 18:01:05.142214
6	394	identify	{'confidence': np.float64(69.03), 'la...	2025-04-22 18:29:17.715271
7	388	identify	{'confidence': np.float64(83.06), 'la...	2025-04-22 18:07:17.125185
8	395	identify	{'confidence': np.float64(88.22), 'la...	2025-04-22 18:29:44.849360
9	375	delete	{'label': '001', 'sample_id': '2'}	2025-04-22 17:51:19.960497
10	376	delete	{'label': '001', 'sample_id': '3'}	2025-04-22 17:51:21.780859
11	1	register	{'label': '001'}	2025-04-22 12:46:06.026345

Figure 5.13: Server Activity Log Database Entries Example

Another requirement was for the GRS' UI to display the log entries. Its 'Statistics & Logs' page was visited and found to display the logs as expected: in descending creation order — shown in Figure 5.14 below.

Recent Activity	
2025-04-22 18:29:44	Identified as 003 (confidence: 88.22%)
2025-04-22 18:29:17	Identified as 002 (confidence: 69.03%)
2025-04-22 18:18:32	Access granted for 003
2025-04-22 18:18:32	Access denied for 003
2025-04-22 18:18:29	Access granted for 003
2025-04-22 18:18:06	Registered new gait sample for 003
2025-04-22 18:07:17	Identified as 002 (confidence: 83.06%)
2025-04-22 18:06:36	Identified as 003 (confidence: 62.03%)

Figure 5.14: Server UI Activity Log Display

This result aligns with what was expected for #FT10, with the system correctly logging and displaying activity.

The product of these ten functionality tests shows the GRS and ACC meet expected functionality as defined in Section 3.4.3.1, demonstrating how GR can be used as an authentication mechanism for access control.

5.3 Performance and Accuracy Evaluation

Tests have been conducted following the performance and accuracy testing methodology defined in Section 3.4.3.2. These tests leveraged the Gait Recognition Server (GRS) identity registration and classification Application Programming Interface (API) endpoints to automate the training and testing of its classification model using a secondary dataset. All tests were conducted on the benchmark machine (laptop) defined in Table 4.1, targeting test #PT2 in Table 3.8.

The CASIA-B dataset is a well-known dataset, created between 2012 and 2014, used for benchmarking in Gait-Recognition (GR) research. It contains 124 subjects captured multiple times from eleven (11) different angles (0deg, 18deg, ..., 180deg). Subjects were captured in three different conditions: normal, wearing a coat, and carrying a bag (Fu et al., 2023).

For these tests, only the normal walking condition, and 90deg angle (perpendicular lateral view) samples were used, resulting in six samples per subject. These samples were split into training and testing sets, with three samples used for training and three

used for testing, with no overlap between the two sets. The final dataset metrics are listed below. Although diversity details are not specified, subjects are known to be of varying age and sex.

- **Subjects:** 124 unique individuals
- **Samples per Subject:** 6 gait sequences
- **Total Samples:** 744
- **Split:** 50% Training (3 samples/subject), 50% Testing (3 samples/subject)

For anonymity, the publically available dataset is preprocessed into binary silhouette sequences (as implemented in this system in Section 4.5.1.2). To prepare the dataset for training and testing, the silhouette normalisation and Gait Energy Image (GEI) creation process was extracted from the GRS' `GaitProcessor` class, and applied to the dataset in a custom Python script. This produced a dataset of 744 GEI images, split into training and testing sets of 372 images each, labelled with the subject's ID number. Following dataset preparation, a script was created to automate the submission of each training sample to the GRS' `/api/register` endpoint, with the label defined as the subject's ID number — see Appendix D.1 Training was successful for all 372 samples, with no failed submissions. The process completed in 16 minutes and 57 seconds, with an average registration time of 2.6 seconds.

The time taken to register each sample was recorded to enable evaluation of test #PT4 in Table 3.8. Figure 5.15 below shows the distribution of registration times for each sample, with a mean of 2.6 seconds and a median of 2.9 seconds.

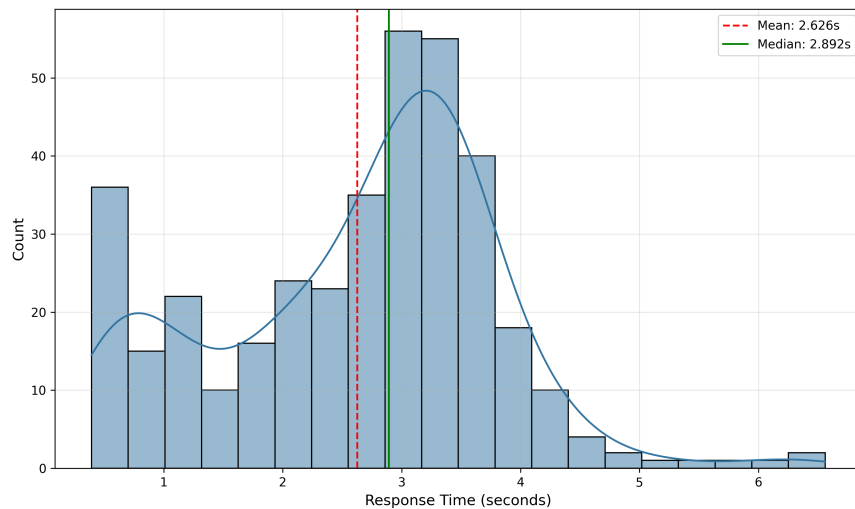


Figure 5.15: GEI Registration Response Time Distribution

It was observed that as the size of the GRS' classification model grew, registration time increased. This is likely reflects the overhead introduced by retraining the model incrementally as new GEI samples are registered.

Figure 5.16 below shows the trend of registration time as the number of samples in the GRS' classification model increased.

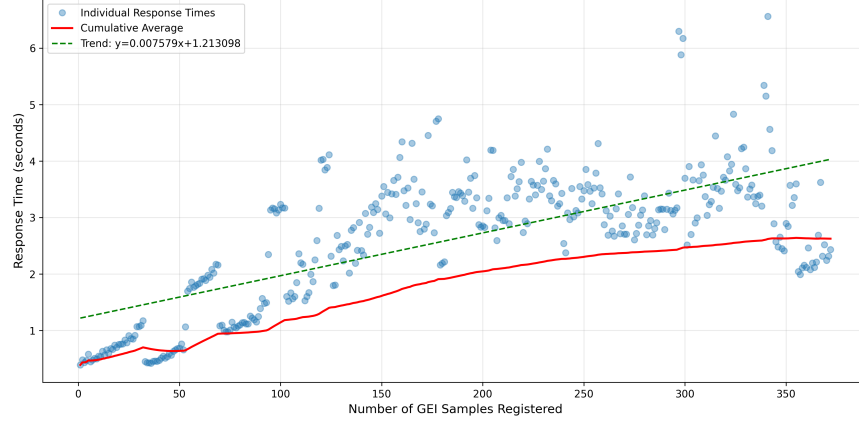


Figure 5.16: GEI Registration Response Time Trend

After training of the GRS' classification model was complete, another script was created to automate the submission of each testing sample to the GRS' `/api/classify` endpoint — see Appendix D.2 The classification results were then evaluated by comparing the predicted subject ID with the true subject ID for each sample.

By leveraging the prediction confidence score, an additional 'confidence threshold' metric was introduced to refine the analysis. This allowed overall accuracy, and accuracy above an acceptable (75%) confidence threshold to be evaluated.

The final results of this accuracy evaluation are shown in Table 5.1 below:

Table 5.1: Classification Accuracy Metric

Metric	Result
Accuracy	0.9624 (96.24%)

The metrics above show that the GRS' classification model is able to correctly classify 96.24% of samples. This however does not account for the model's prediction confidence, which is a critical factor in GR identification for security-sensitive applications, where false positives are not acceptable. To assess accuracy above an 'acceptable confidence', a 75% confidence threshold (also implemented in the Access Control Client (ACC) system's access provision output) was applied to the results.

The overall 'accepted' accuracy of the GRS' classification model after applying this threshold was found to be 72.04%. Of the 372 samples tested, 270 were classified above the confidence threshold (72.57% of the test set), with 268 samples correctly classified, and 2 misclassifications, resulting in an above-threshold precision of 99.26%.

Figure 5.17 below shows the confidence distribution across correct and incorrect classifications with the 75% confidence threshold marked.

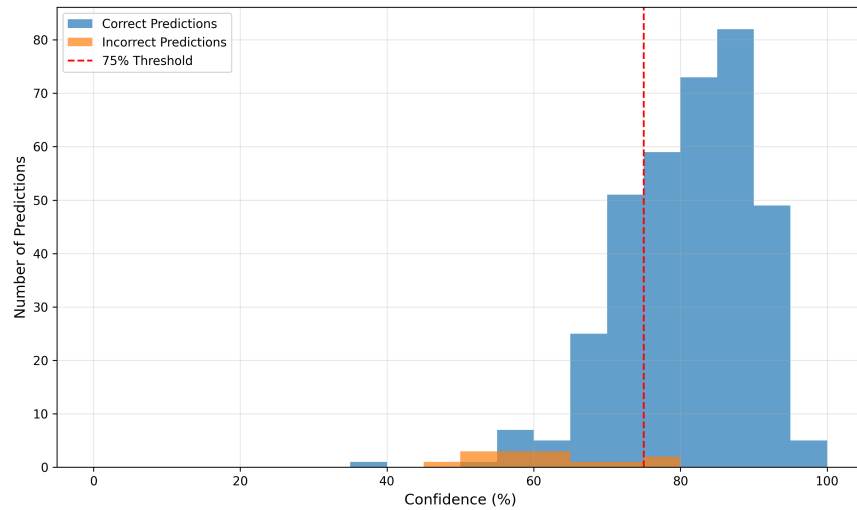


Figure 5.17: GEI Classification Confidence Distribution

To enable evaluation of test #PT3 in Table 3.8, classification request response times were recorded. Figure 5.18 below shows the response times for each sample, with a mean of 11.5 seconds.

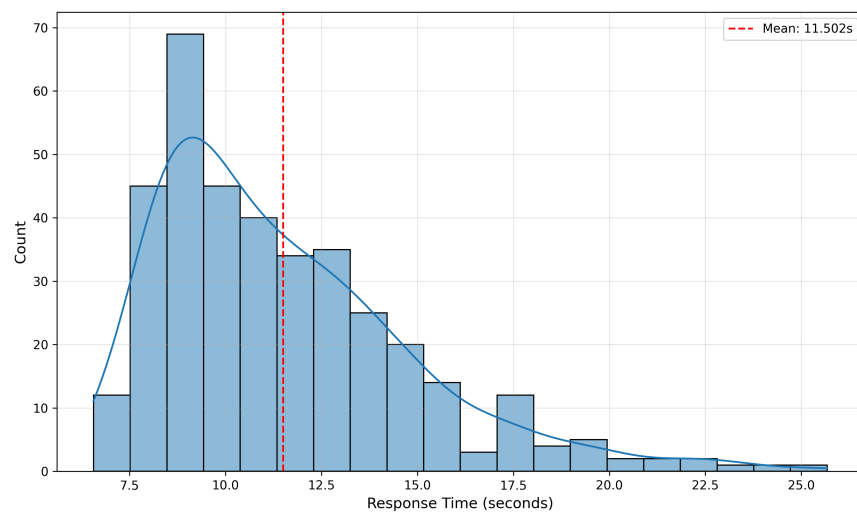


Figure 5.18: GEI Classification Response Time Distribution

Overall, these tests demonstrate that the system meets the performance and accuracy

requirements set out in Section 3.4.3.2:

PT1 — Subject Diversity Tests were conducted using the CASIA-B dataset, containing 124 unique subjects of varying age and sex.

This demonstrates that the system is able to handle a diverse range of subjects, and is not biased towards any particular demographic.

PT2 — Hardware Requirements Tests were conducted on the (consumer-grade) benchmark machine (laptop) defined in Table 4.1.

This demonstrates that the system is able to operate efficiently on hardware without significant processing power.

PT3 — Classification Speed Tests were conducted to evaluate the speed of the system’s classification model.

The mean classification time was found to be 11.5 seconds, which is below the target of at most 15 seconds.

PT4 — Registration Speed Tests were conducted to evaluate the speed of the system’s registration process.

The mean registration time was found to be 2.6 seconds, which is below the target of at most 5 seconds.

Overall, these performance and accuracy findings are positive, demonstrating that the system is able to meet expectations, and also sit at a similar level of accuracy to more popular forms of biometric identification, such as facial and fingerprint recognition which have best-accuracies north of 95%, as discussed in Chapter 2. This proves that the proposal of using GR as an alternative to more popular biometrics is sound.

5.4 User Testing Feedback

Some elements of functional testing were conducted with (6) volunteers, required to test the system’s capability of registering and classifying gait-based identities. All users

were required to read a Participant Information Sheet (PIS) and fill out an Informed Consent Form (ICF) before participating in the testing sessions. Templates of these documents can be seen in Appendix A.2 and Appendix A.3 respectively.

To discern the system's usability and understand the potential for real-world adoption, volunteers were asked to complete a questionnaire following the testing sessions, containing the following questions:

1. How easy was it to understand the instructions during the registration/onboarding process? (Very difficult, Difficult, Neutral, Easy, Very easy)
2. Did you encounter any technical issues or confusion during onboarding? Please specify if yes. (Yes, No)
3. How would you rate the speed of the authentication system? (Very slow, Slow, Acceptable, Fast, Very fast)
4. How convenient was the passive authentication compared to using a physical token, or using facial/fingerprint recognition? (Very inconvenient, Inconvenient, Neutral, Convenient, Very convenient)
5. Did the authentication process feel seamless and non-intrusive? (Not at all, Slightly, Moderately, Mostly, Completely)
6. Would you be willing to use this system in your workplace or daily environment? (Definitely not, Probably not, Not sure, Probably, Definitely)
7. Do you have any privacy, ethical, or security concerns regarding this type of behavioural biometric identification? Please specify if yes. (Yes, No)

The responses from the volunteers are summarised below:

Response 1 Volunteer 1 was a 21 year old male, he claimed to have found the registration process easy, without experiencing any technical issues. He rated the speed of the authentication system as 'Fast', and found the passive authentication process to be 'Convenient'. He found the authentication process to be 'Mostly' seamless and non-intrusive, and stated he would 'Probably' be willing to use the system in a workplace, and had no privacy reservations.

Response 2 Volunteer 2 was a 21 year old male, he found the registration process 'Very easy', and did not experience any technical issues. He rated the speed of the authentication system as 'Acceptable', and found the passive authentication process to

have 'Convenient'. He found the authentication process to be 'Completely' seamless and non-intrusive, stating he would 'Definitely' use the system in a workplace, and had no privacy concerns.

Response 3 Volunteer 3 was a 23 year old female, she found the registration process 'Easy', but stated 'It was unclear how long to wait for the system to respond when waiting for authentication'. She rated the speed of the authentication system as 'Slow', and found the passive authentication process to be 'Neutral'. She stated the authentication process was 'Slightly' seamless and non-intrusive, but admitted she would 'Probably not' be willing to use the system in a workplace, and had privacy concerns: Yes — 'I have concerns about how this type of system could be used for surveillance, the idea of being tracked without knowing makes me uncomfortable'.

Response 4 Volunteer 4 was a 53 year old male, he found the registration process 'Easy', and did not experience any technical issues. He rated the speed of the authentication system as 'Acceptable', and found the passive authentication process to be 'Neutral'. He found the authentication process to be 'Moderately' seamless and non-intrusive, but stated he was 'Not sure' if he would be willing to use the system in a workplace, yet had no privacy concerns.

Response 5 Volunteer 5 was a 23 year old female, she found the registration process 'Easy', but pointed out the fact that her silhouette was incomplete due to the low contrast between her trousers and the background. She rated the speed of the authentication system as 'Acceptable', and rated the passive authentication process as 'Neutral'. She found the authentication process to be 'Moderately' seamless and non-intrusive, but stated she would 'Probably not' be willing to use the system in a workplace, and had no privacy concerns but mentioned that she: 'Was satisfied with the convenience of using a key card'.

Response 6 (Unseen Test Subject) Volunteer 6 was a 56 year old female, she found the registration process 'Easy', and did not experience any technical issues. She rated the speed of the authentication system as 'Acceptable', and rated the passive authentication process as 'Neutral'. She found the authentication process to be 'Mod-

erately' seamless and non-intrusive, but stated she would 'Probably not' be willing to use the system in a workplace, and had no privacy concerns.

These responses convey a mixed but overall positive outlook on Gait-Recognition (GR) based authentication, with most volunteers finding the registration process and subsequent authentication process easy, experiencing no technical issues. However, some did raise concerns about privacy and the potential for surveillance, echoing past sentiments from the survey conducted in Chapter 2. Another volunteer simply stated that she was happy with the convenience of Radio Frequency Identification (RFID) key cards. From a usability and performance perspective, most volunteers rated the authentication speed positively and found the passive authentication process to be convenient.

Overall, these responses, when combined with the previous survey results, show that while there is an interest in GR based authentication, significant concerns about privacy and data handling remain. This makes sense, as technology that can authenticate users based on biometrics without their participation, could also be used for more nefarious purposes, such as non-consensual surveillance.

5.5 Evaluation Summary and Feasibility Discussion

This project has resulted in a working Proof-of-Concept (PoC) of a Gait-Recognition (GR)-based authentication system, with a registration, and passive authentication process. It uses an appearance-based GR approach, capturing movement data and creating Gait Energy Images (GEIs) to represent statistical and spatiotemporal gait features in a single image. A Machine Learning (ML)-based approach was taken to feature extraction and classification, using Convolutional Neural Network (CNN) and Principal Component Analysis (PCA), followed by cosine similarity for final predictions. This allowed for a confidence score to be generated for each prediction, with a 75% threshold set for an 'acceptable' prediction.

Prediction accuracy of the system was found to be PCA 96.24%, with a post-threshold 'acceptable' prediction accuracy of 72.04%, and the above the threshold being 99.26%. These results are promising, with the highest accuracies sitting beside the >90% accuracies of facial, fingerprint, and iris recognition discussed in Chapter 2.

The speed of registration and classification also met targets, with the average iden-

tity registration time sitting at 2.6 seconds, and the average classification time at 11.5 seconds. This could be considered acceptable, as if for example, a GR authentication system was installed in a hallway, a user could be identified and authenticated within the 30 seconds it takes to walk towards the entrance.

Although only a PoC, the final solution is surprisingly functional. The main issue encountered during testing involved the contrast level between the background and test subject, which could lead to inconsistent results. This issue can be compared to problems experienced with facial recognition systems, where lighting level can affect accuracy. Additionally, the scope of the project and tests predominantly focused on lateral movement, with little acknowledgement of other directions. Other angles may be more susceptible to occlusion, affecting accuracy (e.g., frontal views have less variance). Varying clothing and footwear may also have an impact, although this system supports multiple gait samples per identity, so this may be mitigated by collecting more data (e.g., GEIs captured from different angles, at different gradients, or when wearing different clothing). If being implemented in a real-world environment, these issues would need addressing, which may involve tightening tolerances for background subtraction, or using a different GR approach (e.g., model-based), and collecting much more data.

User feedback was generally positive, with volunteers finding the registration process easy to follow, and the passive authentication process convenient. There were some outliers in opinion, the main concerns relating to privacy and the potential for this type of technology to be used for surveillance.

Adoption of GR-based authentication in the real-world seems feasible, with issues pertaining to either inconsistent data collection, or ethics. Solving the first, as discussed, would involve further research and development to improve robustness, while the second would require clear transparency and assurances about how data is handled (e.g., through legislation) — but what if the governments are the ones leveraging the technology?

Overall, this project has successfully demonstrated GR-based authentication, justifying its proposed use in Physical Access Control System (PACS) and possibly other applications.

Chapter 6

Conclusion and Future Work

6.1 Introduction

This chapter will conclude the report, including a summary of the project's success, a self-assessment of the researcher's performance, suggestions for further research, a discussion of Legal, Ethical, Social, and Professional Issues (LESPIs), and a reflection in respect to personal aspirations.

6.2 Conclusion

The aim of this project, set in Chapter 3, was to design, implement, and evaluate the performance of a "prototype access control system that uses Gait-Recognition (GR) to passively identify subjects".

Referring to the results of Chapter 4 — Implementation and Chapter 5 — Results and Discussion, the project can be considered a success, both functional, and personal in respect to the researcher.

6.2.1 Achievements

This project successfully achieved the core aim outlined in the Project Proposal Document (PPD) and refined in Chapters 2 and 3, with all five objectives met as well as all mandatory and desired functional and non-functional requirements. The system's architecture was implemented as planned, with a client-server topology communicating via an Application Programming Interface (API). This supports scalability, with the potential for a multi-client-server setup to be implemented in the future — beneficial in the context of a Physical Access Control System (PACS).

Significant technical achievements of this project include the implementation of a Convolutional Neural Network (CNN) + Principal Component Analysis (PCA) + cosine similarity classification pipeline for identifying users based on a single image representation of their statistical and spatio-temporal gait features — a Gait Energy Image

(GEI). Fully functional identity registration and classification pipelines were also implemented, which enable live and asynchronous demonstration of the systems capabilities, aiding in the justification of implementing such a system in a real-world setting. The manual classification feature and Gait Recognition Server (GRS) 'register' and 'classify' endpoints significantly supporting the evaluation of the system's performance and accuracy.

All of this was achieved while using medium-powered consumer-grade hardware (defined in Section 4.3) for development and evaluation. This proves the system's efficiency and feasibility for cost-effective implementation in a real-world setting, especially when compared to the assumed cost of the only existing example of a GR-based PACS system — Stepscan, 2020.

6.2.2 Criticism

Despite the overwhelming success of the project, several issues were encountered, throughout both development and evaluation. The system also faces several limitations that have not been mitigated, these issues and limitations include:

Overfitting The initial choice of classification approach was a k -Nearest Neighbour (NN) model. Unfortunately, this displayed high sensitivity when tested on small datasets, resulting in overfitting where subjects were commonly misclassified, and confidently so. This necessitated further development time to transition to a CNN + PCA + cosine similarity based approach, which has proven much more effective, as evidenced by the results in Chapter 5. There still remains room for improvement, with a more targeted approach — the pre-trained `EfficientNetB0` model is a generic image classification model, and could be replaced with a model trained on a gait-specific dataset.

Data Sensitivity and Environment Inconsistency Although considered a success, the effectiveness of the silhouette extraction process is dependent on the contrast between the subject and background, meaning in varied environments, and with different subjects and clothing, accuracy is reduced.

Test Diversity While performance testing was conducted using a secondary dataset of over 100 subjects, primary testing only consisted of six (6) subjects, albeit with a

balanced sex ratio: 3 females and 3 males. The small sample size may have resulted in potential issues being missed, meaning the results may not represent the performance of the system in a real-world setting with far more subjects.

Ethical Concerns Most negative-leaning feedback received from the initial public opinion survey (see Section A.1) and post-user testing questionnaire (see Section 3.4.3.3) involved concerns about the potential misuse — especially relating to surveillance — of GR-based identification technology. If adopted in a real-world setting, legislation and policies must be updated to control the use of behavioural biometric factors, like how existing legislation controls the use of Personally Identifiable Information (PII).

6.2.3 Future Work

As it stands, several improvements could be made to this system, these have been briefly covered throughout the report, but include the following:

Multiview Capture As discussed in Chapter 2, state-of-the-art proposals in GR utilise multiple cameras to capture different angles of subjects. Increasing the number of datapoints available would likely have a direct impact on classification precision and robustness. OpenPose, the model-based pose estimation framework discussed in Section 2.3.2.1 supports multi-view input, infact it is required if wanting to produce a 3-dimensional pose estimate. Given suitable hardware is available, an 3D model-based approach may be a more effective demonstration of the potential accuracy of GR. Alternatively, multi-view GEI could be implemented, which would likely still have an positive impact on accuracy.

Hardware Optimization Although development and evaluation was performed on a consumer-grade laptop, future research could focus on optimising the system for deployment on low(er)-powered edge devices (e.g., the Nvidia Jetson Nano). If successful, this may increase interest in adopting GR in a real-world setting due to cost-effectiveness.

Accessibility Enhancements A requirement that was categorised using the MoSCoW prioritisation technique as a 'Won't Have' was to implement back-up access methods as a failsafe for when GR is not possible due to environmental conditions, or

subject impairment (e.g., wheelchair users). Implementing alternative access methods and producing a multi-modal access control architecture would be more comprehensive.

6.2.4 Final Thoughts

This project has validated the sanity in a proposition for a GR-based PACS, enabling user-independent authentication by leveraging behavioural biometrics to assess subjects from a distance.

Although only a Proof-of-Concept (PoC), the system goes far beyond what would be considered an Minimum Viable Product (MVP) and is fully functional, as proven by demonstration of it's capabilities to volunteers and during functional and performance testing.

Hopefully future research and development will be inspired by these findings, with the public sentiment and acceptance of GR increasing.

6.3 Legal, Ethical, Social, and Professional Issues

When submitting the proposal for thus project, a requirement was to consider the associated Legal, Ethical, Social, and Professional Issues (LESPIs). The original impacts in these areas where defined in the Project Proposal Document (PPD) and can be seen in Appendix E.

The following sections cover the LESPIs relating to the now finalised project.

6.3.1 Legal Issues

The core concept of this project is to use Gait-Recognition (GR) to passively identify individuals. This relies on the collection and processing of behavioural biometric data, which falls under the "special category" of data under the GDPR and the Data Protection Act 2018 (GDPR INFO, 2024; Legislation.gov.uk, 2025). This was outlined in the PPD in Section 8.1.

During the development and testing process, appropriate measures were taken to ensure compliance. This included:

- Obtaining informed consent from all participants via a participant information sheet and informed consent form.

- Anonymising collected data wherever possible (e.g., storing Gait Energy Images (GEIs) under generic identifiers and not retaining raw footage).
- Limiting data use to the purposes outlined in the participant information sheet.
- Disposing of ALL primary data securely after testing concluded.

This ensured that research remained ethical and volunteers were not subjected to any unnecessary risks. It also ensured all legal obligations were met by the researcher and supervisor during development and testing.

6.3.2 Ethical Issues

Although arguably not as susceptible to biases as other biometric recognition technologies, GR is still susceptible to biases relating primarily to differences in physical build, correlating mostly with sex and age. Primary and secondary datasets used during testing processes included a mix of males and females, with a not-explicitly-targeted, but varying nonetheless, age range. This reduced the risk of bias which, if present, would have invalidated the results and claimed accuracy figures.

Risk of data misuse was another ethical concern, echoed by feedback received during surveying and post-testing questionnaires. This has been acknowledged during the testing process through participant information sheets and informed consent forms, with all primary data being both anonymised during collection and securely disposed of afterwards.

The potential for nefarious use of GR was not explored, with surveillance falling outside of the scope of this project.

6.3.3 Social Issues

Social considerations relate primarily to the usability of the solution, as well as the potential impact on society. While GR-based authentication enables user-independent authentication, improving convenience, it does not account for individuals with disabilities, as acknowledged in the PPD (Section 8.2.2).

Though not addressed in the proposed or finalised solution, multi-modal authentication was identified as a potential solution to this issue, providing alternate methods of authentication for users with disabilities or when GR is otherwise unavailable.

As mentioned in the ethical issue assessment, the use of GR for surveillance could have

a negative societal impact, invading privacy and potentially leading to a 'panopticon' effect — a concept coined by Michel Foucault in which society is monitored by the state to maintain control (Foucault, M., 1977)

6.3.4 Professional Issues

Throughout planning, development, and testing, this project adhered to the BCS Code of Conduct. Efforts were made to uphold public trust (e.g., informing participants of what data was being collected and how it would be used) and ensure the system was built with security and fairness as priorities, with research conducted with integrity and proper attribution (BCS, 2024).

This project in no way aimed to encourage or enable unethical practices, with the resulting system designed to highlight the potential of GR for improving security and convenience, using technology that can unfortunately enable surveillance — invading privacy.

6.4 Personal Reflection and Aspirations

I am proud of the work I have produced over the course of this project, I feel that I have grown significantly in an academic, professional, and personal capacity.

Gaining experience in a once foreign field of research (both Machine Learning (ML) for which I have partaken in no previous modules, and Gait-Recognition (GR) for which I had no previous exposure to) has been a great experience, and I feel that I have a much deeper understanding of the subject and many attributing concepts than when I started.

My interest in computer security, high-performance computing/data science, software development, and general problem-solving has been further solidified, and I plan to pursue a career that encompasses these areas where possible.

Completing a project that produced technology with a potential ethical and societal impact has been interesting. I have had to consider the potential implications caused by misuse of the technology I have developed from both a philosophical and legal perspective, which is a change of scenery from the technical aspects of my studies.

This final year project has allowed me to apply knowledge gained throughout my studies

and placement year, and has necessitated the exploration of unfamiliar areas — most notably ML, and a significant amount of mathematics. I appreciate the opportunity provided by the university and support received from my supervisor, family, and friends. I look forward to applying what I have learned not only from this project, but from the studies I have completed in parallel in my future career.

Bibliography

- admin (20th July 2024). *Is Gait Recognition the Next Big Thing in Biometric Identity Verification?* Facia.ai. URL: <https://facia.ai/blog/is-gait-recognition-the-next-big-thing-in-biometric-identity-verification/> (visited on 19/01/2025).
- Bari, A. S. M. Hossain and Marina L. Gavrilova (Jan. 2022). ‘KinectGaitNet: Kinect-Based Gait Recognition Using Deep Convolutional Neural Network’. In: *Sensors* 22.7 (7), p. 2631. ISSN: 1424-8220. DOI: 10.3390/s22072631.
- BCS (2024). *BCS Code of Conduct for Members - Ethics for IT Professionals | BCS*. URL: <https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/> (visited on 24/10/2024).
- Bishop, Christopher M. (2006). *Pattern Recognition and Machine Learning*. Information Science and Statistics. New York: Springer. ISBN: 978-0-387-31073-2.
- Boon Edam (2024). *Facial Recognition System for Doors and Turnstiles*. URL: <https://www.boonedam.com/en-gb/products/accessories/facial-recognition-system> (visited on 22/11/2024).
- Boyd, Jeffrey E. and James J. Little (2009). ‘Gait Recognition, Silhouette-Based’. In: *Encyclopedia of Biometrics*. Ed. by Stan Z. Li and Anil Jain. Boston, MA: Springer US, pp. 646–652. ISBN: 978-0-387-73003-5. DOI: 10.1007/978-0-387-73003-5_36.
- Braje, Wendy L. et al. (Dec. 1998). ‘Illumination Effects in Face Recognition’. In: *Psychobiology* 26.4, pp. 371–380. ISSN: 0889-6313. DOI: 10.3758/BF03330623.
- Brennan, Kevin, ed. (2009). *A Guide to the Business Analysis Body of Knowledge(R) (BABOK(R) Guide)*. Toronto, ON, Canada: International Institute of Business Analysis. ISBN: 978-0-9811292-1-1.
- Camilla Ashdown and 2N (4th May 2023). *The History of Door Access Control Systems / 2N*. URL: <https://www.2n.com/en-GB/blog/the-history-of-door-access-control-systems/> (visited on 16/01/2025).

- Cao, Zhe et al. (30th May 2019). *OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields*. DOI: 10.48550/arXiv.1812.08008. URL: <http://arxiv.org/abs/1812.08008> (visited on 28/11/2024). Pre-published.
- Catrana, Andy, Adrian Cosma and Emilian Radoi (24th Jan. 2024). *GaitPT: Skeletons Are All You Need For Gait Recognition*. DOI: 10.48550/arXiv.2308.10623. URL: <http://arxiv.org/abs/2308.10623> (visited on 28/11/2024). Pre-published.
- Chris Burt (17th Oct. 2024). *PopEye to Strengthen EU Border Biometrics with Gait Recognition Integration / Biometric Update*. URL: <https://www.biometricupdate.com/202410/popeye-to-strengthen-eu-border-biometrics-with-gait-recognition-integration> (visited on 08/01/2025).
- Cooper, Holly et al. (2nd July 2022). ‘The Effects of Face Coverings, Own-Ethnicity Biases, and Attitudes on Emotion Recognition’. In: *Cogn. Research* 7.1, p. 57. ISSN: 2365-7464. DOI: 10.1186/s41235-022-00400-x.
- Davis, Roy B (1st Dec. 1997). ‘Reflections on Clinical Gait Analysis’. In: *Journal of Electromyography and Kinesiology* 7.4, pp. 251–257. ISSN: 1050-6411. DOI: 10.1016/S1050-6411(97)00008-4.
- Engati (2024). *Convolutional Neural Network*. Engati. URL: <https://www.engati.com/glossary/convolutional-neural-network> (visited on 07/12/2024).
- Faouzi, Johann and Olivier Colliot (2023). ‘Classic Machine Learning Methods’. In: *Machine Learning for Brain Disorders*. Ed. by Olivier Colliot. New York, NY: Humana. ISBN: 978-1-07-163194-2 978-1-07-163195-9.
- Foucault, M. (1977). *Discipline and Punish: The Birth of the Prison*.
- Fu, Yang et al. (17th Aug. 2023). *CASIA-B Pose*. Version V1. Science Data Bank. DOI: 10.57760/SCIENCEDB.10304.
- GDPR INFO (2024). *General Data Protection Regulation (GDPR) – Legal Text*. General Data Protection Regulation (GDPR). URL: <https://gdpr-info.eu/> (visited on 24/10/2024).
- Ghilom, Milkias and Shahram Latifi (Jan. 2024). ‘The Role of Machine Learning in Advanced Biometric Systems’. In: *Electronics* 13.13 (13), p. 2667. ISSN: 2079-9292. DOI: 10.3390/electronics13132667.
- Guan, Yu, Xingjie Wei and Chang-Tsun Li (1st Jan. 2014). ‘On the Generalization Power of Face and Gait in Gender Recognition:’ in: *International Journal of Digital*

- Crime and Forensics* 6.1, pp. 1–8. ISSN: 1941-6210, 1941-6229. DOI: 10.4018/ijdcf.2014010101.
- Güner Şahan, Pınar, Suhap Şahin and Fidan Kaya Gülağz (1st Sept. 2024). ‘A Survey of Appearance-Based Approaches for Human Gait Recognition: Techniques, Challenges, and Future Directions’. In: *J Supercomput* 80.13, pp. 18392–18429. ISSN: 1573-0484. DOI: 10.1007/s11227-024-06172-z.
- Han, J. and Bir Bhanu (Feb. 2006). ‘Individual Recognition Using Gait Energy Image’. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 28.2, pp. 316–322. ISSN: 1939-3539. DOI: 10.1109/TPAMI.2006.38.
- Hasan, Md. Mehedi et al. (Dec. 2022). ‘Silhouette-Based Gait Recognition Using CNN-RNN’. In: *2022 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*. 2022 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET), pp. 1–4. DOI: 10.1109/ICRPSET57982.2022.10188542.
- Home Office (23rd Mar. 2011). *London 2012: A Safe and Secure Games for All*. GOV.UK. URL: <https://www.gov.uk/government/publications/london-2012-a-safe-and-secure-games-for-all> (visited on 08/02/2025).
- Hongye, Xue and Hao Zhuoya (Sept. 2015). ‘Gait Recognition Based on Gait Energy Image and Linear Discriminant Analysis’. In: *2015 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC)*. 2015 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC), pp. 1–4. DOI: 10.1109/ICSPCC.2015.7338934.
- IFSEC Global (2022). *The 2022 State of Physical Access Control Report*. URL: <https://www.ifsecglobal.com/resources-1/the-2022-state-of-physical-access-control-report/> (visited on 20/10/2024).
- (2024). *State of Physical Access Trend Report 2024*. URL: <https://www.ifsecglobal.com/downloads-resources/the-state-of-physical-access-trend-report-2024/> (visited on 20/10/2024).
- Information Commissioner’s Office (13th Nov. 2024). *Biometric Data Guidance: Biometric Recognition*. URL: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/biometric-data-guidance-biometric-recognition/> (visited on 19/01/2025).

- Iwama, Haruyuki et al. (Oct. 2012). ‘The OU-ISIR Gait Database Comprising the Large Population Dataset and Performance Evaluation of Gait Recognition’. In: *IEEE Transactions on Information Forensics and Security* 7.5, pp. 1511–1521. ISSN: 1556-6021. DOI: 10.1109/TIFS.2012.2204253.
- Kowalczyk, Alexandre (8th June 2015). *SVM - Understanding the Math : The Optimal Hyperplane*. SVM Tutorial. URL: <https://www.svm-tutorial.com/2015/06/svm-understanding-math-part-3/> (visited on 27/01/2025).
- Legislation.gov.uk (2025). *Data Protection Act 2018*. URL: <https://www.legislation.gov.uk/ukpga/2018/12/contents> (visited on 24/04/2025).
- Leslie, David (26th Sept. 2020). *Understanding Bias in Facial Recognition Technologies*. Zenodo. DOI: 10.5281/ZENODO.4050457.
- Liao, Rijun et al. (1st Feb. 2020). ‘A Model-Based Gait Recognition Method with Body Pose and Human Prior Knowledge’. In: *Pattern Recognition* 98, p. 107069. ISSN: 0031-3203. DOI: 10.1016/j.patcog.2019.107069.
- Ling, Haibin and David W. Jacobs (Feb. 2007). ‘Shape Classification Using the Inner-Distance’. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 29.2, pp. 286–299. ISSN: 1939-3539. DOI: 10.1109/TPAMI.2007.41.
- Mallick, Tanwi et al. (2016). ‘Fast Gait Recognition from Kinect Skeletons.’ in: *Proceedings of the 11th Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*. International Conference on Computer Vision Theory and Applications. Rome, Italy: SCITEPRESS - Science and Technology Publications, pp. 340–347. ISBN: 978-989-758-175-5. DOI: 10.5220/0005713903400347.
- Maxsenti, Mike (9th Nov. 2023). *What Is Physical Access Control? - Modern Security Systems*. Geneva. URL: <https://www.getgenea.com/blog/what-is-physical-access-control/> (visited on 19/11/2024).
- Moy, Laura (2019). ‘How Police Technology Aggravates Racial Inequity: A Taxonomy of Problems and a Path Forward’. In: *SSRN Journal*. ISSN: 1556-5068. DOI: 10.2139/ssrn.3340898.
- MyEducator (2024). *MyEducator - Advantages and Disadvantages of KNN*. URL: <https://app.myeducator.com/reader/web/1421a/11/q07a0/> (visited on 05/12/2024).
- National Institute of Standards and Technology (6th July 2004). ‘NIST Study Shows Computerized Fingerprint Matching Is Highly Accurate’. In: *NIST*.

- National Institute of Standards and Technology (13th July 2021). ‘NIST Evaluates Face Recognition Software’s Accuracy for Flight Boarding’. In: *NIST*.
- Nebula M P (2nd July 2024). *Kannur Police ‘follow in Footsteps’ of Accused, Use Forensic Gait Analysis to Uncover Identity*. Kannur police ‘follow in footsteps’ of accused, use forensic gait analysis to uncover identity. URL: <https://www.onmanorama.com/news/kerala/2024/07/02/kannur-police-forensic-gait-analysis.html> (visited on 08/01/2025).
- Neves, João C. et al. (1st Jan. 2017). ‘Chapter 1 - Unconstrained Data Acquisition Frameworks and Protocols’. In: *Human Recognition in Unconstrained Environments*. Ed. by Maria De Marsico, Michele Nappi and Hugo Proença. Academic Press, pp. 1–30. ISBN: 978-0-08-100705-1. DOI: 10.1016/B978-0-08-100705-1.00001-4.
- Nunes, João Ferreira, Pedro Miguel Moreira and João Manuel R. S. Tavares (2019). ‘GRIDDS - A Gait Recognition Image and Depth Dataset’. In: *VipIMAGE 2019*. Ed. by João Manuel R. S. Tavares and Renato Manuel Natal Jorge. Vol. 34. Cham: Springer International Publishing, pp. 343–352. ISBN: 978-3-030-32039-3 978-3-030-32040-9. DOI: 10.1007/978-3-030-32040-9_36.
- Parashar, Anubha et al. (1st Apr. 2024). ‘Advancements in Artificial Intelligence for Biometrics: A Deep Dive into Model-Based Gait Recognition Techniques’. In: *Engineering Applications of Artificial Intelligence* 130, p. 107712. ISSN: 0952-1976. DOI: 10.1016/j.engappai.2023.107712.
- Parliamentary Office of Science and Technology, Vincent Saverat and Simon Brawley (23rd Sept. 2024). *Biometric Data: Misuse, Use, and Collation*. Parliamentary Office of Science and Technology. DOI: 10.58248/PN731.
- Prabhakar, S., S. Pankanti and A.K. Jain (Mar. 2003). ‘Biometric Recognition: Security and Privacy Concerns’. In: *IEEE Security & Privacy* 1.2, pp. 33–42. ISSN: 1558-4046. DOI: 10.1109/MSECP.2003.1193209.
- Sandhu, R.S. and P. Samarati (Sept. 1994). ‘Access Control: Principle and Practice’. In: *IEEE Communications Magazine* 32.9, pp. 40–48. ISSN: 1558-1896. DOI: 10.1109/35.312842.
- Shen, Chuanfu et al. (5th Aug. 2023). *A Comprehensive Survey on Deep Gait Recognition: Algorithms, Datasets and Challenges*. DOI: 10.48550/arXiv.2206.13732. URL: <http://arxiv.org/abs/2206.13732> (visited on 03/12/2024). Pre-published.

- Song, Chunfeng (18th Nov. 2024). *Developfeng/GaitRecognition*.
- Stepscan (15th Apr. 2020). *Gait Authentication Security System / Stepscan®*. URL: <https://stepscan.com/security/> (visited on 08/01/2025).
- Terranova Security (Jan. 2023). *Everything You Need to Know About Biometrics Hacking / Terranova Security*. URL: <https://www.terrano vasecurity.com/blog/hacking-biometrics> (visited on 22/11/2024).
- University of Baghdad et al. (June 2024). ‘High-Accuracy Models for Iris Recognition with Merging Features’. In: *Int. j. adv. appl. sci.* 11.6, pp. 89–96. ISSN: 2313626X, 23133724. DOI: 10.21833/ijaas.2024.06.010.
- Uwazuruike, Allwell et al. (20th Jan. 2025). ‘Police Use of Live Facial Recognition Technology’. In.
- Verkada (2024). *Guide to Secure Physical Access Control Systems*. URL: <https://info.verkada.com/door-access-systems/secure-entry-solutions/> (visited on 22/11/2024).
- Xu, Linhao et al. (9th Jan. 2024). *A Comprehensive Framework for Occluded Human Pose Estimation*. DOI: 10.48550/arXiv.2401.00155. URL: <http://arxiv.org/abs/2401.00155> (visited on 28/11/2024). Pre-published.

Appendix A

Forms and Questionnaires

A.1 Public Opinion Survey

Demographic Information

1. What is your age range?
2. What is your gender?
3. What is your race/ethnicity? (Select all that apply)
4. Which of the following best describes your current workplace environment? (Select all that apply)
5. Do you have any mobility impairments that affect how you interact with access control systems?

Current Experience with Access Control

1. Which of the following physical access control methods do you have experience using? (Select all that apply)
2. On a scale of 1-5, how satisfied are you with your current access control method(s)? (1 = Very dissatisfied, 5 = Very satisfied)
3. Have you ever experienced any of the following issues with access control methods?
4. If you've used biometric authentication (fingerprint, face, iris), have you noticed any inconsistencies in performance based on:
5. Are you aware of, or have you experienced any cases where an access control system has shown bias/discrimination or has failed to work for you while working for others?
6. If you selected Yes to the previous question, please describe the case(s) you have witnessed and/or experienced.

Preferences and Priorities

1. Rate the importance of the following factors in access control systems (1-5 scale):
 - Speed and Convenience
 - Security Level
 - Privacy Protection
 - Not having to carry additional items
 - Reliability in different conditions
2. Which of the following best describes your interaction preferences with access control systems?

Security Perceptions

1. Rate the perceived security level of the following access methods (1-5 scale):
 - RFID/key cards
 - PIN codes
 - Fingerprint recognition
 - Facial recognition
 - Gait recognition

Gait Recognition Specific Questions

1. How comfortable would you be with a system that authenticates you based on your walking style

(gait) as you approach a door?

2. Which of the following would increase your comfort with biometric authentication systems?
3. How would you prefer to be authenticated in the future?
4. Would you be willing to participate in the enrollment process (providing initial data) for a gait recognition system if it meant more convenient access afterward?

Additional Comments

1. Have you experienced any access control issues that weren't covered in this survey?
2. Do you have any additional comments, concerns, or suggestions regarding physical access control systems?

A.2 Participant Information Sheet



Nottingham Trent
University

PARTICIPANT INFORMATION SHEET

Study title

Gait Recognition-based Authentication for User-Independent Physical Access Control

Brief summary

This project explores the development and feasibility of a gait recognition-based authentication system for physical access control, aiming to provide a less obtrusive and more user-independent alternative to traditional biometric and token-based systems.

Who is doing the research and why, and what is involved in the overall study?

I, Rafferty Simms, am a final-year Computer Systems (Cyber Security) Undergraduate at Nottingham Trent University, conducting my final year project on assessing how gait recognition could be implemented within a physical access control system.

Participation in this study is entirely voluntary, and you are able to stop participating at any point without giving reason. Biometric information collected during these tests will be pseudo-anonymised and stored securely following university guidelines and UK legislation for the duration of testing, before being securely disposed of when no longer needed.

What would taking part involve for the participant, and what is the time commitment?

If participating in these tests, you will be expected to 1: walk through a camera's field of view in order to allow your walking style to be captured, and 2: fill out a questionnaire detailing your experience during testing and demonstration of the system using the biometric data you provided. The expected testing time per-subject is 10 minutes, with the questionnaire taking an extra 5-10 minutes depending on the amount of feedback you wish to give.

What are the possible benefits and disadvantages / risks of taking part?

There are no direct or immediate benefits to anyone but the researcher who benefits from the results of these tests as they enable evaluation of his solution.

However, research in this area may lead to the acceptance and adoption of frictionless access control in real-world scenarios in the future – a convenience benefit for society.

A risk assessment has been conducted on the topic of this research and content of these tests, with risks to volunteers found to be negligible.

What if something goes wrong?

Although unlikely, if something does go wrong during testing or in following stages of evaluation with effect to volunteers, the supervising academic member of staff: Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk) should be contacted directly to raise concerns or request additional information.

What will happen if I don't want to carry on with the study?

As previously mentioned, participation in this study is entirely voluntary, and you are able to stop participating at any point without giving reason.

If you would like to withdraw from this research at any point, having pseudo-anonymised data (if not yet deleted) or any submitted questionnaires disposed of, please email me (rafferty.simms2021@ntu.ac.uk) or Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk). As results will be used to provide insights into the performance, accuracy, and acceptability of the system, complete withdrawal after data analysis has begun may not be possible.

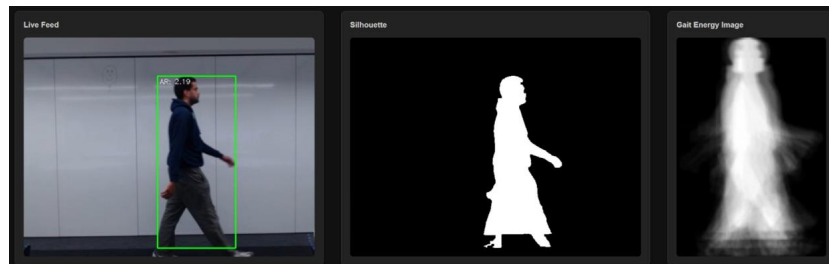
How will my information be kept confidential?

Research is a task that we perform in the public interest. Nottingham Trent University is the data controller. This means that we, as Nottingham Trent University researchers, are responsible for looking after your information and using it properly. We will use the minimum personally identifiable information possible and perform pseudo-anonymisation – samples will be labelled by unique codes. This will be stored safely in NTU data storage provision, with all biometric data destroyed following the conclusion of testing.

The age and gender of the subject will be recorded, for reference when discussing the demographic split of the dataset, as these experiments target diversity to reduce bias.

Further supporting information (full guidance covering further supporting information)

Below is an example of how your biometric gait information will be collected and represented:

**Version control**

Document Version 2, February 3rd, 2025

Contacts

For additional information, or to make a query please email me (rafferty.simms2021@ntu.ac.uk) or Dr. Nemitari Ajenka (nemitari.ajienka@ntu.ac.uk).

A.3 Informed Consent Form



INFORMED CONSENT FORM FOR PARTICIPANTS

Study Title	Gait Recognition-based Authentication for User-Independent Physical Access Control
Name of Investigator(s)	Rafferty Simms

- 1) I, [name of participant] agree to partake as a participant in the above study. I understand that my participation is voluntary and that I am free to withdraw without giving any reason and without my rights being affected
- 2) I understand from the participant information sheet (03/02/2025, V2), which I have read in full, and from my discussion(s) with Rafferty Simms that this will involve me walking through a camera's field of view to allow my movement data to be assessed for identification purposes before filling out a questionnaire. This has been stated to take no more than ten (10) minutes.
- 3) It has also been explained to me by Rafferty Simms that the risks and side effects that may result from my participation are negligible following a risk assessment.
- 4) I confirm that I have had the opportunity to ask questions about the study and, where I have asked questions, these have been answered to my satisfaction.
- 5) I undertake to abide by university regulations and the advice of researchers regarding safety.
- 6) I understand that any personal/sensitive information I have provided will be treated as confidential and in accordance with university policies. Confidential data will only be handled by those working on the study. Anonymised data may be shared with third parties for wider dissemination.

Initial Each Box

Participant name and signature		Date	
Independent witness name and signature	Nemitari Ajenka	Date	
Primary Researcher name and signature	Rafferty Simms	Date	

Appendix B

Risk Assessment Matrix

	Impact				
	Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5
Probability	Almost Certain 5	Medium 5	Very High 15	Critical 20	Critical 25
	Likely 4	Medium 4	High 12	Very High 16	Critical 20
	Moderate 3	Low 3	Medium 9	High 12	Very High 15
	Unlikely 2	Very Low 2	Medium 6	Medium 8	High 10
	Rare 1	Very Low 1	Low 3	Medium 4	Medium 5

Appendix C

Implementation

C.1 Code Extracts

C.1.1 The 'Camera' Class

```
1 class Camera:
2     def __init__(self, camera_id=0):
3         self.camera_id = camera_id
4         self.cap = None
5         self.is_running = False
6         self.current_frame = None
7         self.lock = threading.Lock()
8
9     def start(self):
10        if self.is_running:
11            return
12
13        self.cap = cv2.VideoCapture(self.camera_id)
14        if not self.cap.isOpened():
15            raise ValueError(f"Failed to open camera {self.camera_id}")
16
17        self.is_running = True
18        self.thread = threading.Thread(target=self._capture)
19        self.thread.daemon = True
20        self.thread.start()
21
22    def stop(self):
23        self.is_running = False
24        if self.thread:
25            self.thread.join()
26        if self.cap:
27            self.cap.release()
28
29    def _capture(self):
30        while self.is_running:
31            ret, frame = self.cap.read()
32            if ret:
33                with self.lock:
34                    self.current_frame = frame
35            time.sleep(0.01)
36
37    def get_frame(self):
38        with self.lock:
39            if self.current_frame is None:
```

```

40         return None
41         return self.current_frame.copy()

```

C.1.2 The 'GaitProcessor' Class

```

1 class GaitProcessor:
2     NORMALISED_WIDTH = 88
3     NORMALISED_HEIGHT = 128
4     MIN_CONTOUR_AREA = 500
5     MIN_ASPECT_RATIO = 1.5
6     MAX_ASPECT_RATIO = 3.5
7
8     def __init__(self, initial_frames=30, buffer_size=30, alpha=0.2):
9         self.background_model = None
10        self.initial_frames_count = initial_frames
11        self.background_frames = []
12        self.buffer_size = buffer_size
13        self.alpha = alpha
14        self.aligned_buffer = deque(maxlen=buffer_size) if buffer_size >
15        ↪ 0 else []
16        self.aligned_gei = None
17
18    def initialise_background(self, frame):
19        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
20        self.background_frames.append(gray)
21
22        if len(self.background_frames) >= self.initial_frames_count:
23            self.background_model = np.median(self.background_frames,
24            ↪ axis=0).astype(
25                np.uint8
26            )
27            self.background_frames = []
28            return True
29
30        return False
31
32    def reset_background(self):
33        self.background_model = None
34        self.background_frames.clear()
35        self.aligned_gei = None
36        self.aligned_buffer.clear()
37
38    def process_frame(self, frame):
39        if self.background_model is None:
40            return None, None
41
42        silhouette, bounding_box = self._extract_silhouette(frame)
43
44        if bounding_box is not None:
45            self._draw_bounding_box(frame, bounding_box)

```

```

44
45         aligned_silhouette = self._align_silhouette(silhouette,
46             ↪ bounding_box)
47         if aligned_silhouette is not None:
48             self._update_gei(aligned_silhouette)
49
50     if self.aligned_gei is not None:
51         gei = (self.aligned_gei * 255).astype(np.uint8)
52     else:
53         gei = np.zeros(
54             (self.NORMALISED_HEIGHT, self.NORMALISED_WIDTH),
55             ↪ dtype=np.uint8
56         )
57
58     return silhouette, gei
59
60 def _extract_silhouette(self, frame):
61     gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
62     gray = cv2.GaussianBlur(gray, (3, 3), 0)
63
64     diff = cv2.absdiff(self.background_model, gray)
65     _, binary = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)
66
67     silhouette = cv2.morphologyEx(
68         binary, cv2.MORPH_CLOSE, np.ones((5, 5), np.uint8)
69     )
70
71     contours, _ = cv2.findContours(
72         silhouette, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE
73     )
74     large_contours = [
75         cnt for cnt in contours if cv2.contourArea(cnt) >
76         ↪ self.MIN_CONTOUR_AREA
77     ]
78
79     filtered_silhouette = np.zeros_like(silhouette)
80     cv2.drawContours(filtered_silhouette, large_contours, -1, 255,
81         ↪ -1)
82
83     bounding_box = None
84     if large_contours:
85         largest_contour = max(large_contours, key=cv2.contourArea)
86         x, y, w, h = cv2.boundingRect(largest_contour)
87
88         aspect_ratio = h / w
89         if self.MIN_ASPECT_RATIO <= aspect_ratio <=
90             ↪ self.MAX_ASPECT_RATIO:
91             bounding_box = (x, y, w, h)
92
93     return filtered_silhouette, bounding_box
94
95 def _align_silhouette(self, silhouette, bounding_box):

```

```

91     x, y, w, h = bounding_box
92
93     person_region = silhouette[y : y + h, x : x + w]
94
95     ty, tx = np.where(person_region > 0)
96
97     if len(ty) == 0 or len(tx) == 0:
98         return None
99
100     sy, ey = ty.min(), ty.max() + 1
101     sx, ex = tx.min(), tx.max() + 1
102
103     silhouette_h = ey - sy
104     silhouette_w = ex - sx
105
106     if silhouette_h <= silhouette_w:
107         return None
108
109     cx = int(tx.mean())
110     cenX = silhouette_h // 2
111     start_w = (silhouette_h - silhouette_w) // 2
112
113     if max(cx - sx, ex - cx) < cenX:
114         start_w = cenX - (cx - sx)
115
116     square_silhouette = np.zeros((silhouette_h, silhouette_h),
117     ↪ np.uint8)
118     square_silhouette[:, start_w : start_w + silhouette_w] =
119     ↪ person_region[
120     sy:ey, sx:ex
121     ]
122
123     resized_silhouette = cv2.resize(
124     square_silhouette,
125     (self.NORMALISED_HEIGHT, self.NORMALISED_HEIGHT),
126     interpolation=cv2.INTER_AREA,
127     )
128
129     offsetX = 20
130     normalised_silhouette = resized_silhouette[
131     :, offsetX : offsetX + self.NORMALISED_WIDTH
132     ]
133
134     return normalised_silhouette.astype(float) / 255.0
135
136 def _update_gei(self, aligned_silhouette):
137     self.aligned_buffer.append(aligned_silhouette)
138
139     if self.buffer_size == 0 or len(self.aligned_buffer) ==
140     ↪ self.buffer_size:
141         if self.aligned_gei is None:
142             self.aligned_gei = np.mean(self.aligned_buffer, axis=0)

```

```

140         else:
141             new_gei = np.mean(self.aligned_buffer, axis=0)
142             self.aligned_gei = (
143                 1 - self.alpha
144             ) * self.aligned_gei + self.alpha * new_gei

```

C.1.3 Client-Side Identity Classification Thread

```

1 def classification_thread():
2     while True:
3         if state["frame_counter"] >= FRAME_INTERVAL:
4             current_gei = state["gei_buffer"]
5             last_saved_gei = state["last_saved_gei"]
6
7             if current_gei is not None:
8                 if is_different(current_gei, last_saved_gei):
9
10                    _, buffer = cv2.imencode(".jpg", current_gei)
11                    gei_base64 =
12                    ↪ base64.b64encode(buffer).decode("utf-8")
13
14                    state["last_saved_gei"] = current_gei
15
16                    response, message =
17                    ↪ api_client.classify_gei(gei_base64)
18                    print(response)
19                    if not response:
20                        print(f"Failed to classify GEI: {message}")
21                    else:
22                        socketio.emit("status", response)
23
24                    state["frame_counter"] = 0
25
26                time.sleep(1)

```

C.1.4 SQLAlchemy Database Models

```

1 class Identity(db.Model):
2     __tablename__ = "Identities"
3     id = db.Column(db.Integer, primary_key=True)
4     label = db.Column(db.String(50), nullable=False)
5     gait_samples = db.relationship(
6         "GaitSample", backref="identity", lazy=True,
7         ↪ cascade="all, delete-orphan"
8     )
9     access_rule = db.relationship(
10         "AccessRule",
11         backref="identity",

```



```

11         lazy=True,
12         uselist=False,
13         cascade="all, delete-orphan",
14     )
15
16
17 class GaitSample(db.Model):
18     __tablename__ = "GaitSamples"
19     id = db.Column(db.Integer, primary_key=True)
20     identity_id = db.Column(db.Integer, db.ForeignKey("Identities.id"),
21                             ↪ nullable=False)
22     gei_image = db.Column(db.Text, nullable=False)
23     features = db.Column(db.LargeBinary, nullable=False)
24     created_at = db.Column(db.DateTime, default=datetime.now)
25
26 class AccessRule(db.Model):
27     __tablename__ = "AccessRules"
28     identity_id = db.Column(
29         db.Integer, db.ForeignKey("Identities.id"), primary_key=True
30     )
31     rule = db.Column(db.Boolean, nullable=False, default=False)
32
33
34 class GaitModel(db.Model):
35     __tablename__ = "GaitModels"
36     id = db.Column(db.Integer, primary_key=True)
37     pca_components = db.Column(db.LargeBinary, nullable=False)
38     mean_vector = db.Column(db.LargeBinary, nullable=False)
39     n_components = db.Column(db.Integer, nullable=False)
40     created_at = db.Column(db.DateTime, default=datetime.now)
41     updated_at = db.Column(db.DateTime, default=datetime.now,
42                             ↪ onupdate=datetime.now)
43
44 class ActivityLog(db.Model):
45     __tablename__ = "ActivityLogs"
46     id = db.Column(db.Integer, primary_key=True)
47     activity_type = db.Column(db.String(20), nullable=False)
48     details = db.Column(db.Text, nullable=True)
49     created_at = db.Column(db.DateTime, default=datetime.now)

```

C.1.5 Activity Log Retrieval and Formatting

```

1 recent_activity =
2     ↪ ActivityLog.query.order_by(ActivityLog.created_at.desc()).all()
3
4 formatted_logs = []
5 for log in recent_activity:
6     timestamp = log.created_at.strftime("%Y-%m-%d %H:%M:%S")

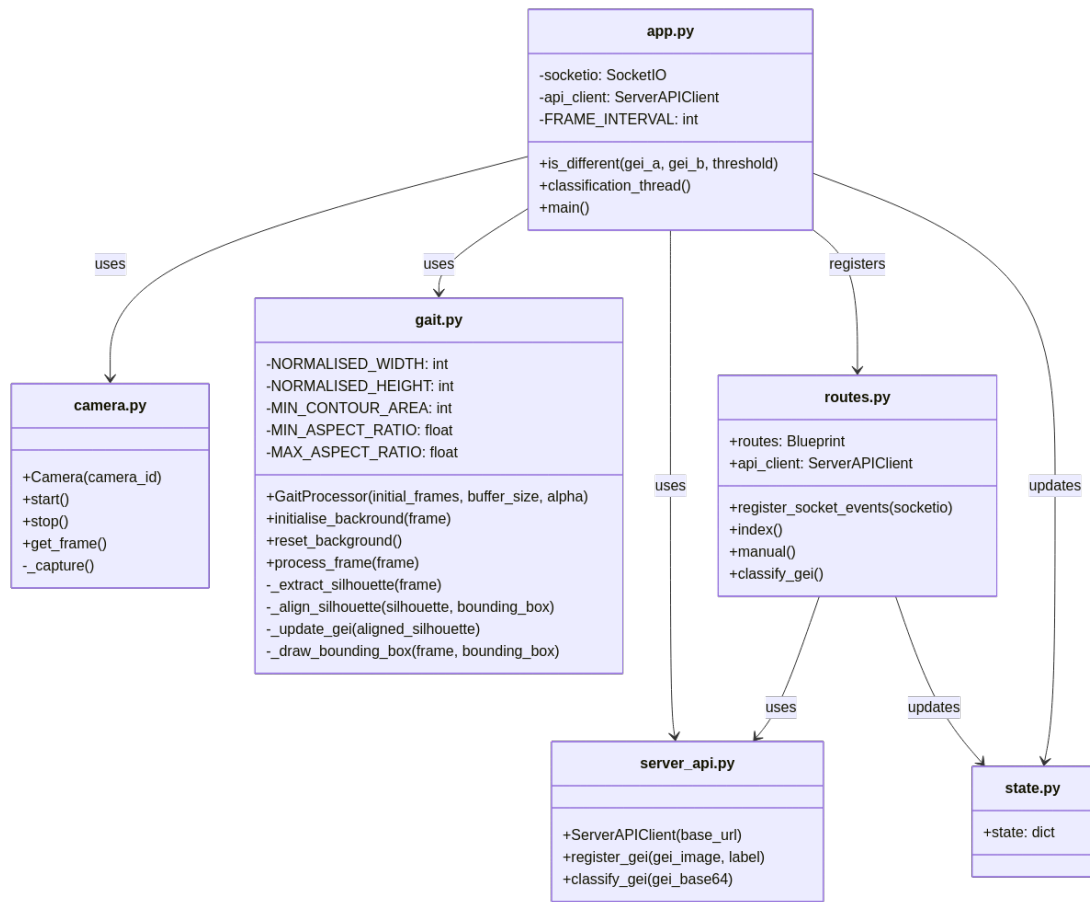
```

```

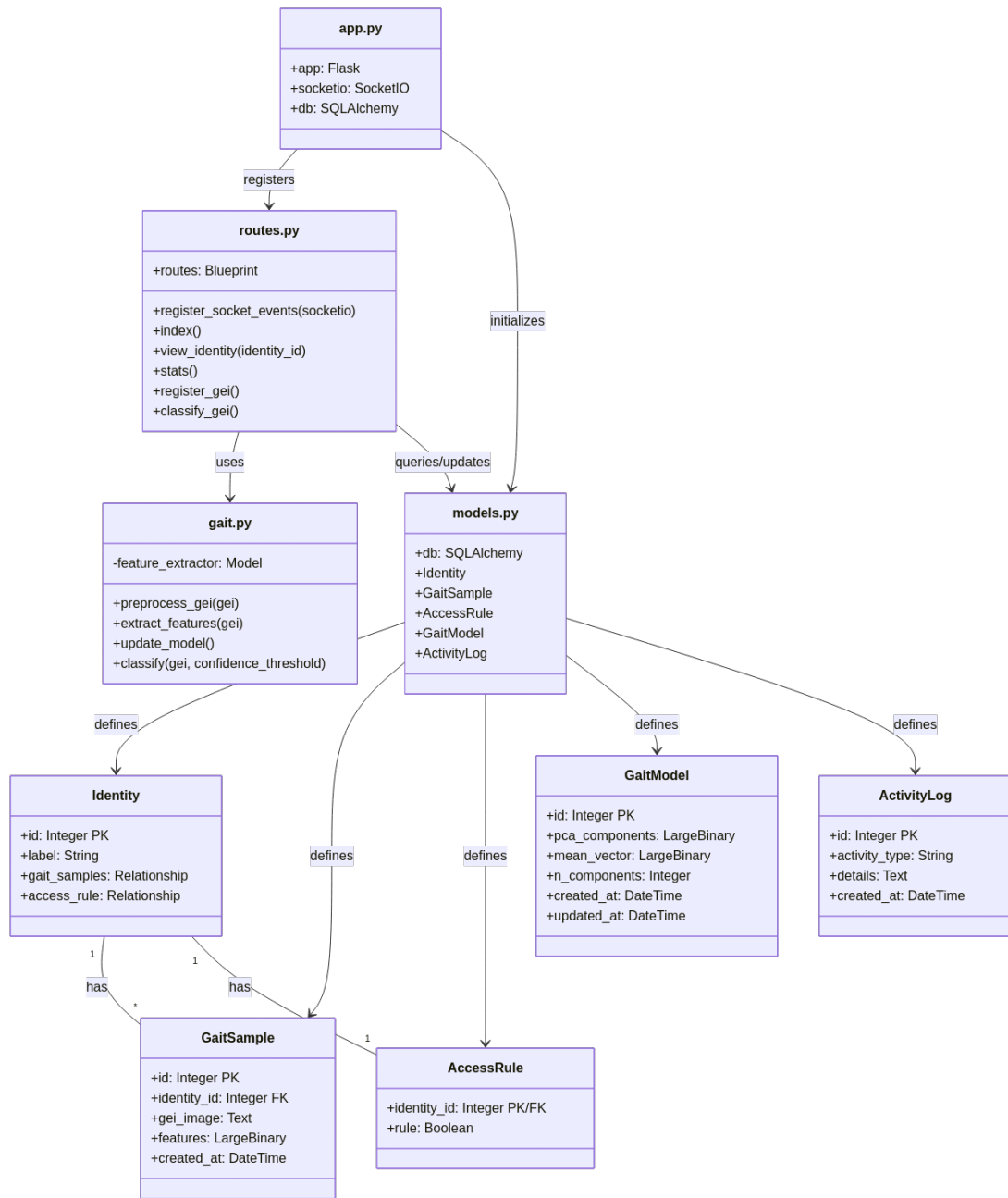
6     details = eval(log.details) if log.details else {}
7
8     if log.activity_type == "register":
9         label = details.get("label", "Unknown")
10        message = f"Registered new gait sample for {label}"
11    elif log.activity_type == "delete":
12        label = details.get("label", "Unknown")
13        if details.get("type") == "identity":
14            message = f"Deleted identity: {label}"
15        else:
16            message = f"Deleted gait sample for {label}"
17    elif log.activity_type == "access_change":
18        label = details.get("label", "Unknown")
19        new_state = "granted" if details.get("new_state") else "denied"
20        message = f"Access {new_state} for {label}"
21    elif log.activity_type == "identify":
22        label = details.get("label", "Unknown")
23        confidence = details.get("confidence", 0)
24        message = f"Identified as {label} (confidence: {confidence}%)"
25    else:
26        message = f"Unknown activity"
27
28    formatted_logs.append(
29        {"timestamp": timestamp, "message": message, "type":
30         ↪ log.activity_type}
31    )

```

C.2 Client Code Architecture Diagram



C.3 Server Code Architecture Diagram



Appendix D

Training and Testing Automation

D.1 Automated Registration Submission Function

```
1 LOG_FILE = "registration_times.csv"
2
3 def submit_gei_to_api(gei_dir, api_url):
4     gei_files = [f for f in os.listdir(gei_dir)]
5
6     if not os.path.exists(LOG_FILE):
7         with open(LOG_FILE, "w") as f:
8             f.write("timestamp,file,subject_id,response_time,status\n")
9
10    for gei_file in tqdm(gei_files):
11        gei_path = os.path.join(gei_dir, gei_file)
12        subject_id = gei_file[:3]
13
14        with open(gei_path, "rb") as img_file:
15            encoded_image =
16                ↪ base64.b64encode(img_file.read()).decode("utf-8")
17
18        data = {"gei": encoded_image, "label": subject_id}
19
20        start_time = time.time()
21        response = requests.post(api_url, json=data)
22        end_time = time.time()
23        response_time = end_time - start_time
24
25        timestamp = datetime.datetime.now().isoformat()
26
27        if response.status_code == 200:
28            status = "success"
29        else:
30            status = f"error_{response.status_code}"
31
32        with open(LOG_FILE, "a") as log_file:
33            ... Write Log Entry
```

D.2 Automated Classification Query Function

```
1 def test_gei_classification(test_dir, api_url):
2     gei_files = [f for f in os.listdir(test_dir)]
3     results = []
4
5     for gei_file in tqdm(gei_files):
6         gei_path = os.path.join(test_dir, gei_file)
7
8         true_label = gei_file[:3]
9
10        with open(gei_path, "rb") as img_file:
11            encoded_image =
12                ↪ base64.b64encode(img_file.read()).decode("utf-8")
13
14        data = {"gei": encoded_image}
15
16        start_time = time.time()
17        response = requests.post(api_url, json=data)
18        end_time = time.time()
19        response_time = end_time - start_time
20
21        if response.status_code == 200:
22            result = response.json()
23            predicted_label = result.get("person")
24            confidence = result.get("confidence", 0)
25
26            results.append(
27                {
28                    "file": gei_file,
29                    "true_label": true_label,
30                    "predicted_label": predicted_label,
31                    "confidence": confidence,
32                    "correct": predicted_label == true_label,
33                    "above_threshold": confidence >= 75.0,
34                    "response_time": response_time,
35                }
36            )
37        else:
38            ... Error Entry
39
40    with open("classification_results.json", "w") as out:
41        json.dump(results, out)
```

Appendix E

Project Proposal Document LESPIs

8 – Legal, Social, Ethical, and Professional Issues

8.1 – Legal Issues (Data Protection)

Laws such as the GDPR in the EU, or Data Protection Act in the UK exist to ensure best practices are followed when working with personal data (GDPR INFO, 2024).

9

As this project involves potentially collecting and working with personal biometric information (movement data) these laws must be adhered to and data stored securely, only used for its stated purpose (communicated in participant consent forms) and, when no longer needed, disposed of.

In the GDPR, biometric data is specially categorised if being processed to uniquely identify individuals. This purpose of this project is to do just that, therefore, explicit consent is needed for use of the data (or published with consent) (ICO, 2024).

Although cameras/sensors will be used to capture information, resulting datasets will not consist of images but coordinate data points, somewhat anonymising data and making it not personally identifiable which helps towards meeting these legal expectations.

8.2 – Social Issues

Social issues in the context of a machine learning project generally involve the privacy of subjects whose data will be used and the impact the resulting system may have on a community or in a workplace (Bacciu *et al.*, 2019).

8.2.1 – Surveillance and Privacy

Although the intended purpose of this system has good intentions, the technology produced could be repurposed and used for surveillance. For example, an organisation may be able to set this up to recognise individuals wherever they are, to automatically track how long someone spends in the cafeteria for example.

This could also be used to identify users in areas they aren't permitted, if a record is kept of who is allowed in what room depending on where sensors/cameras are setup.

The privacy concerns resulting from this can create a sense of uneasiness with users wondering what their data is being used for and when and where they are being tracked. This somewhat relates to the concept of the panopticon, a prison design where inmates are under constant observation by an entity they cannot see, resulting in less people acting out or breaking rules (Ball, Haggerty and Lyon, 2012).

8.2.2 – Accessibility and Usability

This system uses gait to uniquely identify individuals, but what happens if a disabled person, who cannot walk, needs to authenticate?

Situations like the above must be considered and the system must be designed in a way where fail safes exist (e.g., alternate methods of authentication) that still allow people unable to take advantage of core functionality to access what they need.

8.3 – Ethical Issues

Combatting ethical issues in the context of a machine learning project generally involves eliminating bias, acquiring consent to use human's data, and preventing misuse (Raji *et al.*, 2020).

8.3.1 – Machine Learning Bias

Some forms of machine learning-based biometrics have well known biases, for example, face recognition technology often struggles to differentiate individuals with darker complexions (Raji *et al.*, 2020).

In the context of this project, differences such as sex, age or disabilities could influence how someone walks and thus how their gait is assessed. If primary research is conducted, it is a good idea to collect a diverse set of data to train the machine learning model to hopefully combat this issue, making it more likely for recognition to work as intended.

8.3.2 – Risk of Misuse

Relating to 8.2.1 – Surveillance and Privacy, there is a risk that the technology produced because of this project would be misused. If implemented in a corporate environment, the most likely case of misuse would be the surveillance of employees. Depending on the level of transparency organisations have, this could be further complicated by data being collected without consent. As mentioned in 8.1 – Legal Issues (Data Protection) biometric data collected could be of special classification if it can be traced back to a person, if this is the case, organisations must acquire explicit consent for data collection.

In other contexts, this type of technology may be misused to monitor individuals in specific economic areas, perform mass surveillance, monitor gait-related medical issues, or make decisions based on gait data without human intervention (Kindt, 2013).

8.4 – Professional Issues

The BCS Code of Conduct outlines four key principles on producing work that best serves public interest, ensuring high work is produced to a high standard (BCS, 2024).

8.4.1 – Public Interest

The technology resulting from this project could have several impacts on society both positive and negative. For example, introducing gait recognition can help improve security overall – a positive. However, as mentioned in 8.2 – Social Issues, the potential to monitor activity covertly could make people uneasy. Discrimination (intentional or unintentional) could happen because of improperly calibrated biometric systems; targeting a certain demographic of individuals in a community with no accountability should not happen.

To act within public interest in mind, subjects whose data is collected by this system must be explicitly informed about what data is collected and given the option to opt out. The potential for inaccuracies or errors must be accepted and the system owner must accept responsibility for any issues (e.g., data protection, discrimination).

8.4.2 – Professional Competence and Integrity

It is critical that sufficient research is conducted throughout this project to ensure work produced is of a high standard. The gait recognition and access control system should be robust and reliable, testing must be conducted based on expectations, and the accuracy of the machine learning model should be regularly evaluated to check for errors leading to possible misidentification or discrimination.

In addition to ensuring system accuracy and reliability, the integrity and security of any data used or produced by the system must be kept. Reliable and secure data storage must be used to prevent data loss or unauthorised access.

Both the creators and implementers of the resulting access control system must have adequate knowledge of the data privacy laws associated with the types of information collected by the gait recognition system – discussed in 8.1 – Legal Issues (Data Protection), not having this knowledge could lead to accidental breaches in regulation, affecting any of the individuals whose data had been captured and potentially access maliciously.

8.4.3 – Duty to Relevant Authority

When developing his project, respect in reference to Nottingham Trent University is a must, breaches to data protection regulations or other ethical violations may affect the institution as well as the project owner (BCS, 2024).

Any concerns related to the legal handling of data, or to ethics must be raised to the most appropriate authority to ensure best practices remain followed and laws are not broken, this may be done through communication with the project supervisor who gives guidance throughout and may be able to identify unforeseen risk and advise on next steps.

In relation to the gait recognition system and collection of data through primary research and testing, subjects must be informed and give consent for their data to be collected, if there are further concerns with this, the relevant authority can advise/assist.

8.4.4 – Duty to the Profession

As a computing professional, the project owner must operate with integrity and be as transparent as possible. The intended uses of the gait recognition system must be clearly stated and not strayed from, and any data collected must only be used for the stated purpose. This project will consist of a significant amount of research, including the use of previously published papers. To contribute back, new findings should also be published for use by others, if possible, with credit and references used where applicable.

Responsibility and accountability for actions taken throughout the project belong to the project owner, thus any issues relating to breaches in data protection or improper ethics would be traced back. Operating with full transparency may curb any influence to change aspects of the project in bad faith; either due to a conflict of interest or for other reasons.